PMBOK

OVERVIEW OF EDITION 6

250621

Structure

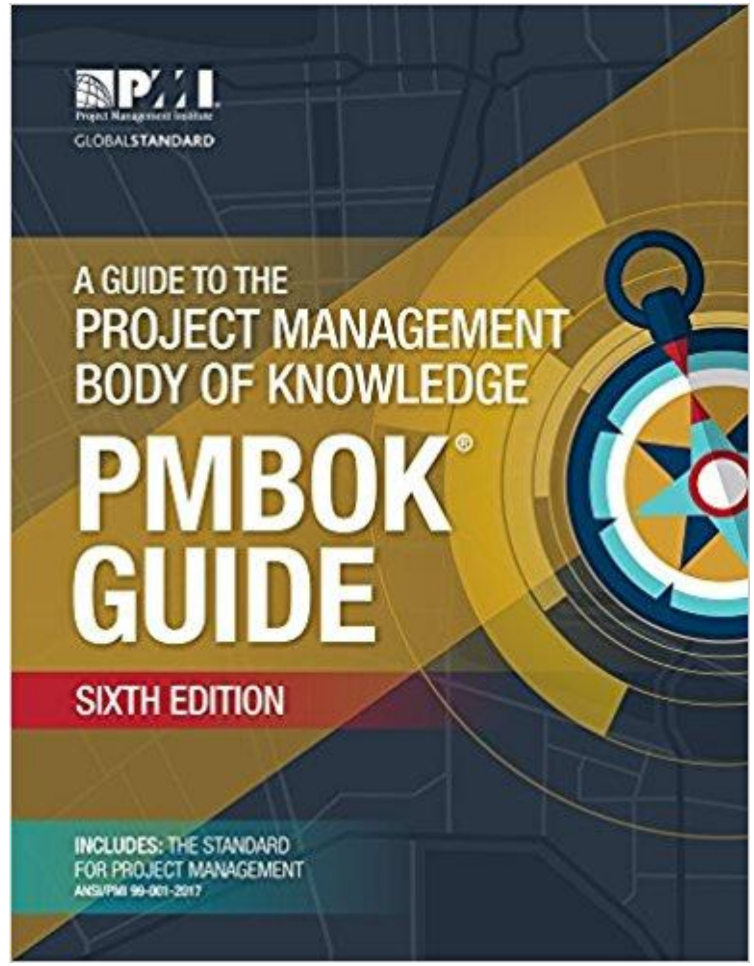
Knowledge Areas

PMBOK (adapted from Wiki)

- The Project Management Body of Knowledge is a set of standard terminology and guidelines
- It is described in ”A Guide to the Project Management Body of Knowledge” (PMBOK Guide)
- The Guide results from work overseen by the Project Management Institute (PMI)
- PMBOK approach is consistent with management standards such as ISO 9000 and the Software Engineering Institute's CMMI.
- PMI offers CAPM (Certified Associate in Project Management) and PMP (Project Management Professional) certifications

PMBOK EVOLUTION (adapted from Wiki)

- 1996: Edition 1: 180 pages
- 2000: Edition 2: 200 pages
- 2004: Edition 3: major changes.
- 2008: Edition 4: A new knowledge area – Project Stakeholder Management
- 2013: Edition 5: 600 pages
- **2017: Edition 6: 700 pages**
- 2021: Edition 7: 370 pages (replacing the 10 knowledge areas with 8 domains and 12 principles)



PMBOK KNOWLEDGE AREAS (Edition 6)

Area	Comment
Scope	Definition of the project scope, plan and lifecycle
Schedule	Project scheduling and progress control
Cost	Project budget / funding
Quality	Definition of Quality Plan and related controls
HR/ resources	Selection and management of the project staff
Communications	Communications & information inside & outside the project
Risk	Identification and control of project risks
Procurement	Management of project supplies
Stakeholder	Identification & management of project stakeholders
Integration	*Integrated management of all project areas; project initiation and closure*

PMBOK provides a guide for each major project area, called KNOWLEDGE AREA, which applies to any kind of project, including Construction (build a metro), Mechanical (build a ship) and Digital (develop software). Depending on the domain and size of a project, each Knowledge Area may be more or less relevant.

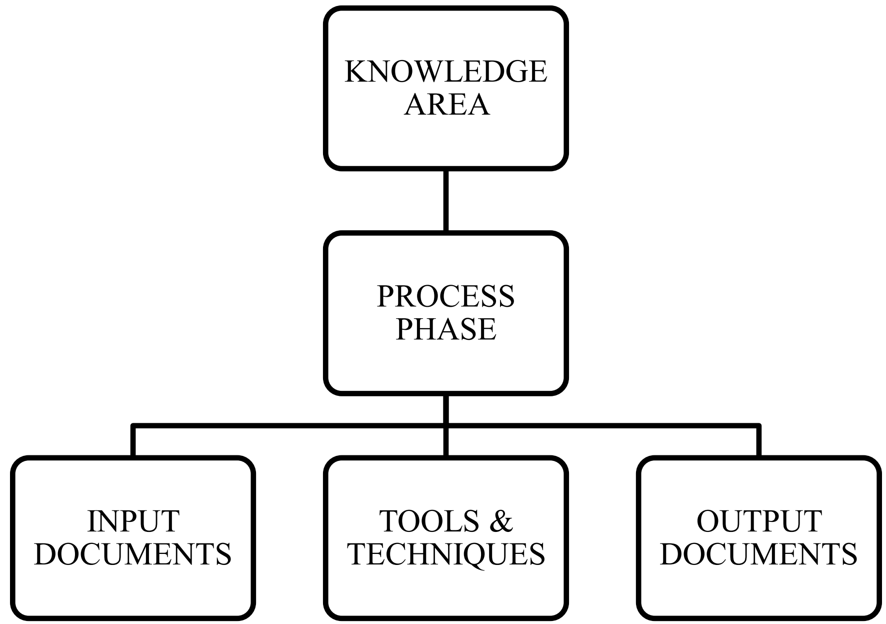
PMBOK Edition 5 versus Edition 6

	Edition 5 (2013)	Edition 6 (2017)	Comment
Pages	616	756	Including Indexes, Glossary, etc.
Knowledge areas	10	10	«External» areas (i.e. cost, stakeholder, resources) are enhanced Terminology is standardized, but becomes too generic
Agile approach	----	Ca 100 pages	A how-to-do guide by PMI and Agile Alliance A generic outline, without any historical analysis
Lifecycle selection	----	Ca 30 pages	Almost a primer No case studies, no research references

- Edition 6
  - Keeps the structure on the10 knowledge areas,
  - Adds an illustration of Life Cycles and Agile Approach
  - Extends some knowledge areas
  - Unified terminology in a flat genericity
  - Becomes very complex

CONTENT OF KNOWLEDGE AREAS

- PMBOK describes a process for each Knowledge Area
- For example, the Area «Project Scope Management» includes the following process phases:
  1. Plan Scope Management
  2. Collect requirements
  3. Define Scope
  4. Create WBS
  5. Validate Scope
  6. Control Scope
- For each process phase, PMBOK lists
  - input documents
  - tools and techniques
  - output documents



MANAGEMENT PROCESSES

Management Processes cover a management lifecycle

1. Initiating: define a new project or phase
2. Planning: establish the scope of the project, refine the objectives, and define the course of action required to attain the objectives
3. Executing
4. Monitoring and Controlling: track, review, and regulate the progress and performance of the project; identify areas in which changes to the plan are required; initiate corresponding changes.
5. Closing: finalize all activities to formally close the project or phase.

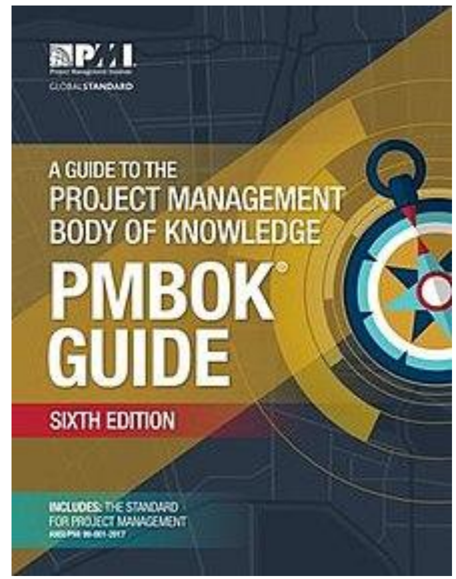
PMBOK architecture: Management Processes

	Initiate	Plan	Execute	Control	Close
Integration					
Scope					
Time					
Cost					
Quality					
HR					
Communi-cations					
Risk					
Procure-ment					
Stakeholder					

- *All Knowledge Areas follow the management cycle Plan/Execute/Control-Monitor*
- *The Project Integration Area summarizes all project management activities, and, therefore, adds 2 phases i.e.*
  - *Initiate the project or phase*
  - *Close the project or phase*

PMBOK FIT

- Fits large projects because guides almost all project management aspects
- Unfits minor projects, since it requires:
  - A structured and heavy project organization with formal rules,
  - A strong customization o select the practices to be used in a specific project,
  - A dedicated project office.



PMBOK
EDITION <u>6</u> OVERVIEW

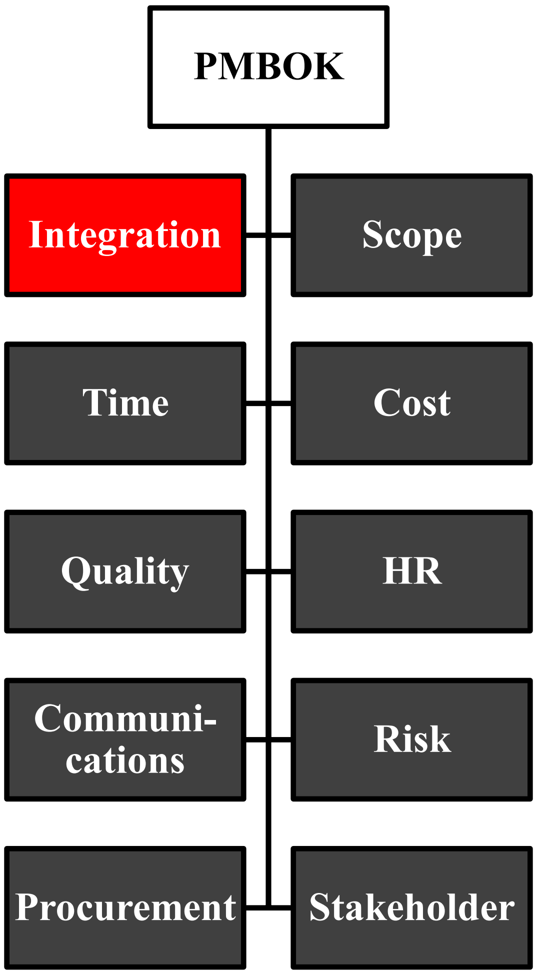
Structure

<u>Knowledge Areas</u>

Knowledge Areas

Project Integration

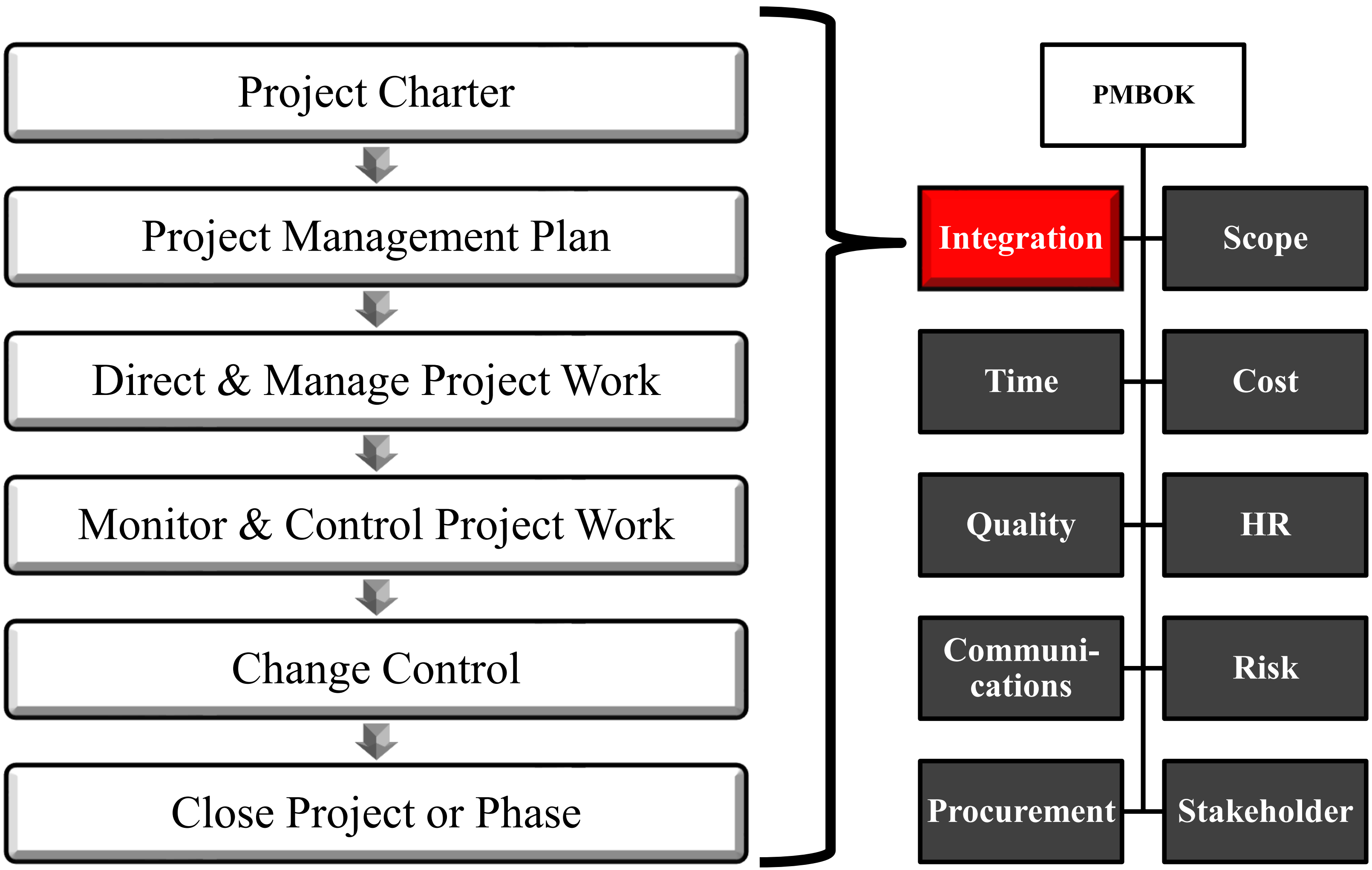
Project Integration Management



- This lecture addresses the orchestration of the management knowledge areas during the project lifecycle.
- In PMBOK, this orchestration is called "Project Management Integration" and includes a series of processes that govern the launch, work, and closure of a project.

Key slides are marked *

Project Integration Management: processes *



Project Integration Management: processes within the project life cycle *

Project Charter

Project Management Plan

Launching actions performed before the project begins (reviewed as needed at the end of a milestone / phase / work package)

Direct & Manage Project Work

Monitor & Control Project Work

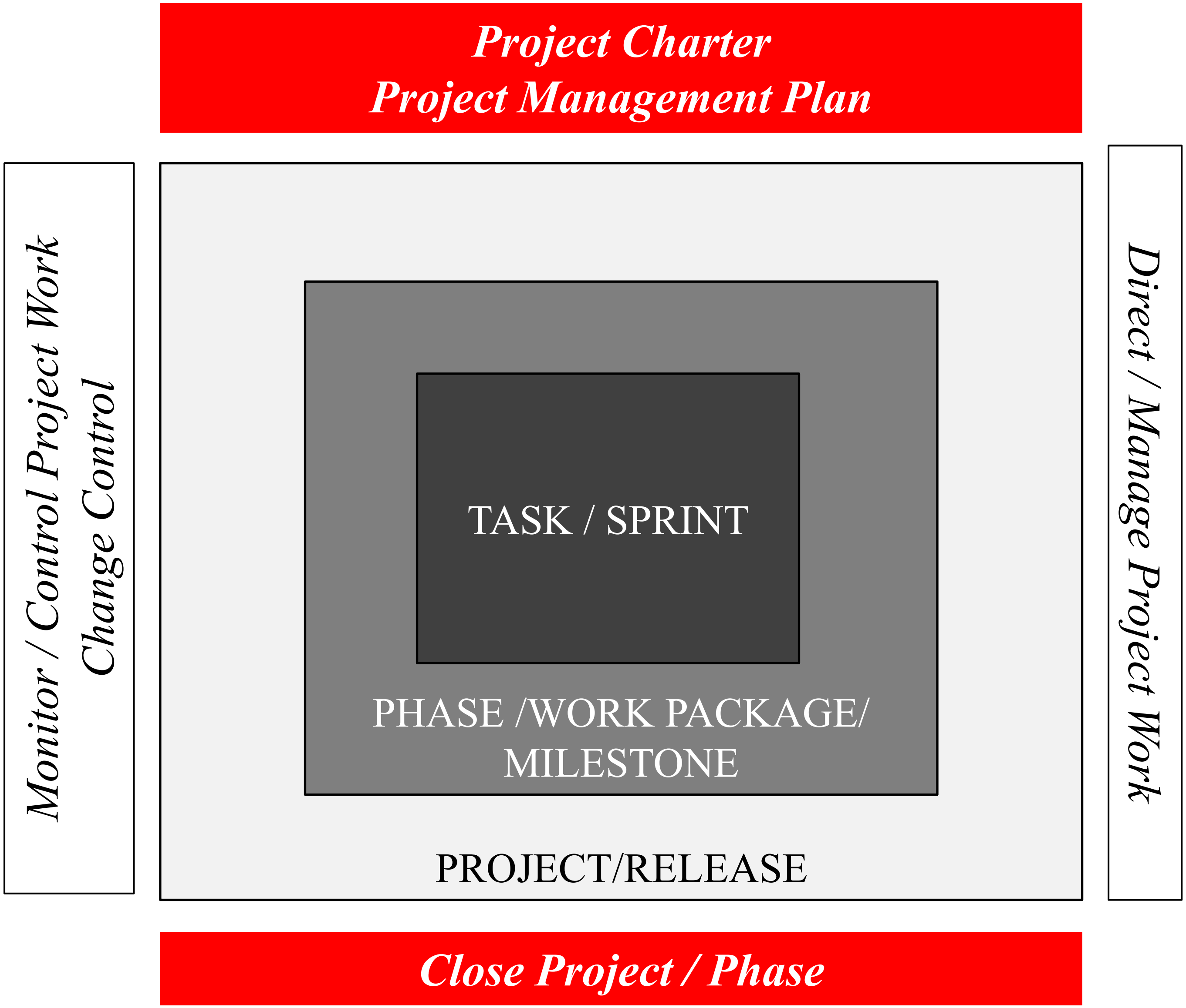
Change Control

Guidance actions, performed during the project work: by day, period (month, quarter etc.), event, task /sprint

Close Project or Phase

Closure actions, performed at the end of the project, phase, milestone, or work package.

The Chinese box of Project/Phase/Task/Sprint *



PROJECT INTEGRATION MANAGEMENT

(*INTEGRATED PROJECT MANAGEMENT*)

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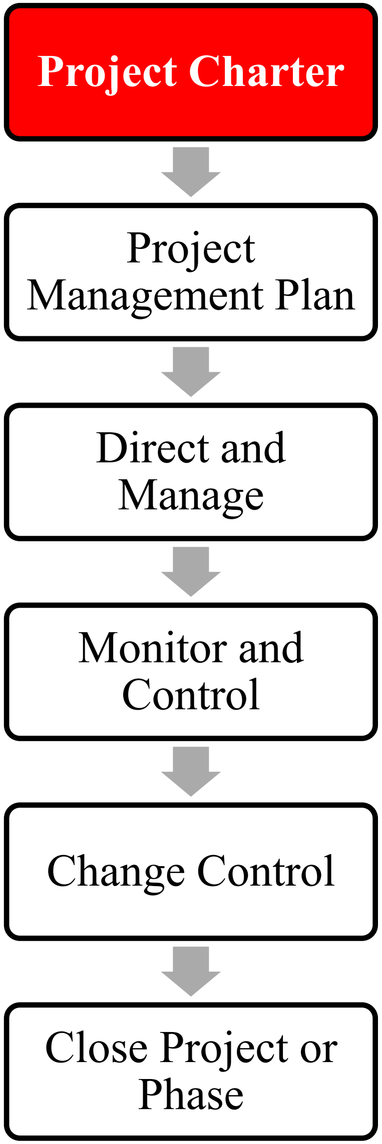
PMBOK'S FRAMEWORK

PROJECT INTEGRATION PROCESSES

APPENDIX

1. Develop Project Charter *



- It produces a document that authorizes the project and provides the project manager with the needed authority.
- The project charter establishes a partnership between the performing and requesting organizations.
- An **approved** project charter formally initiates the project.
- A project manager is identified and assigned as early in the project as is feasible, preferably while the project charter is being developed and always before the project is planned.
- The project charter should be authorized by the sponsoring entity.

1. Develop Project Charter *
Standard Contents (condensed from PMBOK ed.6 pg. 81)

- **Project Charter**
- Project Management Plan
- Direct and Manage
- Monitor and Control
- Change Control
- Close Project or Phase

Concept

1. **Project purpose**
2. **Measurable project objectives and success criteria**
3. **Hi-level requirements**
4. **Hi-level project description, boundaries, and key deliverables**

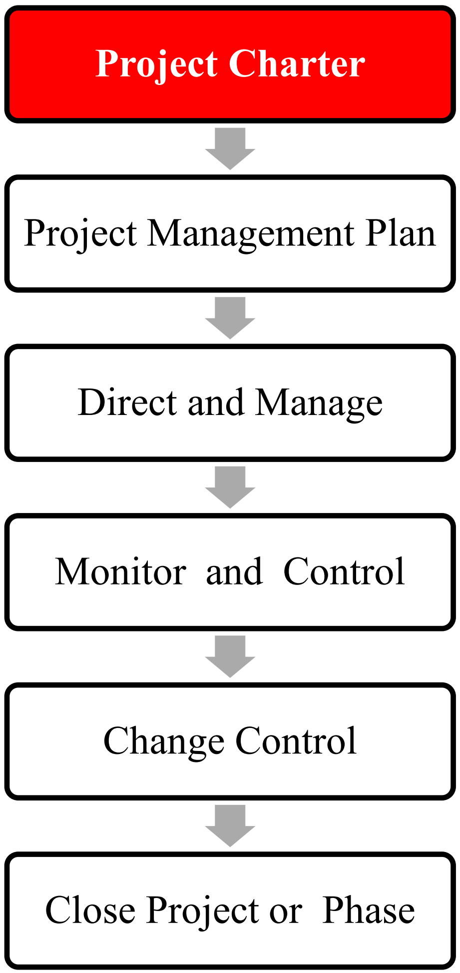
Plan & Budget

5. **Summary milestone schedule**
6. **Pre-approved financial resources (=budget)**
7. **Overall project risk**

Responsibility & Authority

8. **Key stakeholder list**
9. **Project approval requirements**
10. **Project exit criteria**
11. **Assigned project manager**
12. **Name and authority of the project sponsor**

1. Develop Project Charter: Standard Contents



- A project charter can be assimilated to a SLA (Service Level Agreement) where the customer organization buys a service from an external or internal supplier at pre-defined service level conditions.
- SLA includes time, price, penalties
- With external suppliers:
  - A typical example is a project, made by a systems integrator as IBM or Accenture, for a corporate customer, e.g., a car maker.
  - A formal contract is the typical way.
  - In the contract, the supplier responds to a Request For Proposal (RFP) issued by a corporate customer.
- With internal suppliers (e.g., Corporate IT Department):
  - The charter defines internal agreements within an organization to assure proper delivery.

Project Charter Roadmap

Customer (the user corporation)	**Supplier (usually a system integrator)**
Informal discussion in Management (Vice-Presidents and/or Demand Managers and /or CIO)	
RFP (Request For Proposal) • Subject • Budget and Deadlines • References	
	Proposal Preparation : Success stories, Qualifications, References
	Proposal presentation (demo, visits)
Proposal assessment (by CIO and /or Demand Manager)	
Selection (decision by CIO and/or CEO and/or Management Committee)	
Definition of the Project charter (= contract)	Signature of the Project Charter

1. Develop Project Charter

Project Charter

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Project Management Plan

↓

Direct and Manage

↓

Monitor and Control

↓

Change Control

↓

Close Project or Phase

Input

- Project statement of work
- Business Case
- Agreements
- Enterprise Environmental Factors
- Organizational Process Assests

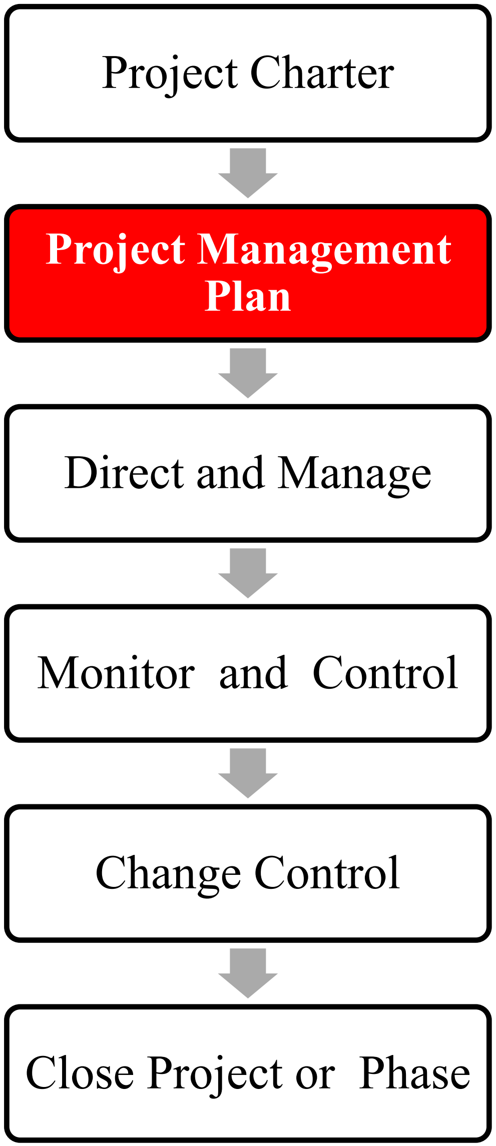
Tools & Techniques

- Expert Judgment
- Facilitation Techniques

Output

- Project Charter

2. Develop Project Management Plan



- This process defines, prepares, and coordinates all subsidiary plans and integrates them into a comprehensive project management plan.
- The project management plan defines how the project should be
  - executed,
  - monitored and controlled,
  - closed.

2. Develop Project Management Plan

Project Charter

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Project Management Plan

↓

Direct and Manage

↓

Monitor and Control

↓

Change Control

↓

Close Project or Phase

Input

- Project charter
- Miscellaneous Intputs
- Enterprise Environmental Factors
- Organizational Process Assests

Tools & Techniques

- Expert Judgement
- Facilitation Techniques

Output

- Project Management Plan

2. Develop Project Management Plan: standard contents /1 *

- Project Charter
- **Project Management Plan**
- Direct and Manage
- Monitor and Control
- Change Control
- Close Project or Phase

Project Management Plan

- *Subsidiary Management Plans*
- *Baselines*
- *Additional Components*

2. Develop Project Management Plan: Subsidiary Management Plans *

Project Charter

Project Management Plan

Direct and Manage

Monitor and Control

Change Control

Close Project or Phase

Subsidiary Management Plans = management plans by project area

Scope

Requirements

Schedule

Cost

Quality

Resource

Communications

Risk

Procurement

Stakeholder Engagement

2. Develop Project Management Plan: Baselines *

- Project Charter
- **Project Management Plan**
- Direct and Manage
- Monitor and Control
- Change Control
- Close Project or Phase

Baselines

(Initial statements of the project profile)

- *Scope (Requirements to be implemented)*
- *Schedule (Gantt/PERT)*
- *Cost (Project Budget)*

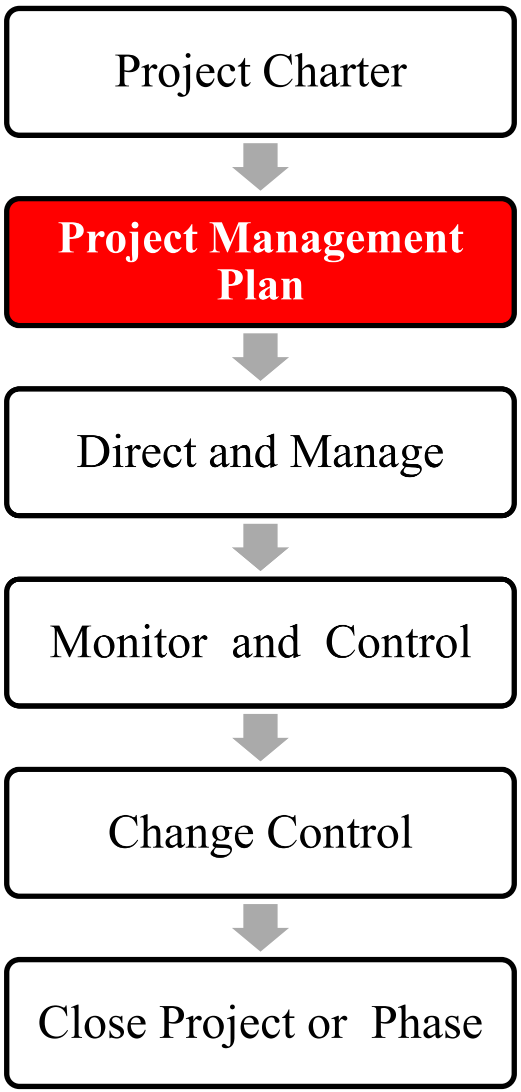
2. Develop Project Management Plan: Additional Components *

- Project Charter
- **Project Management Plan**
- Direct and Manage
- Monitor and Control
- Change Control
- Close Project or Phase

Additional Components

- *Change management plan*
- *Configuration Management Plan*
- *Performance Measurement Baseline*
- *Project Life Cycle*
- *Development Approach*
- *Management Reviews*

2. Develop Project Management Plan: Additional Components (cont.)



1. Change management plan: change procedure description
2. Configuration Management Plan: describes how to register and update the configuration of the digital product (to ensure its consistency)
3. Performance Measurement Baseline: how to compare execution against plans
4. Project Life Cycle: the series of phases (or milestones or work packages) of the project
5. Development Approach: describes the approaches to be followed - predictive (waterfall), iterative (fill-gap), agile, hybrid.
6. Management Reviews: identifies the points where project managers and relevant stakeholders will review project progress (in fact it coincides with project milestones)

3. Direct and Manage Project Work *

Project Charter

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Project Management Plan

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Direct and Manage

↓

Monitor and Control

↓

Change Control

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Close Project or Phase

- It is the process of leading and performing the work defined in the project management plan and implementing approved changes to achieve the project's objectives.
- Process activities include almost every aspect of the project

Integration:
3. Direct and Manage Project Work

- Project Charter
- Project Management Plan
- **Direct and Manage**
- Monitor and Control
- Change Control
- Close Project or Phase

Input

- Project Management Plan
- Approved Change Requests
- Enterprise Environmental Factors
- Organizational Process Assests

Tools & Techniques

- Expert Judgement
- Project Management System
- Meetings

Output

- Deliverables
- Work Perfomance Data
- Change Requests
- Project Management Plan updates
- Project Document Updates

4. Monitor and Control Project Work *

Project Charter

Project Management Plan

Direct and Manage

Monitor and Control

Change Control

Close Project or Phase

- Monitoring is performed throughout the project; it includes
  - collecting, measuring, and distributing performance information,
  - assessing measurements and trends to effect process improvements.
- Control determines
  - corrective or preventive actions or re-planning
  - following up to determine whether the actions taken resolved the performance issue.

4. Monitor and Control Project Work

Project Charter

Project Management Plan

Direct and Manage

Monitor and Control

Change Control

Close Project or Phase

Input

- Project Management Plan
- Schedule Forecasts
- Cost Forecasts
- Validated Changes
- Work Perfomance Data
- Enterprise Environmental Factors
- Organizational Process Assests

Tools & Techniques

- Expert Judgement
- Analytical Techniques
- Project Management Information System
- Meetings

Output

- Change Requests
- Work Perfomance Reportsd
- Project Management Plan updates
- Project Document updates

5. Perform Integrated Change Control *

Project Charter

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Project Management Plan

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Direct and Manage

↓

Monitor and Control

↓

Change Control

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Close Project or Phase

- It is the process of reviewing all change requests; approving changes and managing changes to deliverables, organizational process assets, project documents, and the project management plan; and communicating their disposition.
- It reviews all requests for changes or modifications to project documents, deliverables, baselines, or the project management plan and approves or rejects the changes.
  - Changes may be requested by any stakeholder involved with the project.
  - Change requests are subject to the process specified in the change control and configuration control systems.
  - Those change request processes may require information on estimated time impacts and estimated cost impacts.

5. Perform Integrated Change Control

Project Charter

Project Management Plan

Direct and Manage

Monitor and Control

Change Control

Close Project or Phase

Input

- Project Management Plan
- Work Perfomance Reports
- Change Requests
- Enterprise Environmental Factors
- Organizational Process Assests

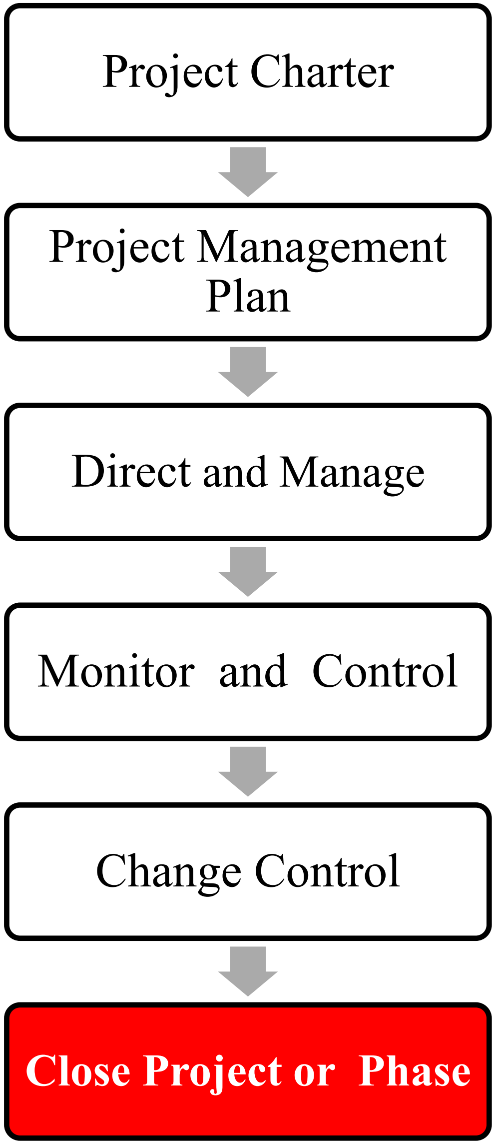
Tools & Techniques

- Expert Judgement
- Meetings
- Change Control Tools

Output

- Approved Change Requests
- Change Log
- Project Management Plan updates
- Project Document Updates

6. Close Project or Phase *



- It is the process of finalizing all activities to formally complete the whole project or a project phase.
- When closing the project, the project manager reviews all prior information from the previous phase closures to ensure that all project work is complete, and the project has met its objectives.
- Since project scope is measured against the project management plan, the project manager reviews the scope baseline to ensure completion
- The Close Project or Phase process also establishes the procedures to investigate and document the reasons for actions taken if a project is terminated before completion.
- To successfully achieve this, the project manager needs to engage all the proper stakeholders in the process.

6. Close Project or Phase

Project Charter

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Project Management Plan

↓

Direct and Manage

↓

Monitor and Control

↓

Change Control

↓

Close Project or Phase

Input

- Project Management Plan
- Accepted Deliverables
- Organizational Process Assets

Tools & Techniques

- Expert Judgement
- Analytical Techniques
- Meetings

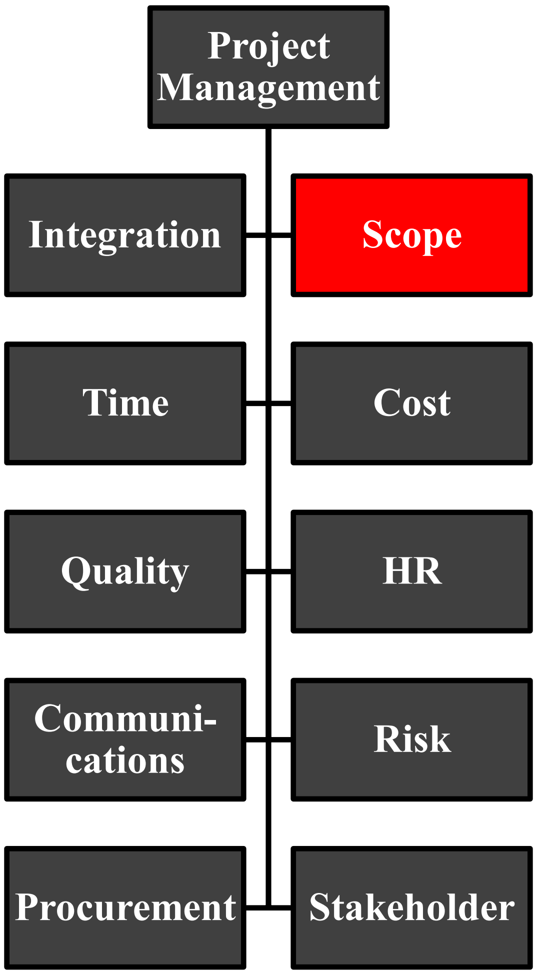
Output

- Final Product, service, or service transition
- Organizational Process Assets updates

Knowledge Areas

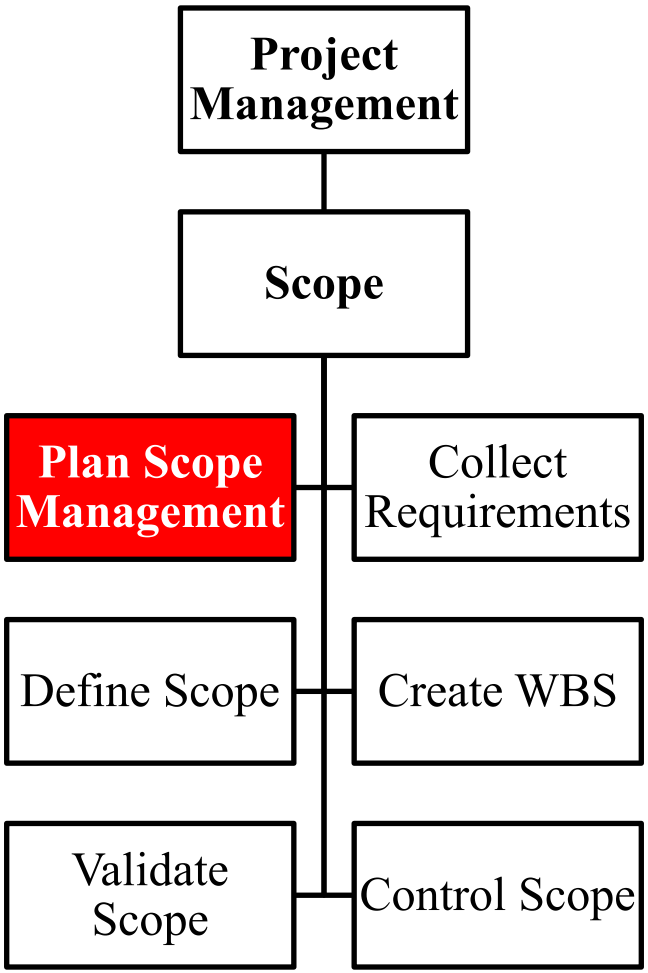
Project Scope

PMBOK: 2. Project Scope Management



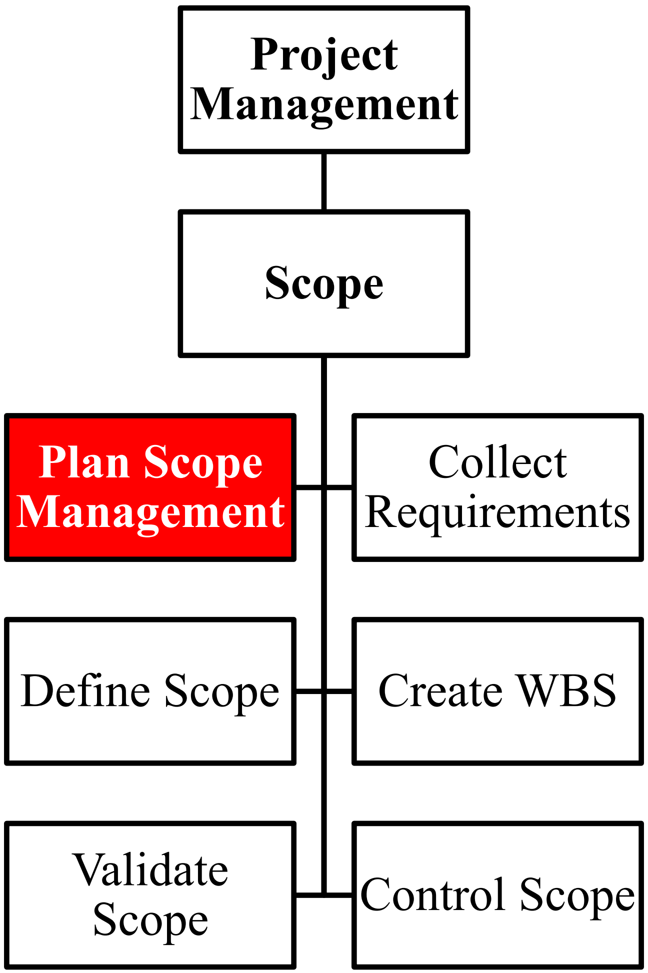
- Project Scope Management includes the processes required to ensure that the project includes all and only the work required to complete the project successfully.
- Managing the project scope primarily concerns defining and controlling what is / is not included in the project.

Scope: 1. Plan Scope Management



- It is the process of creating a scope management plan that documents how the project scope will be defined, validated, and controlled.
- The scope management plan describes how the scope will be defined, developed, monitored, controlled, and verified.
- The development of the scope management plan and the detailing of the project scope begin with
  - the analysis of information contained in the project charter,
  - the latest approved subsidiary plans of the project management plan,
  - historical information contained in the organizational process assets,
  - any other relevant enterprise environmental factors.

Scope: 1. Plan Scope Management



Input

- Project Management Plan
- Project Charter
- Enterprise Environmental Factors
- Organizational Process Assets

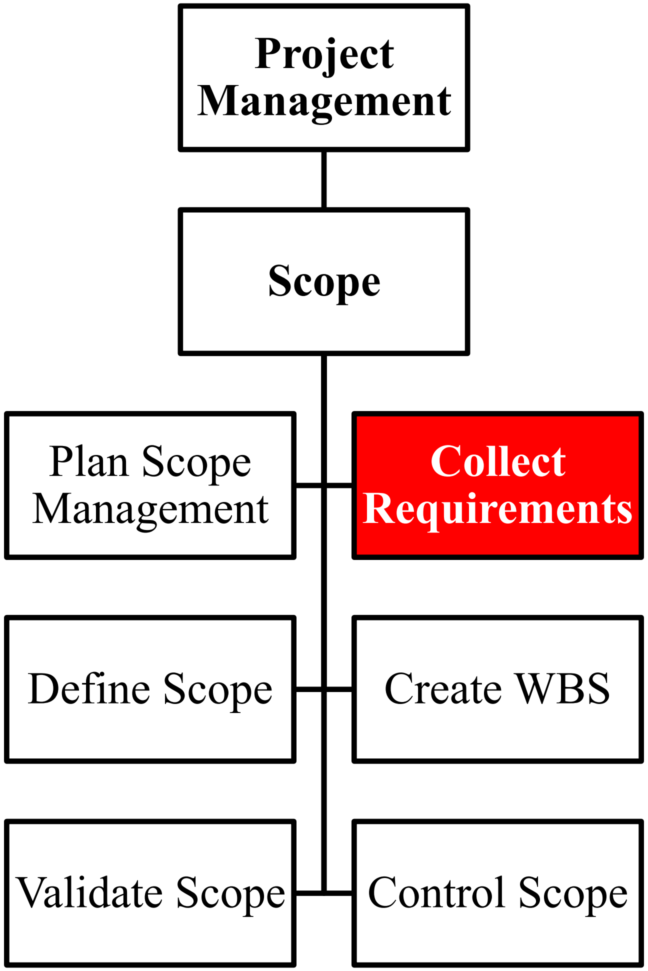
Tools & Techniques

- Expert Judgement
- Meetings

Output

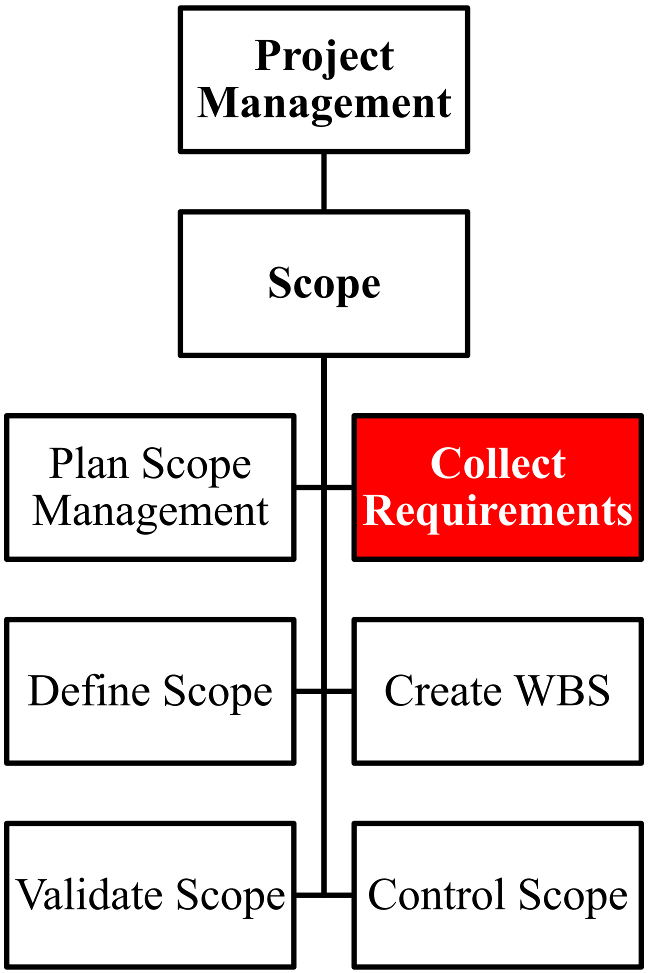
- Scope Management Plan
- Requirements Analysis Management Plan

Scope: 2. Collect Requirements



- It is the process of determining, documenting, and managing stakeholder needs and requirements to meet project objectives.
- Requirements include conditions or capabilities that are to be met or result to satisfy an agreement, or other formally imposed specification.
- Requirements include the quantified and documented needs and expectations of the sponsor, customer, and other stakeholders.
- Requirements need to be elicited, analyzed, and recorded in enough detail to be included in the scope baseline and to be measured once project execution begins.
- Requirements become the foundation of the WBS.

Scope: 2. Collect Requirements



Input

- Scope Management Plan
- Requirements Analysis Management Plan
- Stakeholder Management Plan
- Stakeholder Register
- Project Charter

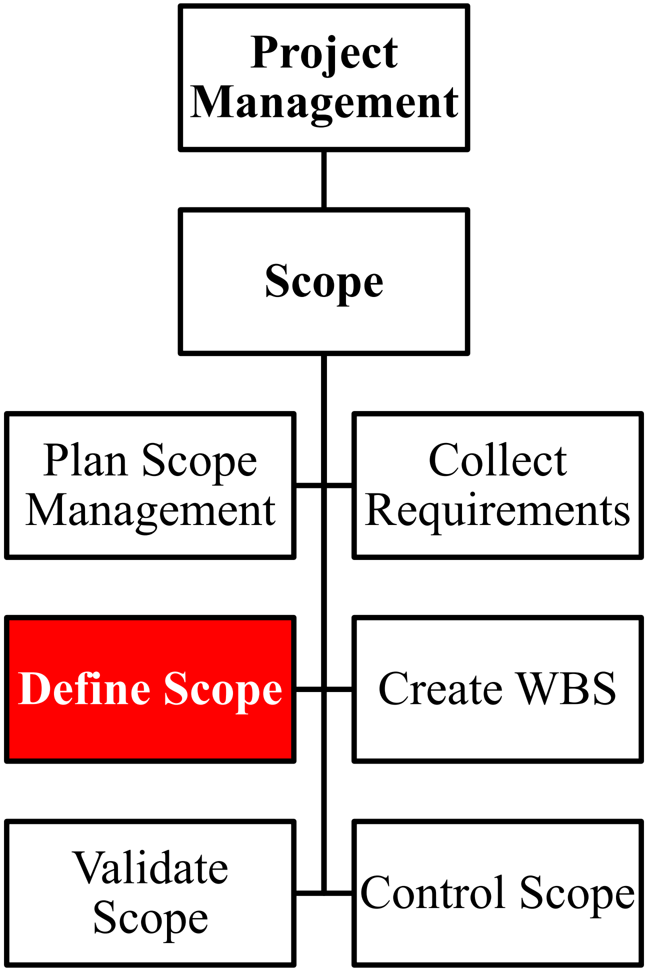
Tools & Techniques

- Interviews
- Focus Groups
- Facilitated Workshops
- Group Creativity Techniques
- Group Decision Making Techniques
- Questionnaires and Surveys
- Observations
- Protototypes
- Benchmarking
- Context Diagrams
- Document Analysis

Output

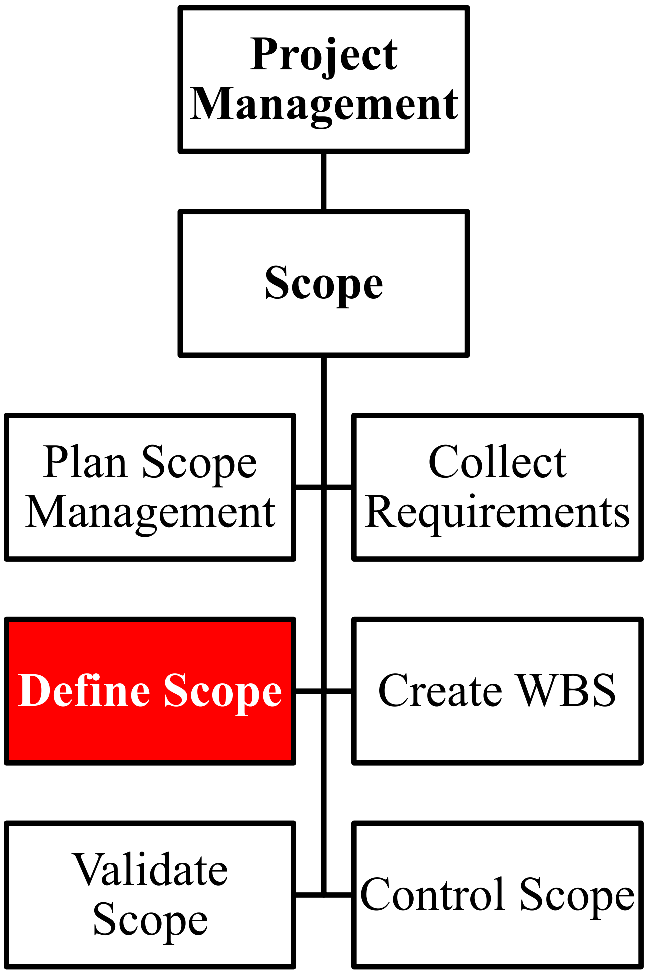
- Requirements Documentation
- Requirements Traceability Matrix

Scope: 3. Define Scope



- It is the process of developing a detailed description of the project and product.
- Since some requirements identified in Collect Requirements may not be included in the project, the process selects the final project requirements and then develops a detailed description of the project and product, service, or result.
- During project planning, the project scope is defined and described with greater specificity as more information about the project is known.
- Existing risks, assumptions, and constraints are analyzed for completeness and added or updated as necessary.
- The process may be iterative. In iterative life cycle projects, a high-level vision will be developed for the overall project, but the detailed scope is determined one iteration at a time and the detailed planning for the next iteration is carried out as work progresses on the current project scope and deliverables.

Scope: 3. Define Scope



Input

- Scope Management Plan
- Project Charter
- Requirements Documentation
- Organizational Process Assets

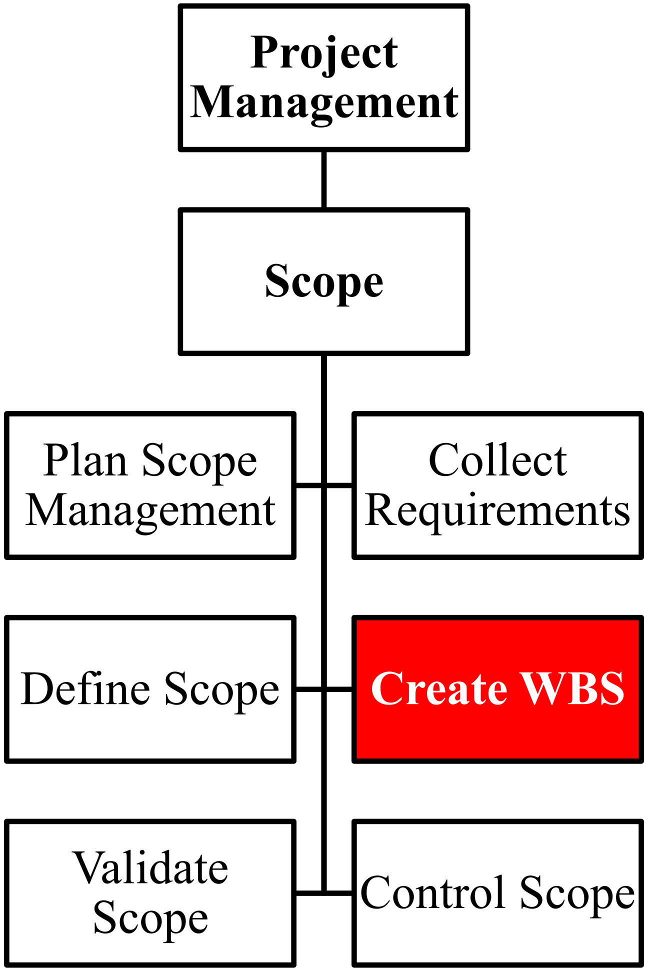
Tools & Techniques

- Expert Judgement
- Ppoduct Analysis
- Alternative Generation
- Facitatated Workshops

Output

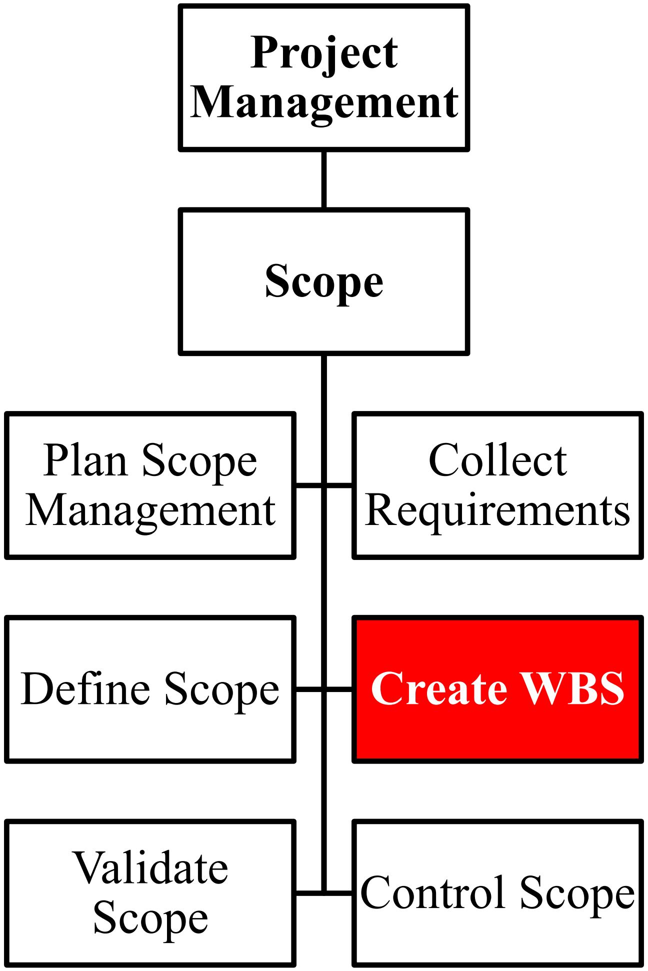
- Project Scope Statement
- Project Documents updates

Scope: 4. Create WBS



- It is the process of subdividing project deliverables and project work into smaller components.
- WBS is a hierarchical decomposition of the project work.
- WBS organizes and defines the total scope of the project, and represents the work specified in the approved project scope statement.
- A work package can be used to group the activities where work is scheduled and estimated, monitored, and controlled.

Scope: 4. Create WBS



Input

- Scope management plan
- Project scope statement
- Requirements documentation
- Enterprise environmental factors
- Organizational process assets

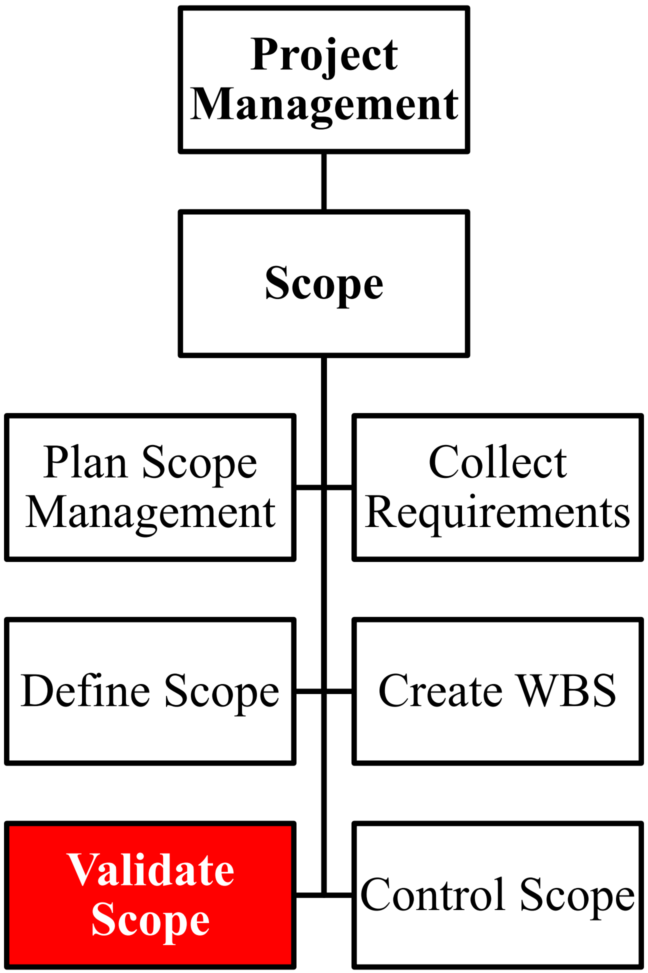
Tools & Techniques

- Decomposition
- Expert Judgement

Output

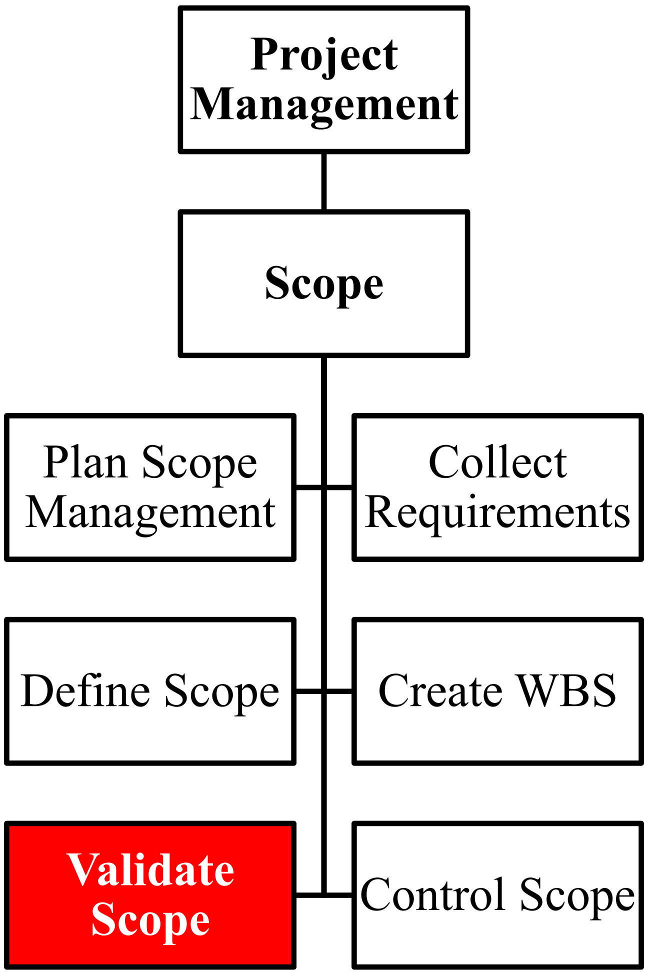
- Scope Baseline
- Project Documents updates

Scope: 5 Validate Scope



- It is the process of formalizing acceptance of the completed project deliverables.
- The verified deliverables obtained from the Control Quality process are reviewed with the customer or sponsor to ensure that they are completed satisfactorily and have received formal acceptance of the deliverables by the customer or sponsor.

Scope: 5. Validate Scope



Input

- Project management plan
- Requirements documentation
- Requirements traceability matrix
- Verified deliverables
- Work performance data

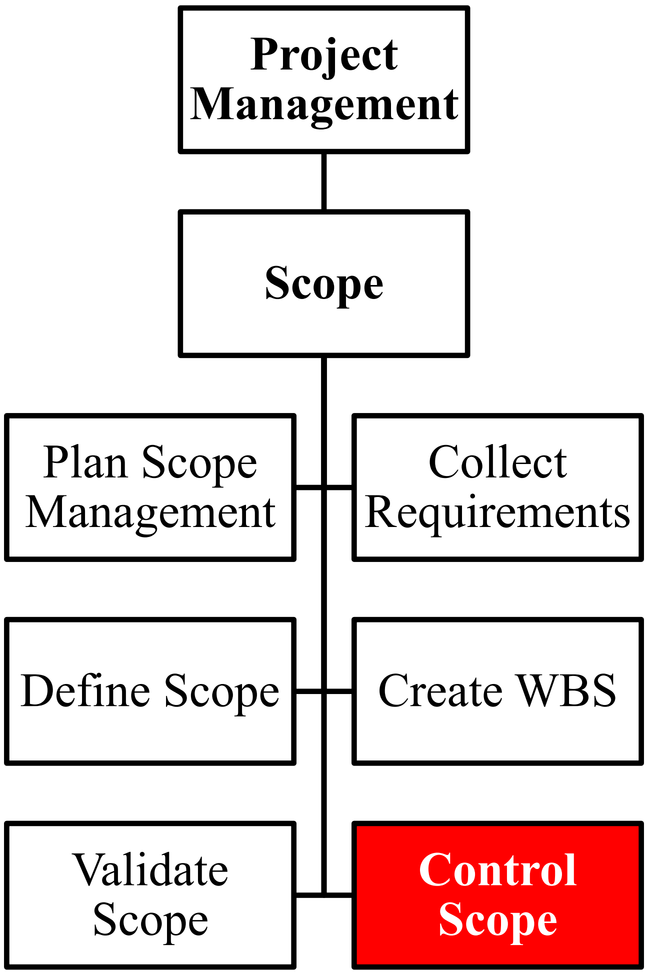
Tools & Techniques

- Inspection
- Group Decision Making Techniques

Output

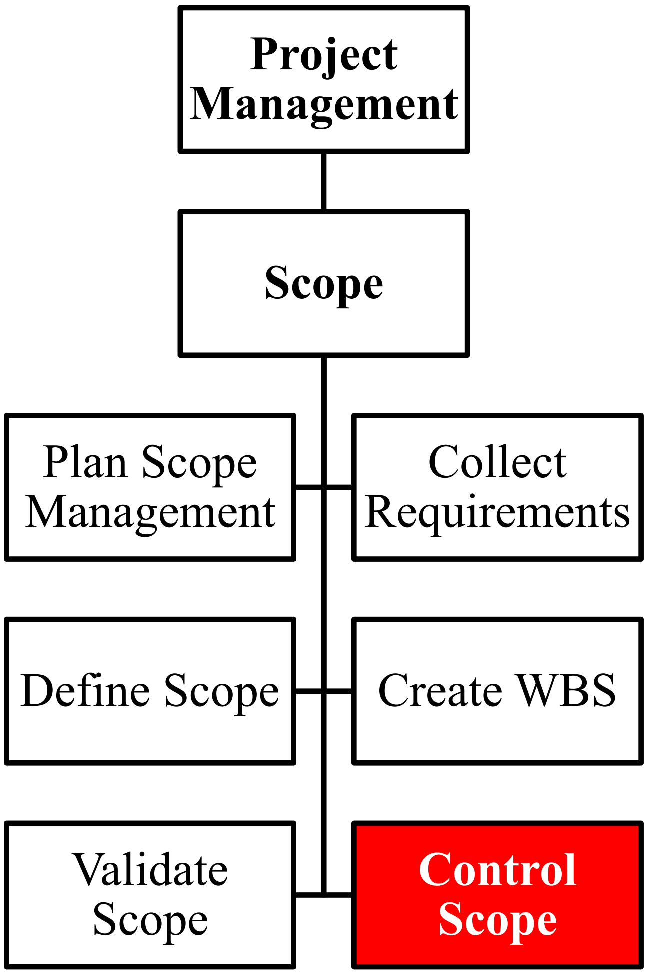
- Accepted deliverables
- Change requests
- Work performance information
- Project document updates

Scope: 6. Control Scope



- It is the process of monitoring the status of the project and product scope and managing changes to the scope baseline.
- It ensures that all requested changes and recommended corrective or preventive actions are processed through the Perform Integrated Change Control process.
- Change is inevitable; therefore, change control process is mandatory.

Scope: 6. Control Scope



Input

- Project management plan
- Requirements documentation
- Requirements traceability matrix
- Work performance data
- Organizational process assets

Tools & Techniques

- Variance Analysis

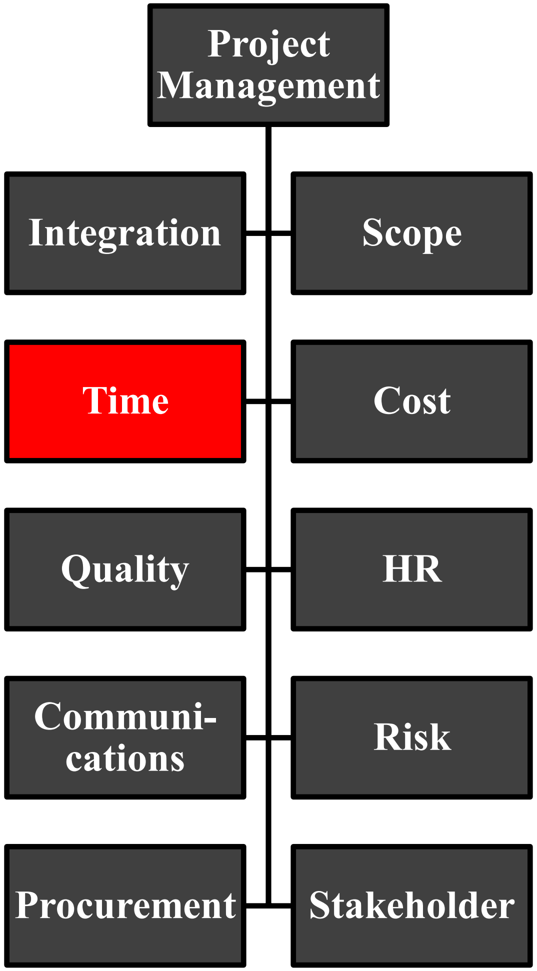
Output

- Work performance information
- Change requests
- Project management plan updates
- Project documents updates
- Organizational process assets updates

Knowledge Areas

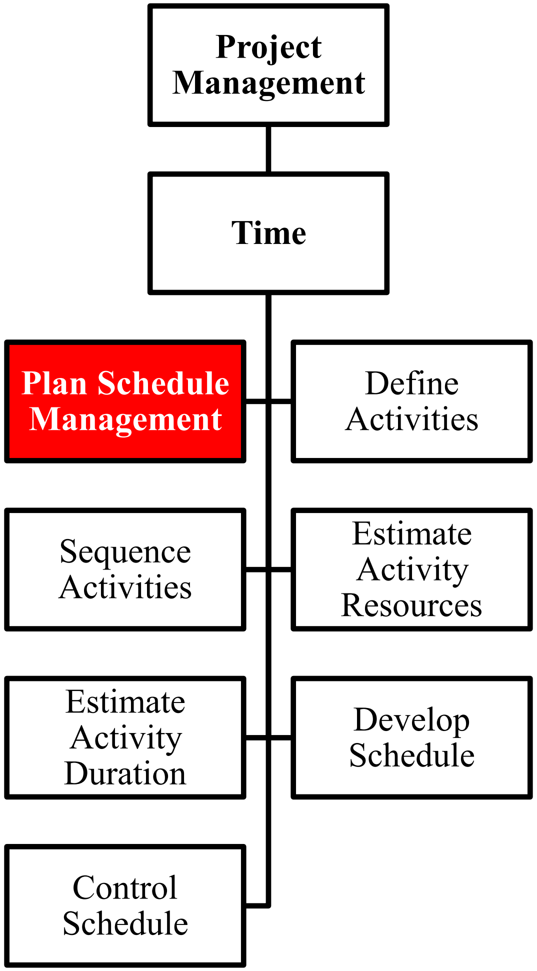
Project Time (Schedule)

PMBOK : 3. Project Time Management / Project Schedule Management



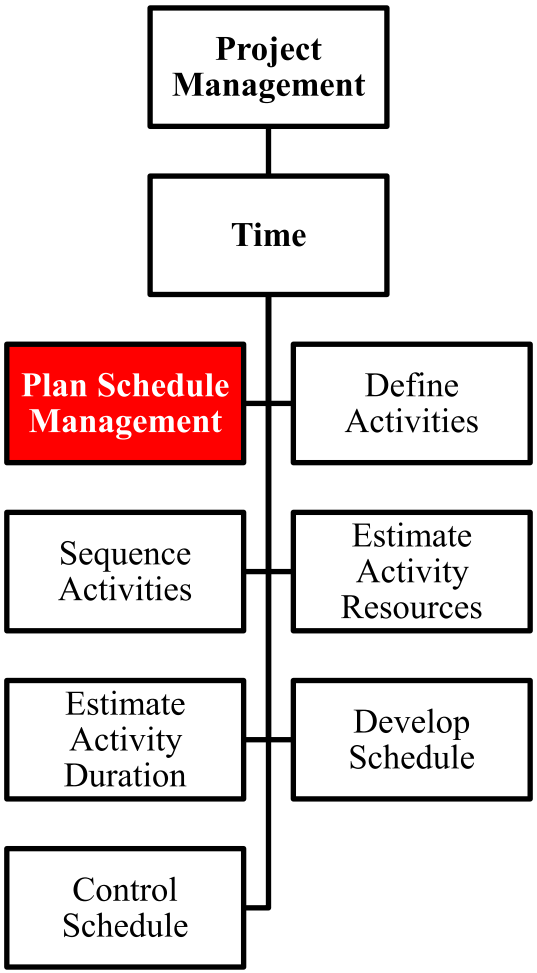
- Project Time Management includes the processes required to manage the timely completion of the project

Time:1. Plan Schedule Management



- It is the process of establishing the policies, procedures, and documentation for planning, developing, managing, executing, and controlling the project schedule.
- The schedule management plan is a component of the project management plan.
- The schedule management plan may be formal or informal, highly detailed or broadly framed, based upon the needs of the project, and includes appropriate control thresholds.
- The schedule management plan defines how schedule contingencies will be reported and assessed.
- The schedule management plan may be updated to reflect a change in the way the schedule is managed.

Time: 1. Plan Schedule Management



Input

- Project Management Plan
- Project Charter
- Enterprise Environmental Factors
- Organizational Process Assets

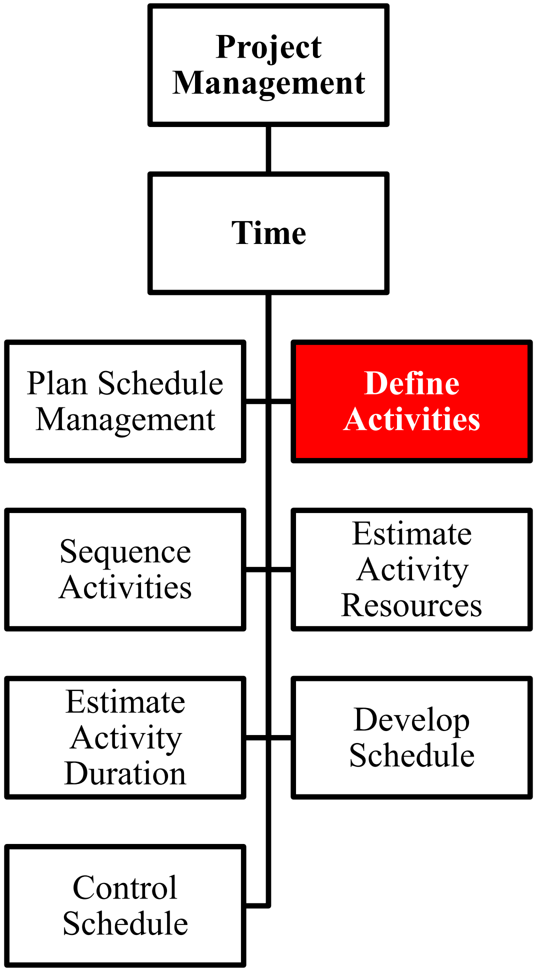
Tools & Techniques

- Expert Judgement
- Analytical Techniques
- Meetings

Output

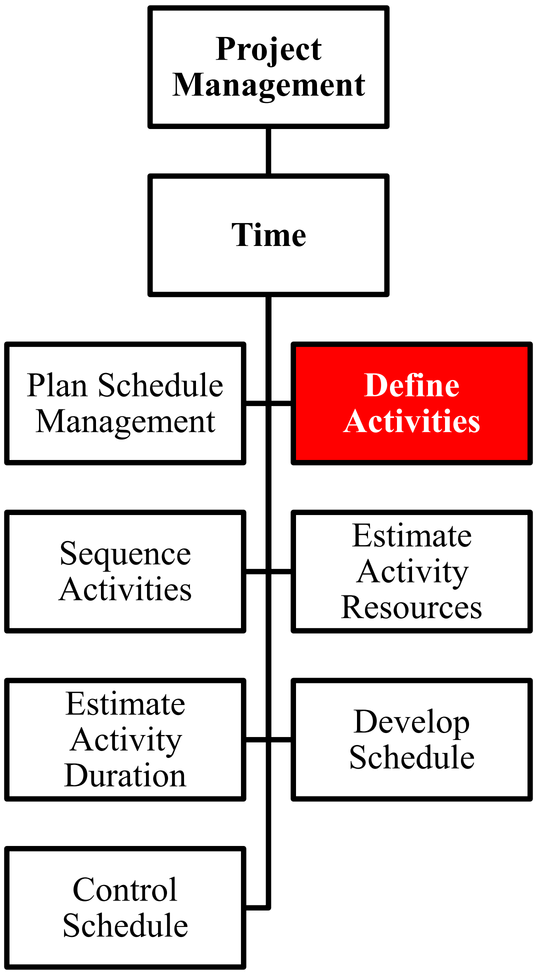
- Schedule Management Plan

Time: 2. Define Activities



- It is the process of identifying and documenting the specific actions to be performed to produce the project deliverables.
- Implicit in this process are defining and planning the schedule activities such that the project objectives will be met.
- The Create WBS process identifies the deliverables at the lowest level in the WBS / work package.
- Work packages are typically decomposed into smaller components called activities that represent the work effort required to complete the work package.

Time: 2. Define Activities



Input

- Schedule management plan
- Scope baseline
- Enterprise environmental factors
- Organizational process assets

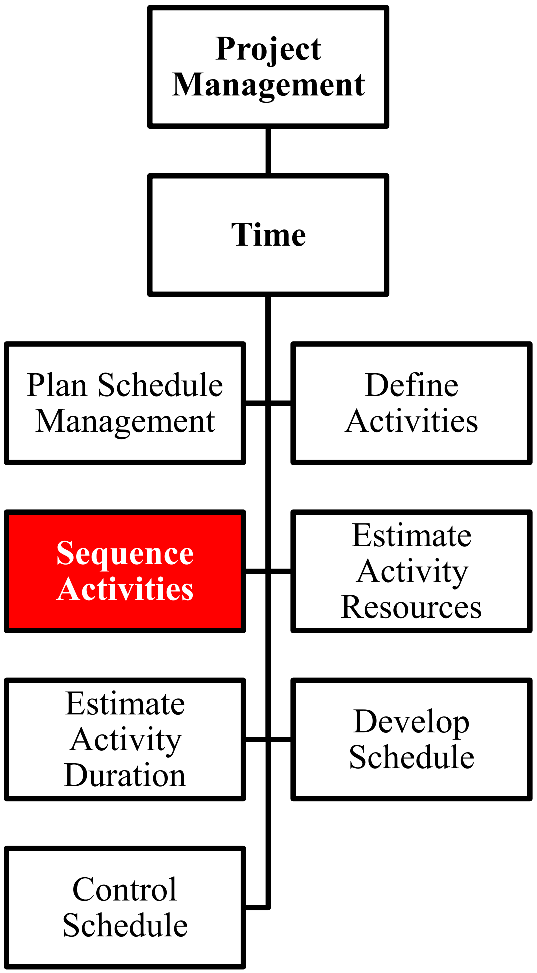
Tools & Techniques

- Decomposition
- Rolling wave planning
- Expert judgment

Output

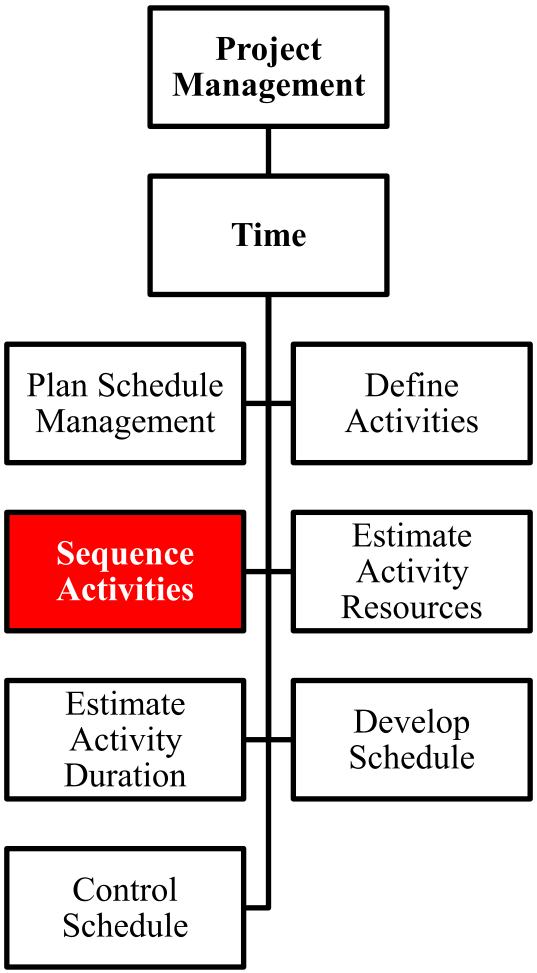
- Activity list
- Activity attributes
- Milestone list

Time: 3. Sequence Activities



- It is the process of identifying and documenting relationships among the project activities.
- Every activity and milestone except the first and last should be connected
  - to at least one predecessor with a finish-to-start or start-to-start logical relationship
  - at least one successor with a finish-to-start or finish-to finish logical relationship.
- Logical relationships should create a realistic project schedule. It may be necessary to use lead or lag time between activities.
- Sequencing can be performed by project management software or manual or automated techniques.

Time: 3. Sequence Activities



Input

- Schedule management plan
- Activity List
- Activity Attributes
- Milestone Lists
- Project scope statement
- Enterprise environmental factors
- Organizational process assets

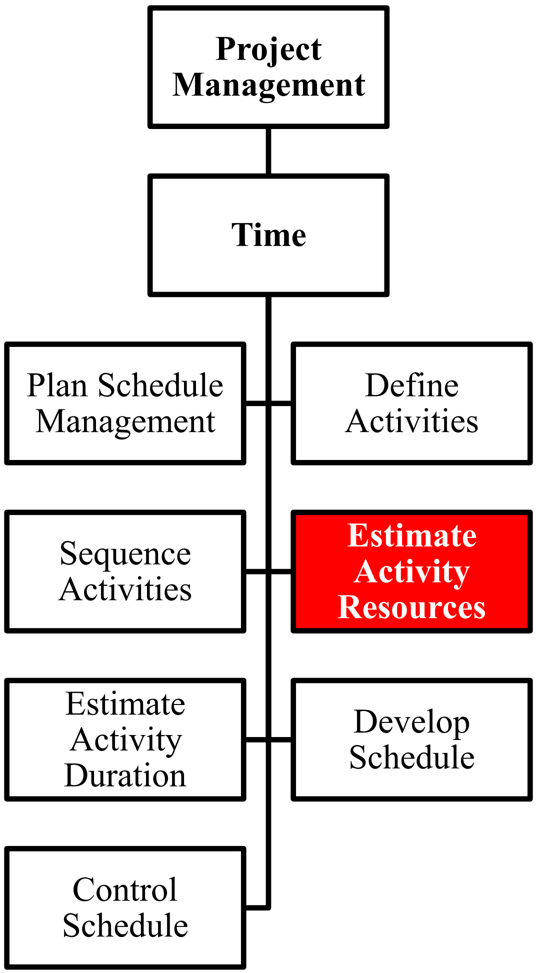
Tools & Techniques

- Precedence diagramming method (PDM)
- Dependency determination
- .Leads and lags

Output

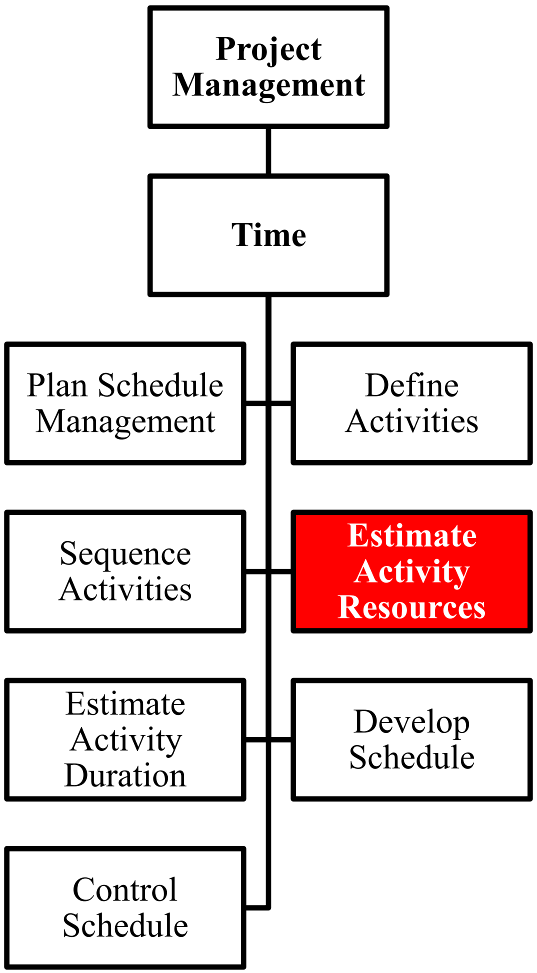
- Project schedule network diagrams
- Project documents updates

Time: 4. Estimate Activity Resources



- It is the process of estimating the type and quantities of material, human resources, equipment, or supplies required to perform each activity.
- The Estimate Activity Resources process is closely coordinated with the Estimate Costs process. For example:
  - A construction project team will need to be familiar with local building codes.
  - Such knowledge is often readily available from local sellers.
  - However, if the local team lacks experience, the additional cost for a consultant may be the most effective way to secure knowledge of the local building codes.

Time: 4. Estimate Activity Resources



Input

- Schedule management plan
- Activity List
- Activity Attributes
- Resource Calendars
- Risk Register
- Activity Cost Estimates
- Milestone Lists
- Enterprise environmental factors
- Organizational process assets

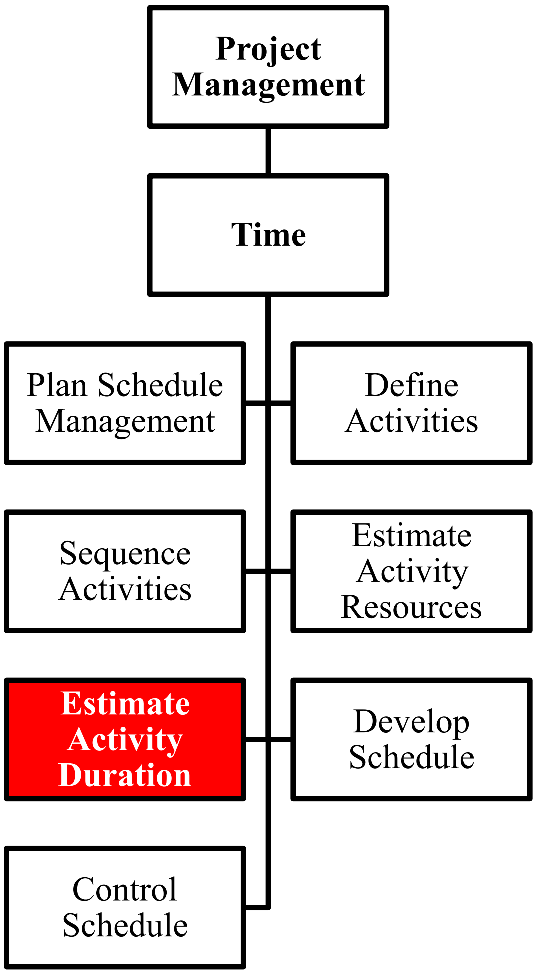
Tools & Techniques

- Expert Judgement
- Alternative Analysis
- Published Estimating Data
- Bottom-up estimating
- Project Management Software

Output

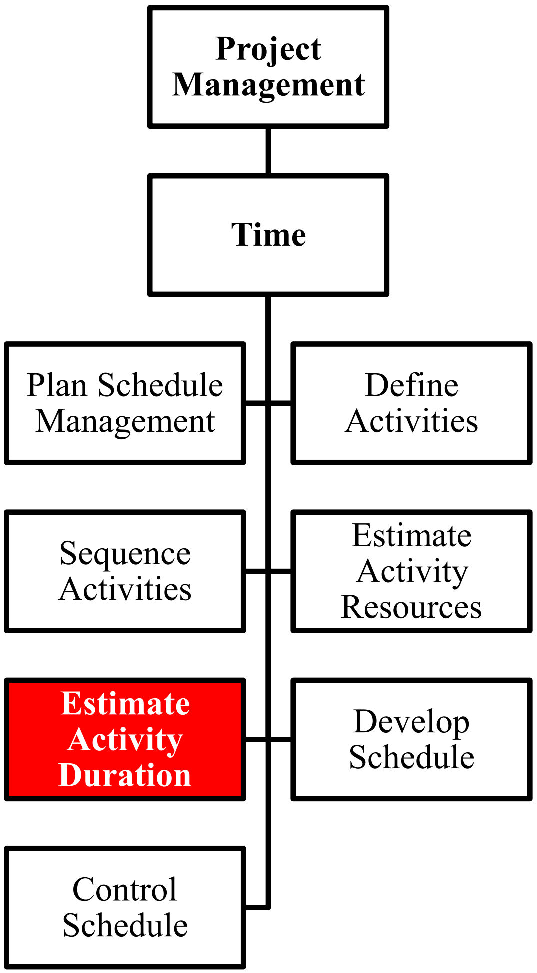
- Activity Resource Requirements
- Resource Breakdown Structure
- Project Documents Updates

Time: 5. Estimate Activity Duration



- It is the process of estimating the number of work periods needed to complete individual activities with estimated resources
- Estimating activity durations uses information on activity scope of work, required resource types, estimated resource quantities, and resource calendars.
- The duration estimate is progressively elaborated, based on quality and availability of the input data. Thus, the estimate can be progressively more accurate.

Time: 5. Estimate Activity Duration



Input

- Schedule management plan
- Activity list
- Activity attributes
- Activity resource requirements
- Resource calendars
- Project scope statement
- Risk register
- Resource breakdown structure
- Enterprise environmental factors
- Organizational process assets

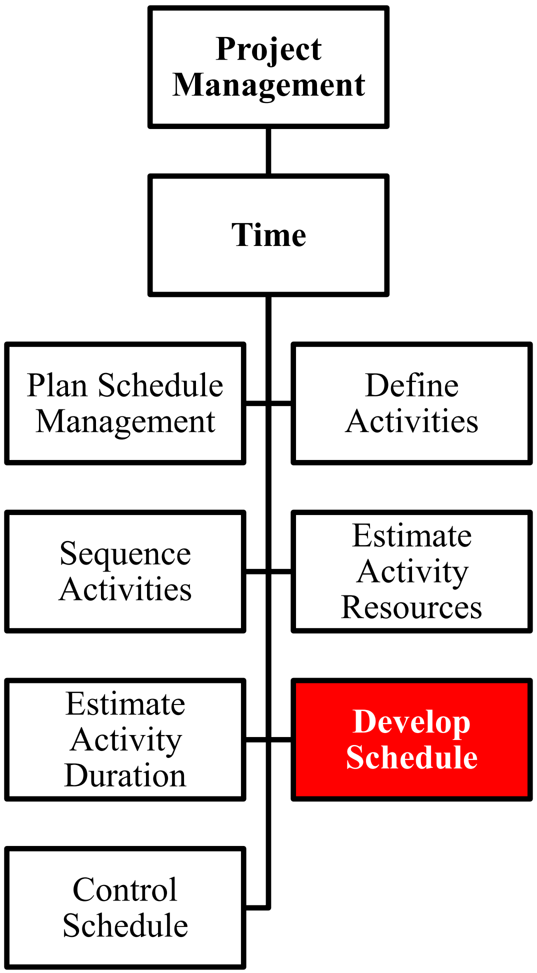
Tools & Techniques

- Expert judgment
- Analogous estimating
- Parametric estimating
- Three-point estimating
- Group decision-making techniques
- Reserve analysis

Output

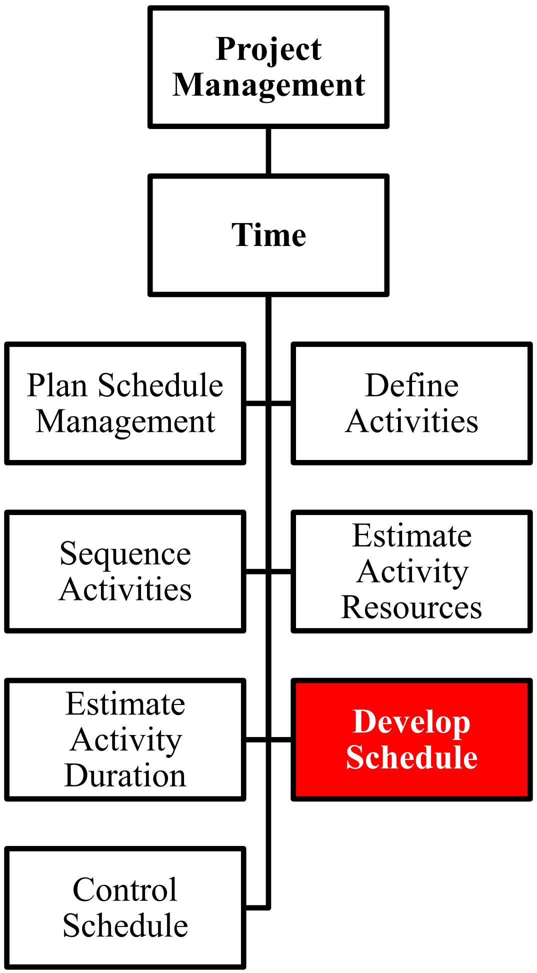
- Activity duration estimates
- Project documents updates

Time: 6. Develop Schedule



- It is the process of analyzing activity sequences, durations, resource requirements, and schedule constraints to create the project schedule model
- Developing an acceptable project schedule is often an iterative process. The schedule model is used to determine the planned start and finish dates for project activities and milestones.
- Once the activity start and finish dates have been determined, project staff review their activities and confirm that start and finish dates do not conflict with resource calendars or other projects or tasks.
- As work progresses, revising the project schedule model continues throughout the project.

Time: 6. Develop Schedule



Input

- Schedule management plan
- Activity list
- Activity attributes
- Project schedule
- network diagrams
- Activity resource requirements
- Resource calendars
- Activity duration estimates
- Project scope statement
- Risk register
- Project staff assignments
- Resource breakdown structure
- Enterprise environmental factors
- Organizational process assets

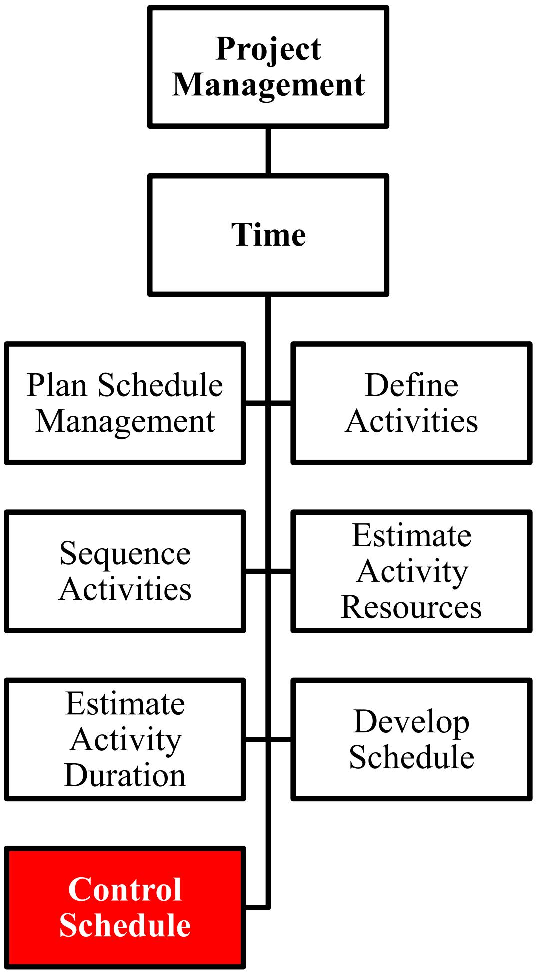
Tools & Techniques

- Schedule network analysis
- Critical path method
- Critical chain method
- Resource optimization techniques
- Modeling techniques
- Leads and lags
- Schedule compression
- Scheduling tool

Output

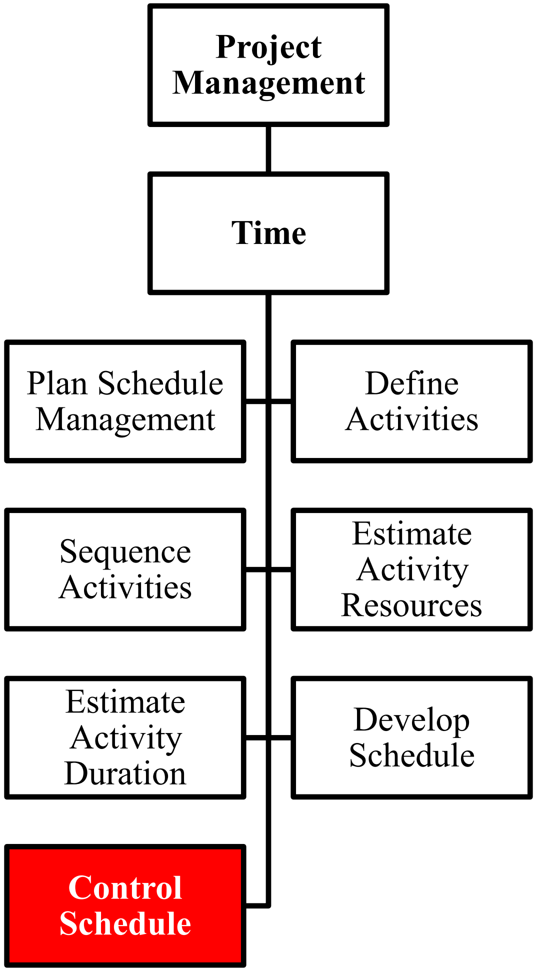
- Schedule baseline
- Project schedule
- Schedule data
- Project calendars
- Project management plan updates
- Project documents updates

Time: 7. Control Schedule



- It is the process of monitoring the status of project activities to update project progress and manage changes to the schedule baseline to achieve the plan
- Updating the schedule model requires knowing the actual performance to date.
- Any change to the schedule baseline can only be approved through the Perform Integrated Change Control process

Time: 1. Plan Schedule Management



Input

- Project management plan
- Project schedule
- Work performance data
- Project calendars
- Schedule data
- Organizational process assets

Tools & Techniques

- Performance reviews
- Project management software
- Resource optimization techniques
- Modeling techniques
- Leads and lags
- Schedule compression
- Scheduling tool

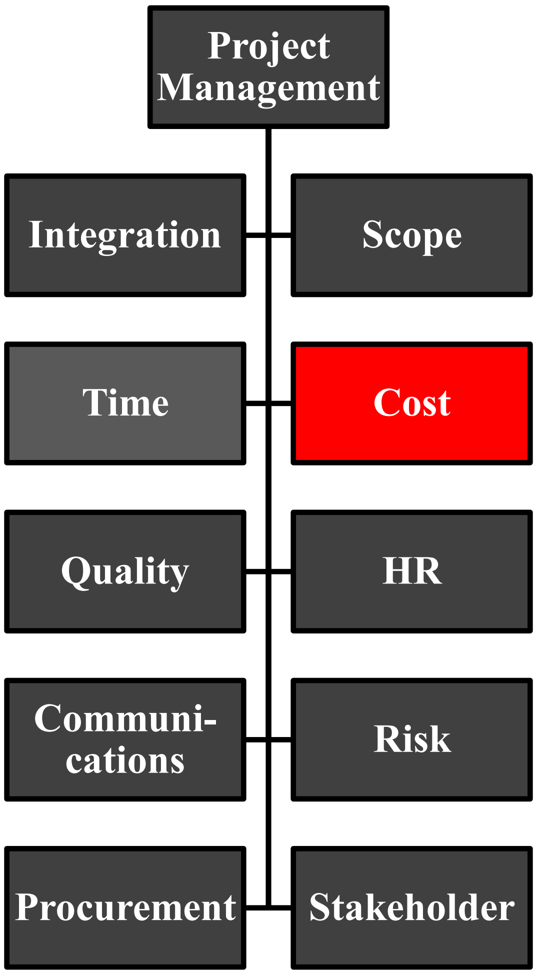
Output

- Work performance information
- Schedule forecasts
- Change requests
- Project management plan updates
- Project documents updates
- Organizational process assets updates

Knowledge Areas

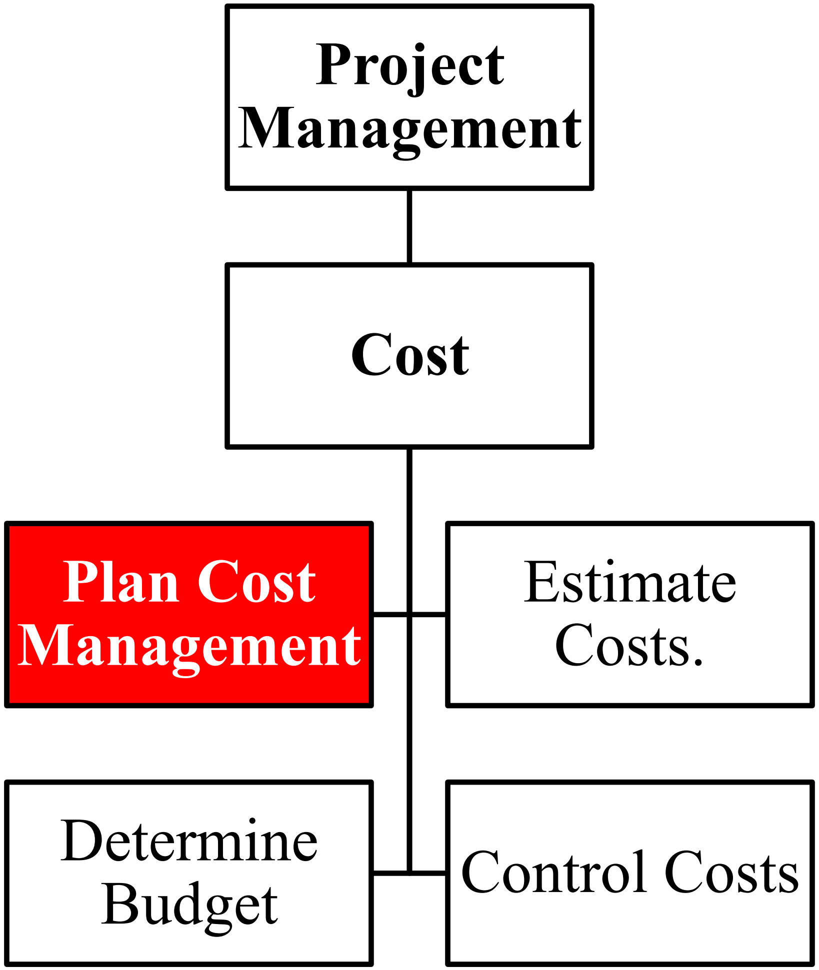
Project Cost

PMBOK: 4. Project Cost Management



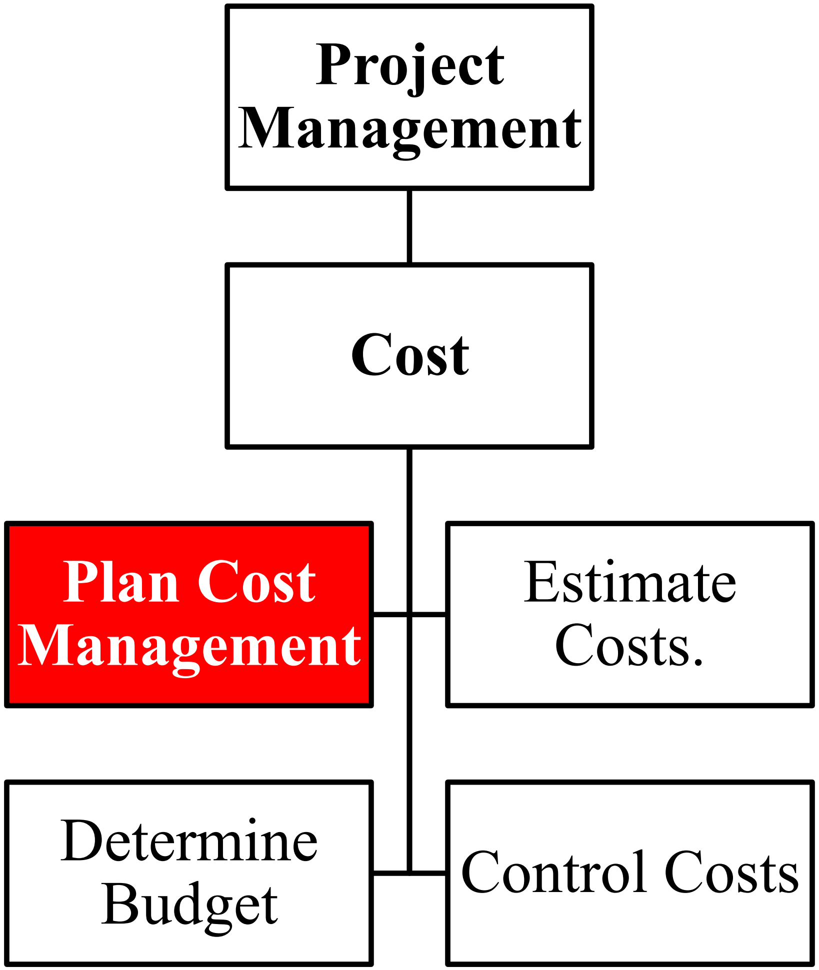
- Project Cost Management includes the processes involved in planning, estimating, budgeting, financing, funding, managing, and controlling costs so that the project can be completed within the approved budget.

Cost: 1. Plan Cost Management



- It is the process that establishes policies, procedures, and documentation for planning, managing, expending, and controlling project costs.
- The cost management process and their associated tools and techniques are documented in the cost management plan.
- The cost management plan is a component of the project management plan

Cost: 1. Plan Schedule Management



Input

- Project management plan
- Project charter
- Enterprise environmental factors
- Organizational process assets

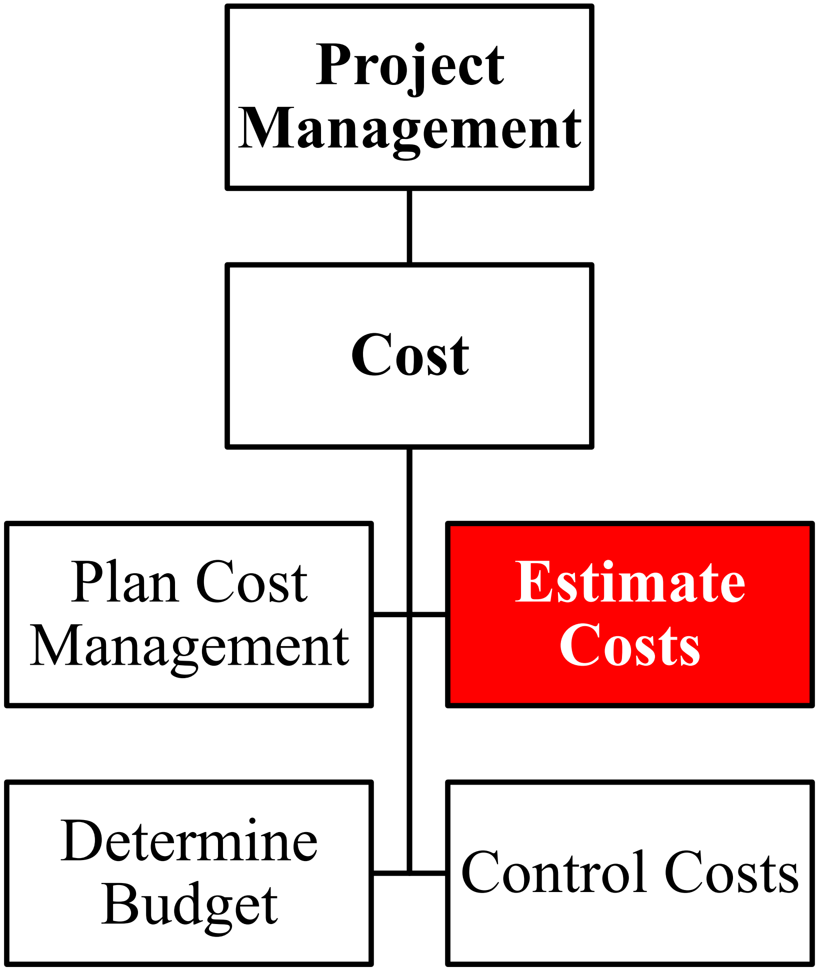
Tools & Techniques

- Expert Judgement
- Analytical Techniques
- Meetings

Output

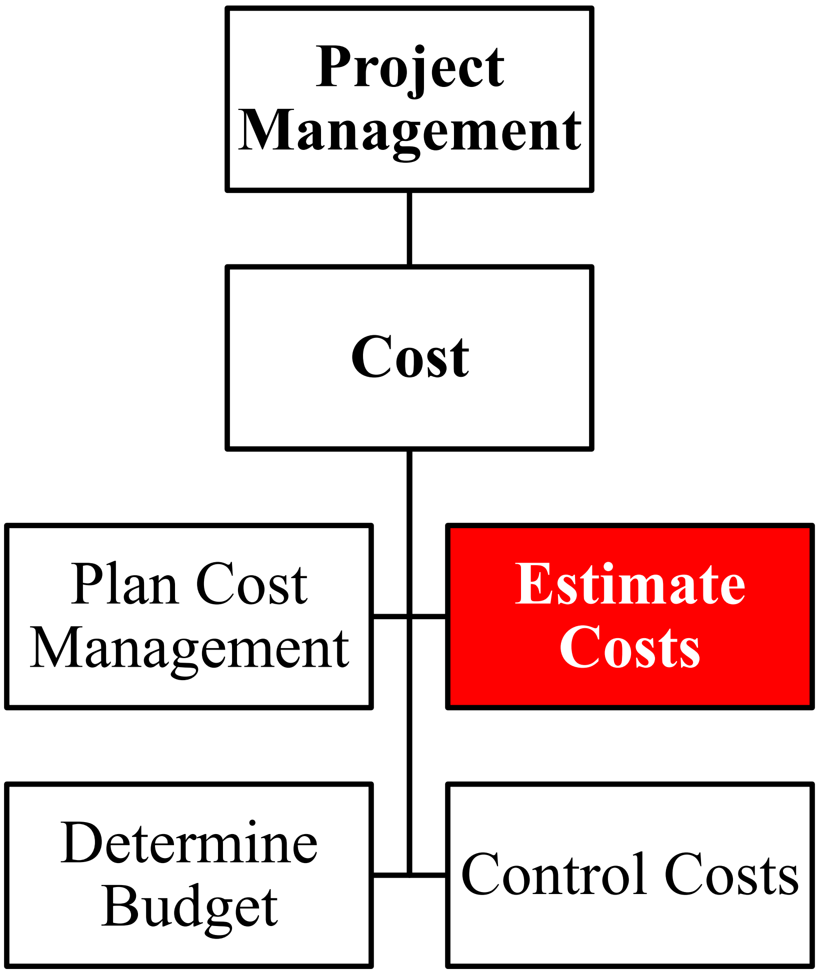
- Cost Management Plan

Cost: 2. Estimate Costs



- It is the process that establishes policies, procedures, and documentation for planning, managing, expending, and controlling project costs.
- The cost management plan is a component of the project management plan

Cost: 2. Estimate Costs



Input

- Cost management plan
- Human resource management plan
- Scope baseline
- Project schedule
- Risk register
- Enterprise environmental factors
- Organizational process assets

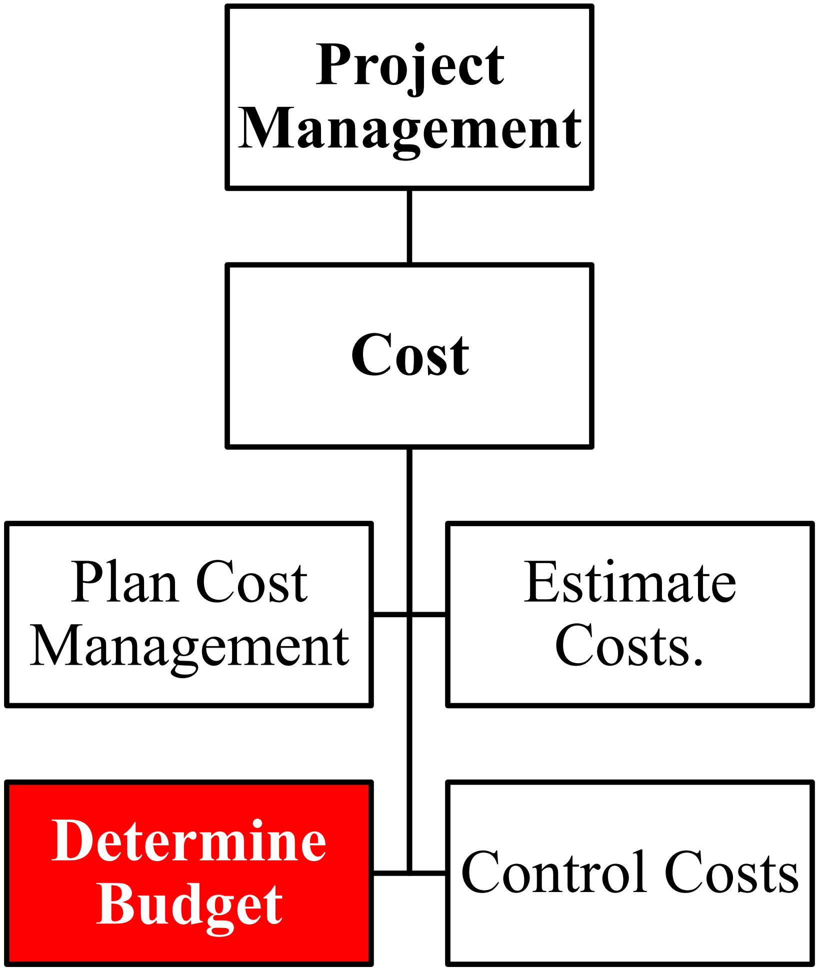
Tools & Techniques

- Expert judgment
- Analogous estimating
- Parametric estimating
- Bottom-up estimating
- Three-point estimating
- Reserve analysis
- Cost of quality
- Project management software
- Vendor bid analysis
- Group decision-making techniques

Output

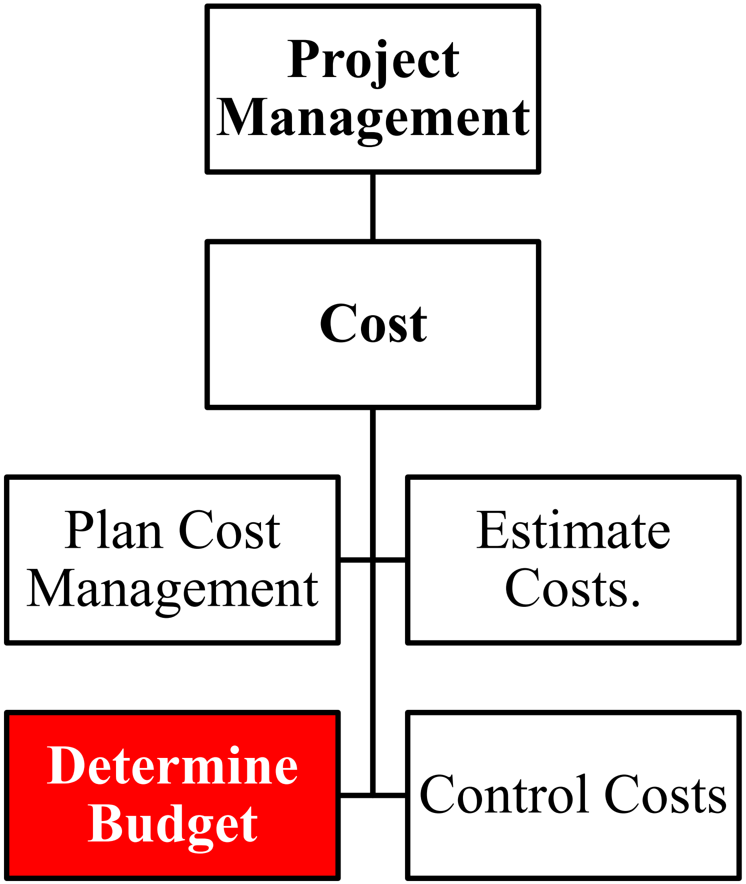
- Activity cost estimates
- Basis of estimates
- Project documents updates

Cost: 3. Determine Budget



- Project Budget includes all costs of the project
- It can be aggregated by cost item and/or organization and /or activity:
  - Cost item: manpower, services, purchasing, etc.
  - Organization: software house, consultant, vendor, etc.
  - Activity: analysis, development, project management
- It is determined early in the project and periodically by the control cost activity

Cost: 3. Determine Budget



Input

- Cost management plan
- Scope baseline
- Activity cost estimates
- Basis of estimates
- Project schedule
- Resource calendars
- Risk register
- Agreements
- Organizational process assets

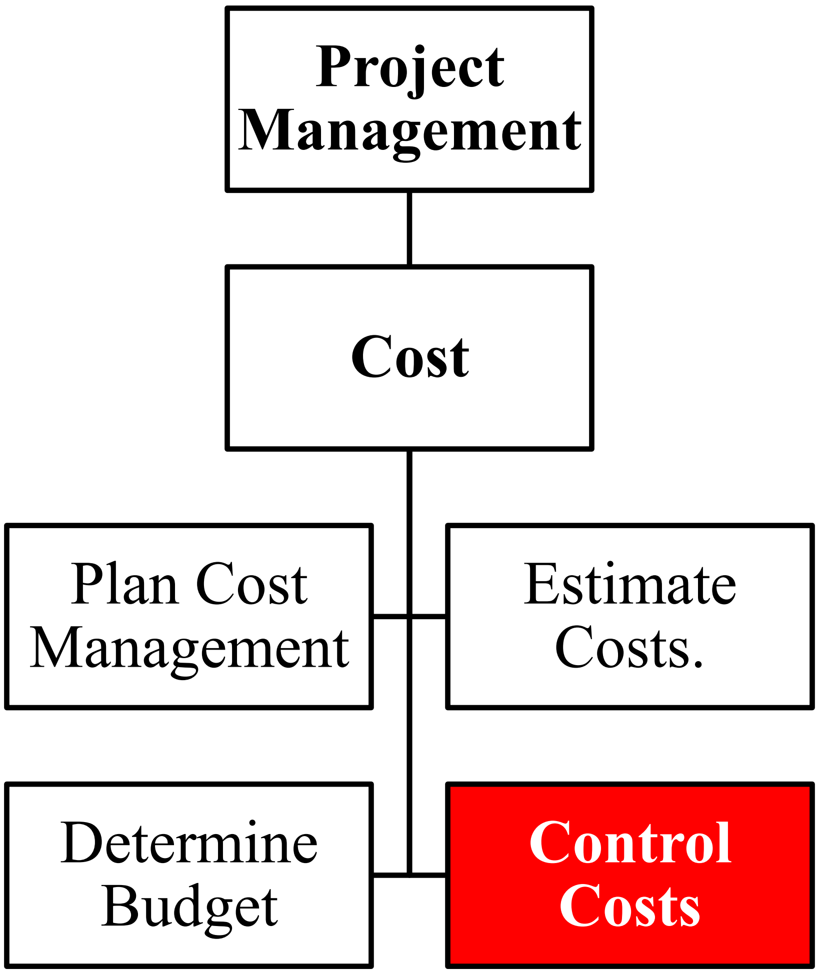
Tools & Techniques

- Cost aggregation
- Reserve analysis
- .Expert judgment
- Historical relationships
- Funding limit reconciliation

Output

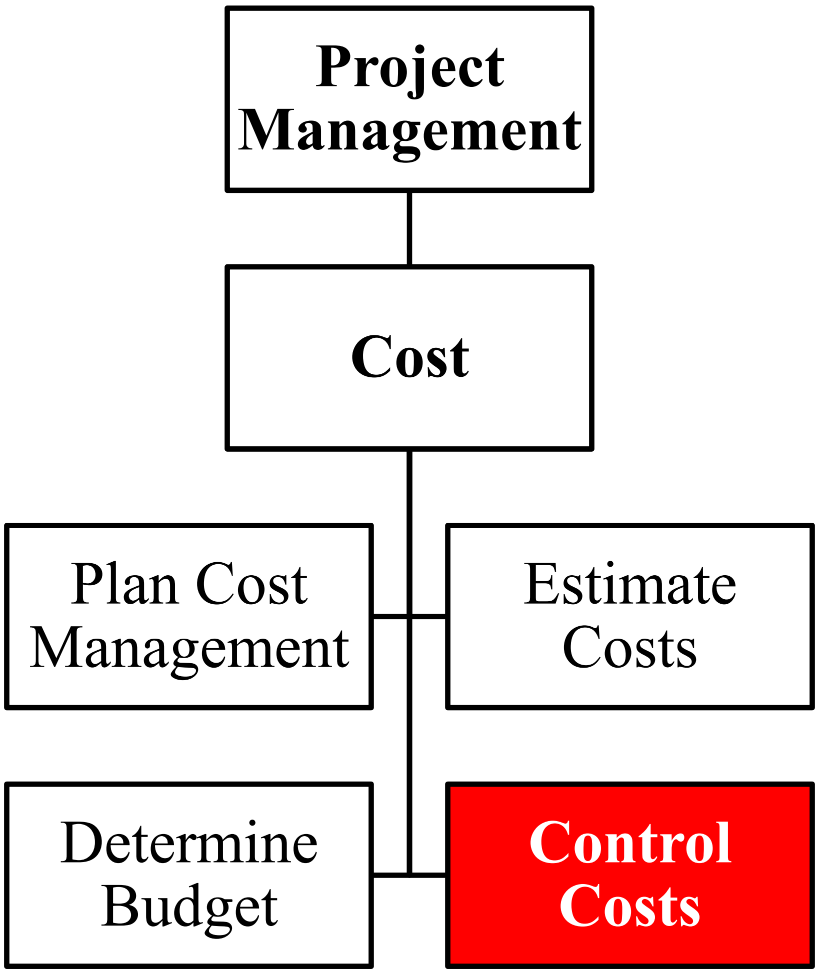
- Cost baseline
- Project funding requirements
- Project documents updates

Cost: 4. Control Cost



- It is the process of monitoring the status of the project to update the project costs and managing changes to the cost baseline
- Updating the budget requires knowledge of the actual costs spent to date.
- The expenditure is monitored by considering to the value of work being accomplished (EARNED VALUE)
- Cost control analyzes the relationship between the consumption of project funds to the earned value.

Cost: 4. Control Cost



Input

- .Project management plan
- .Project funding requirements
- Work performance data
- Organizational process assets

Tools & Techniques

- Earned value management
- Forecasting
- To-complete performance index (TCPI)
- Performance reviews
- Project management software
- Reserve analysis

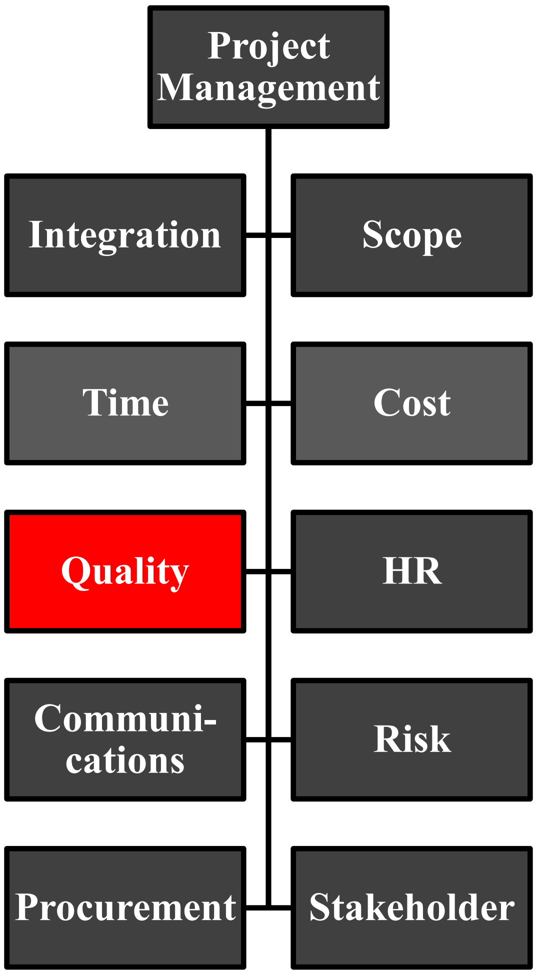
Output

- Work performance information
- Cost forecasts
- Change requests
- Project management plan updates
- Project documents updates
- Organizational process assets updates

Knowledge Areas

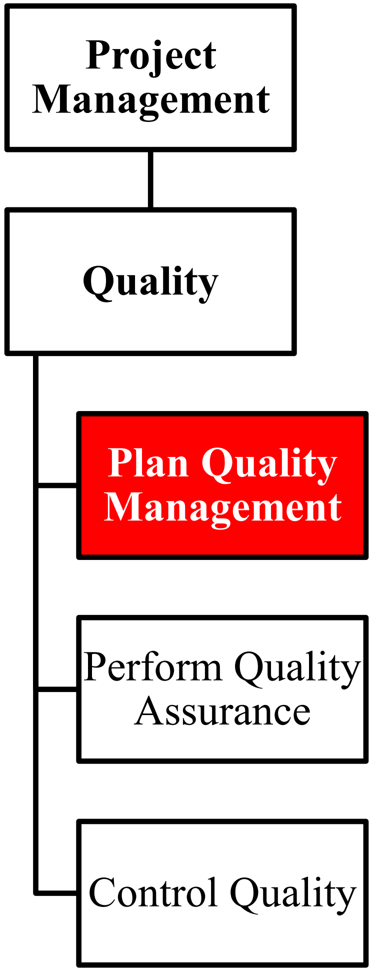
Project Quality

PMBOK: 4. Project Quality Management



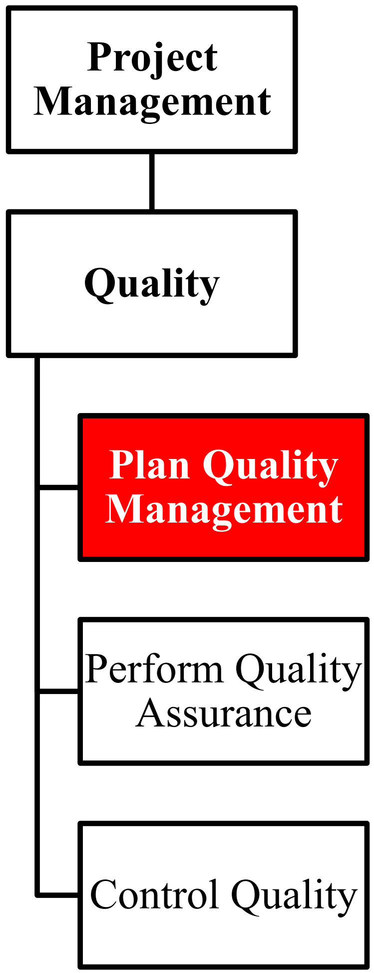
- Project Quality Management includes the processes and activities of the performing organization that determine quality policies, objectives, and responsibilities.
- Project Quality Management ensures that the project requirements, including product requirements, are met and validated.

Quality : 1. Plan Quality Management



- It is the process of
  - identifying quality requirements and/or standards for the project and its deliverables,
  - and documenting how the project complies with relevant quality requirements.
- Quality planning should be performed in parallel with the other planning processes.
- For example, proposed changes in the deliverables to meet identified quality standards may require cost or schedule adjustments and a detailed risk analysis of their impact on plans.

Quality: 1. Plan Quality Management



Input

- Project management plan
- Stakeholder register
- Risk register
- Requirements documentation
- Enterprise environmental factors
- Organizational process assets

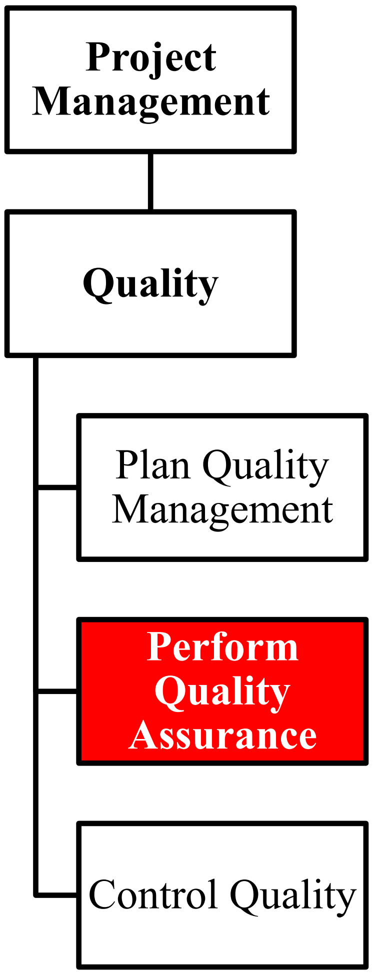
Tools & Techniques

- Cost-benefit analysis
- Cost of quality
- Seven basic quality tools
- Benchmarking
- Design of experiments
- Statistical sampling
- Additional quality planning tools
- Meetings

Output

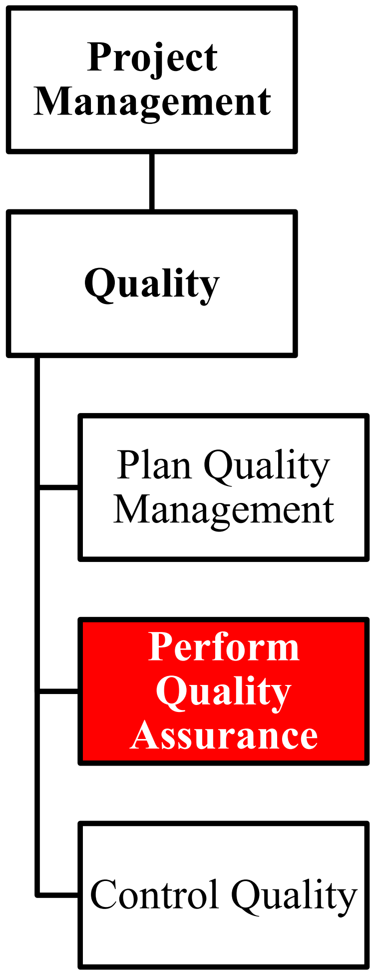
- Quality management plan
- Process improvement plan
- Quality metrics
- Quality checklists
- Project documents updates

Quality: 2. Perform Quality Assurance



- It is the process of auditing the quality requirements and the results from quality control measurements, in order to ensure that appropriate quality standards and operational definitions are used.
- In project management, the prevention and inspection aspects of quality assurance should have a demonstrable influence on the project. Quality assurance work will fall under the conformance work category in the cost of quality framework.

Quality: 2. Perform Quality Assurance



Input

- Quality management plan
- Process improvement plan
- Quality metrics
- Quality control measurements
- Project documents

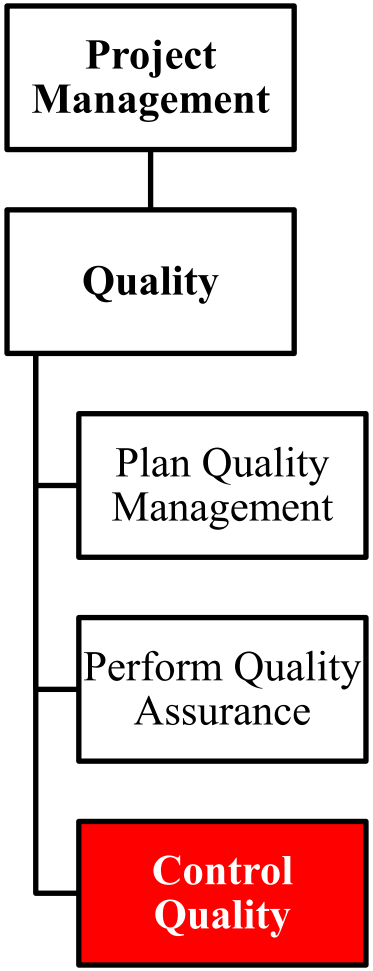
Tools & Techniques

- Quality management and control tools
- Quality audits
- Process analysis

Output

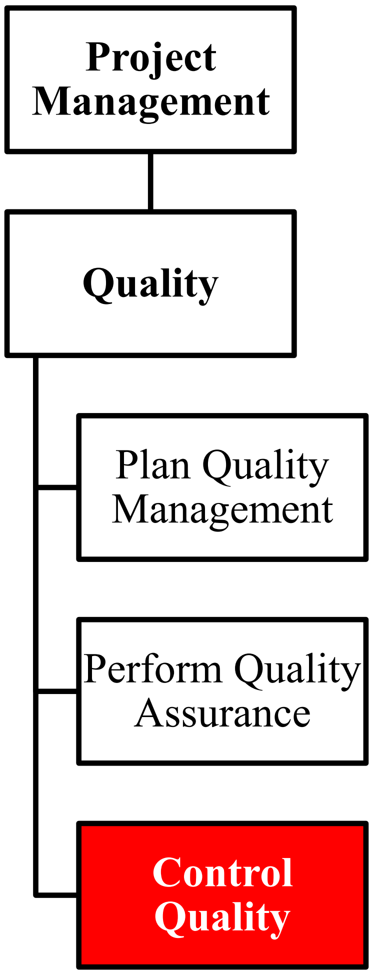
- Change requests
- Project management plan updates
- Project documents updates
- Organizational process assets updates

Quality: 3. Control Quality



- It is the process of monitoring and recording results of executing the quality activities to assess performance and recommend necessary changes.
- Quality assurance should be used during the project phases to provide confidence that the stakeholder's requirements will be met
- Quality control should be used during the project executing and closing phases to formally demonstrate, that the sponsor and/or customer's acceptance criteria have been met.

Quality: 3. Control Quality



Input

- Project management plan
- Quality metrics
- Quality checklists
- Work performance data
- Approved change requests
- Deliverables
- Project documents
- Organizational process assets

Tools & Techniques

- Seven basic quality tools
- Statistical sampling
- Inspection
- Approved change requests review

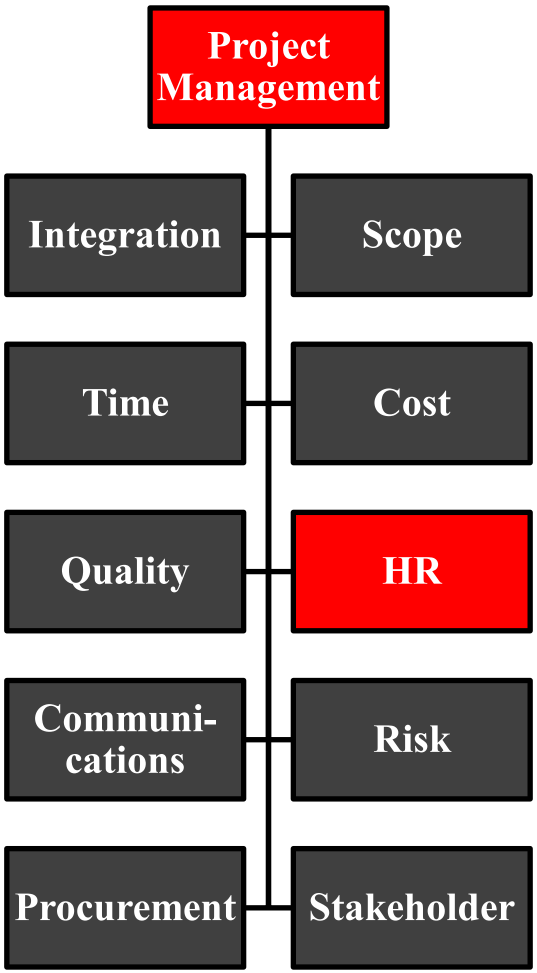
Output

- Quality control measurements
- Validated changes
- Verified deliverables
- Work performance information
- Change requests
- Project management plan updates
- Project documents updates
- Organizational process assets updates

Knowledge Areas

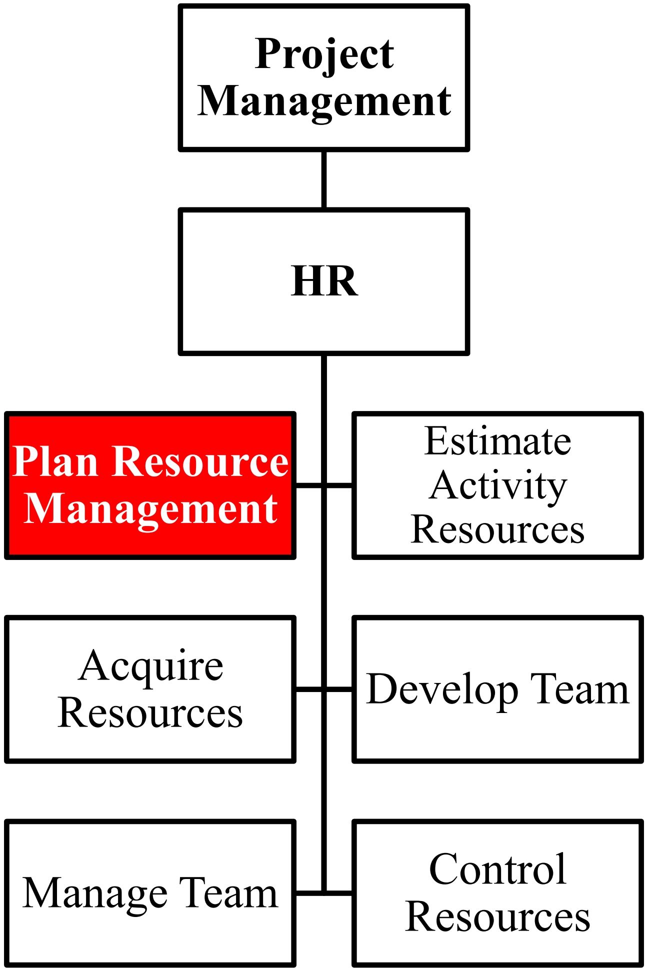
Project Resources

PMBOK: 6. Human Resources Management



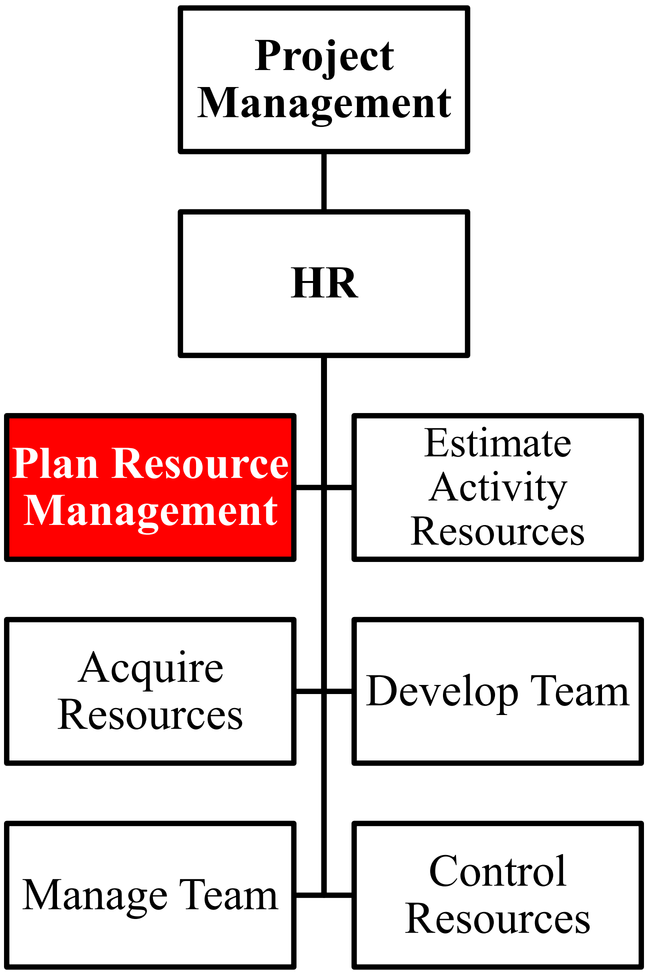
- Project Human Resource Management includes the processes that organize, manage, and lead the project team.
- Project team members may have varied skill sets, may be assigned full or part-time, and may be added or removed from the team during the project.
- Participation of team during planning adds their expertise to the process and strengthens their commitment to the project.
- This area was innovated in edition 6

HR: 1.Plan Human Resource Management



- It defines how to estimate, acquire, manage, and use team and physical resources.
- Resource planning ensures that sufficient resources are available.
- Project resources may include team members, suppliers, materials, equipment, services, and facilities.
- It contains the staffing plan, identifies training needs, etc.

HR: 1. Plan Human Resource Management



Input

- Project charter
- Project management plan
- Project documents
- Activity resource requirements
- Enterprise environmental factors
- Organizational process assets

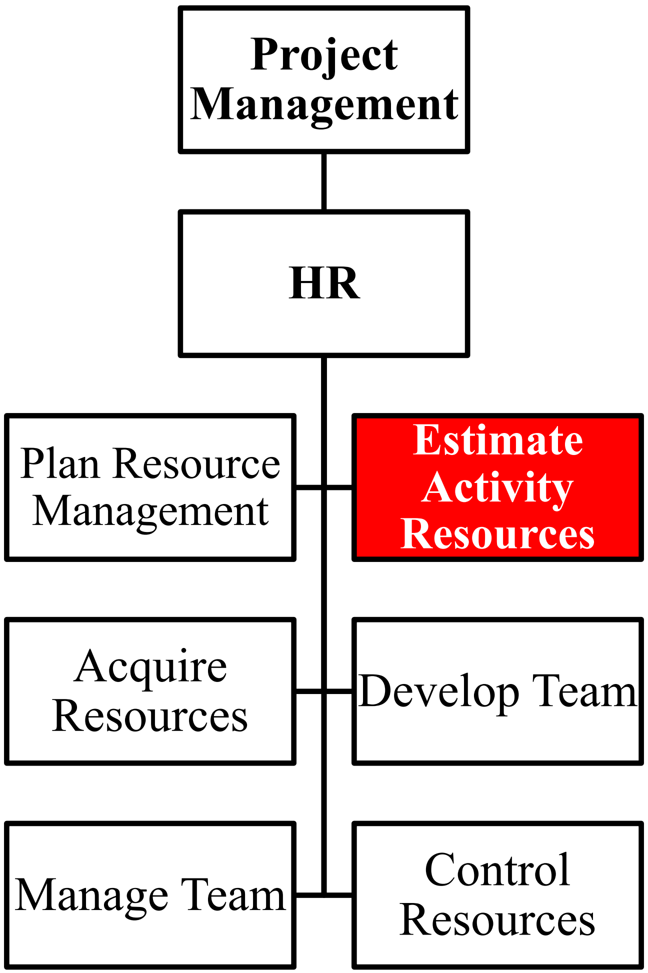
Tools & Techniques

- Expert judgment
- Data representation (Responsibility grids, OBS etc.)
- Organizational theory
- Meetings

Output

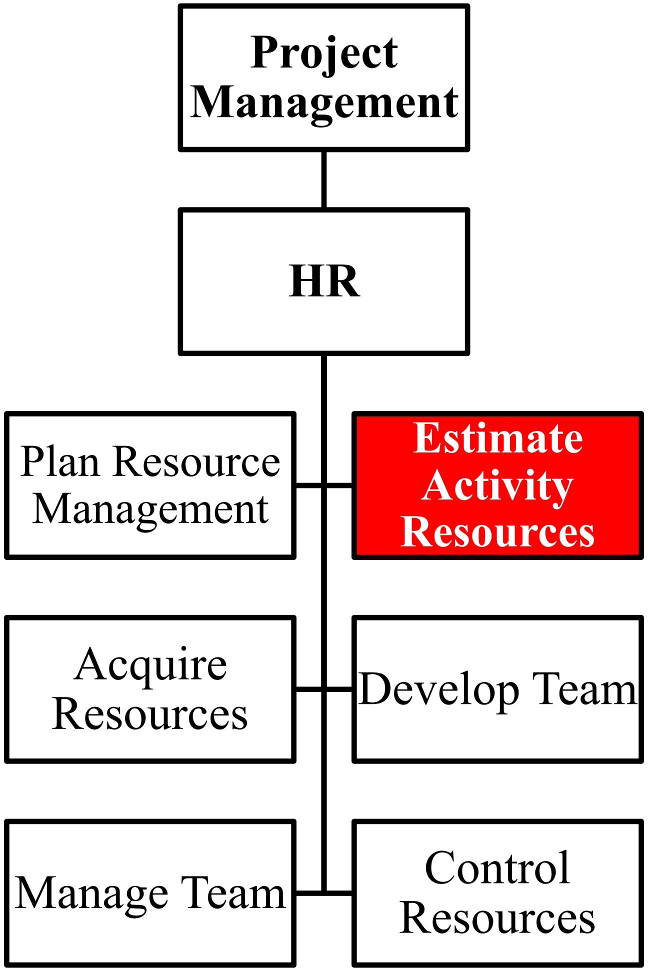
- Resource management plan
- Team Charter
- Project Documents Updates

HR: 2. Estimate Activity Resources



- It is the process of estimating team resources, and the type and quantities of materials, equipment and supplies needed.

HR: 2. Estimate Activity Resources



Input

- Project Management Plan
- Project documents
- Enterprise environmental factors
- Organizational process assets

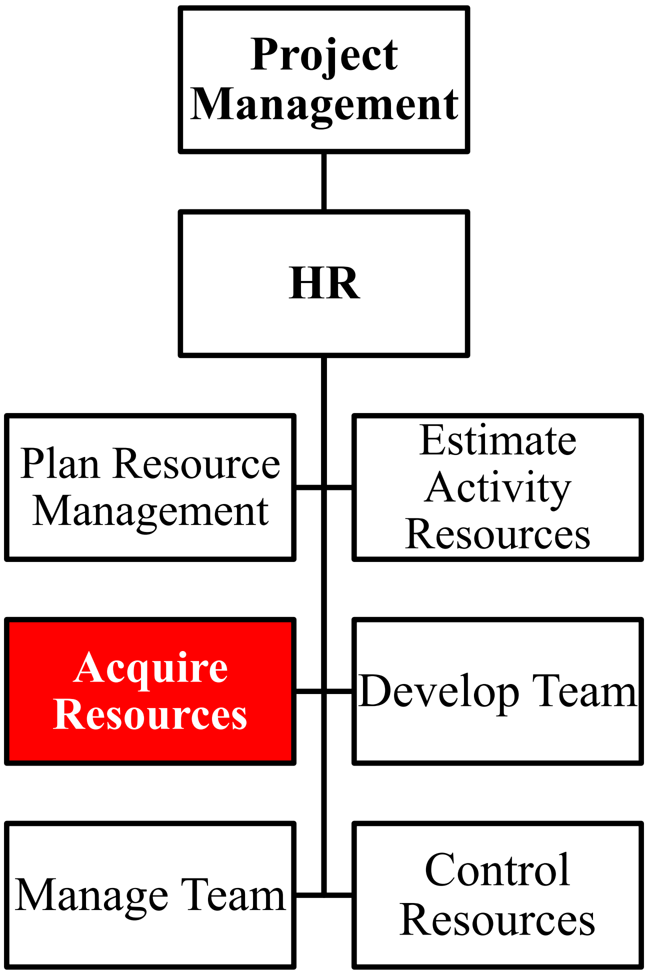
Tools & Techniques

- Expert judgement
- Bottom-up estimating
- Analogous estimating
- Parametric estimating
- Data Analysis
- Project Management Information System
- Meetings

Output

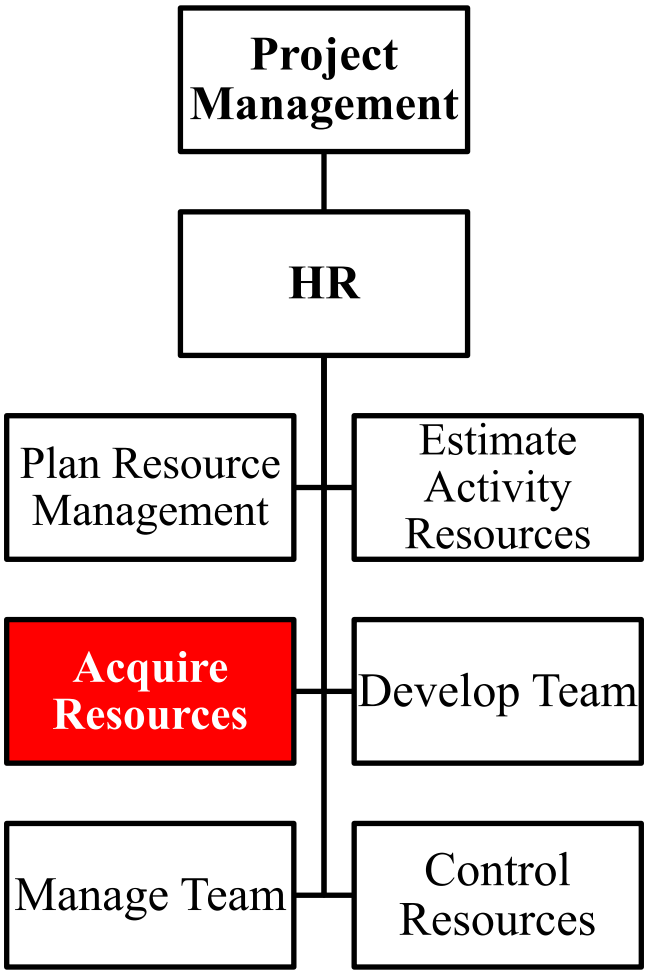
- Resource requirements
- Basis of estimates
- Resource breakdown structure
- Project documents updates

HR: 3. Acquire Resources



- It is the process of obtaining team members, facities, equipment, supplies and other resources needed

HR: 3. Acquire Resources



Input

- Project Management Plan
- Project documents
- Enterprise environmental factors
- Organizational process assets

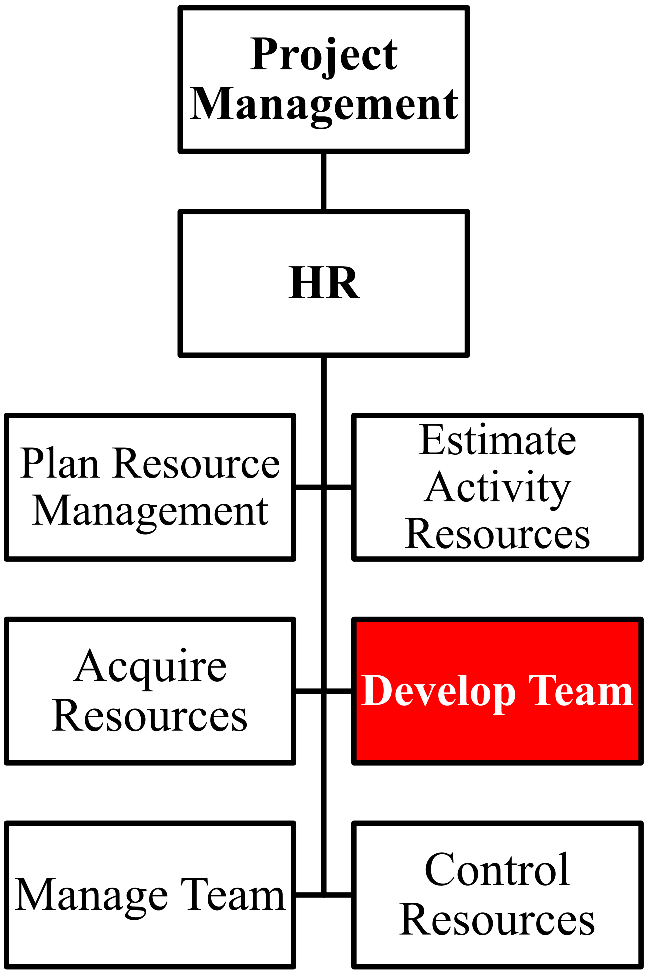
Tools & Techniques

- Decision Making
- Interpersonal and Team Skills
- Pre-assignment
- Virtual teams

Output

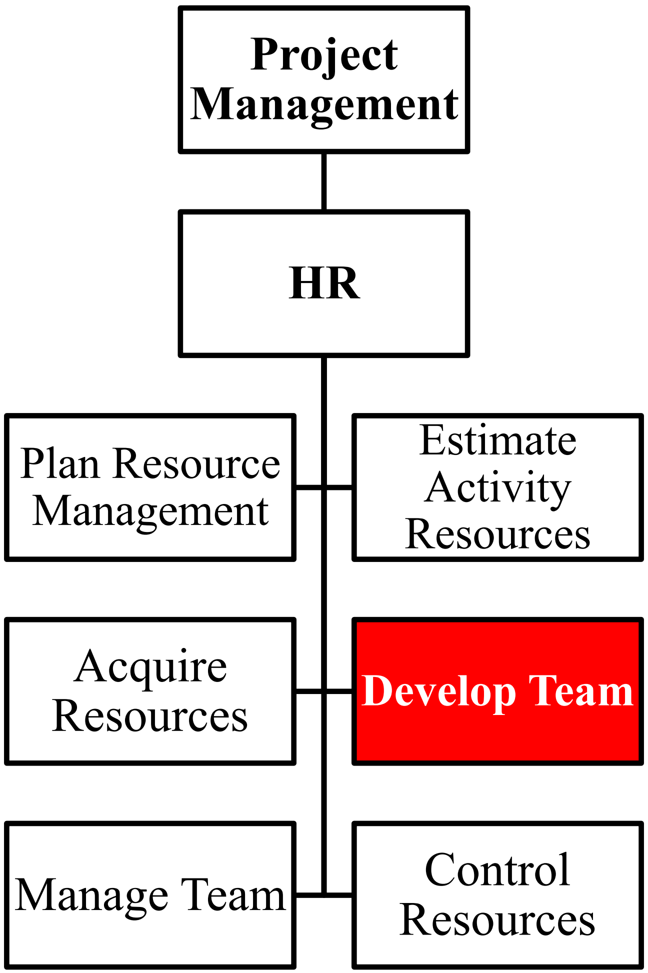
- Physical resource assignments
- Project team assignments
- Resource calendar
- Project management plan updates
- Project documents updates
- Enterprise environmental factors updates
- Organizational process assets upadates

HR: 3. Develop Team



- It is the process of improving competencies, team member interaction, and overall team environment to enhance project performance.
- Project managers should identify, build, maintain, motivate, lead, and inspire project teams to achieve high team performance and to meet the project's objectives.
- Teamwork is a critical factor for project success, and developing effective project teams is one of the primary responsibilities of the project manager.
- The project manager should request management support and/or influence the appropriate stakeholders to acquire the resources needed to develop effective project teams.

HR: 3. Develop Team



Input

- Human resource management plan
- Project documents
- Enterprise environmental factors
- Organizational process assets

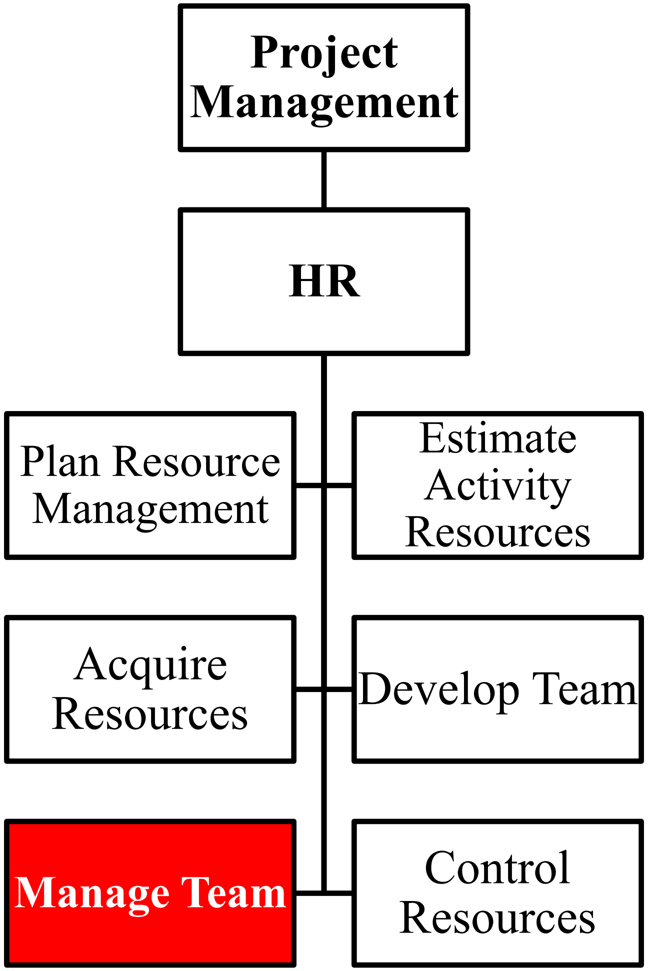
Tools & Techniques

- Co-location
- Virtual teams
- Interpersonal skills
- Communication technology
- Interpersonal and team skills
- Recognition and rewards
- Training
- Individual and team assessments
- Meetings

Output

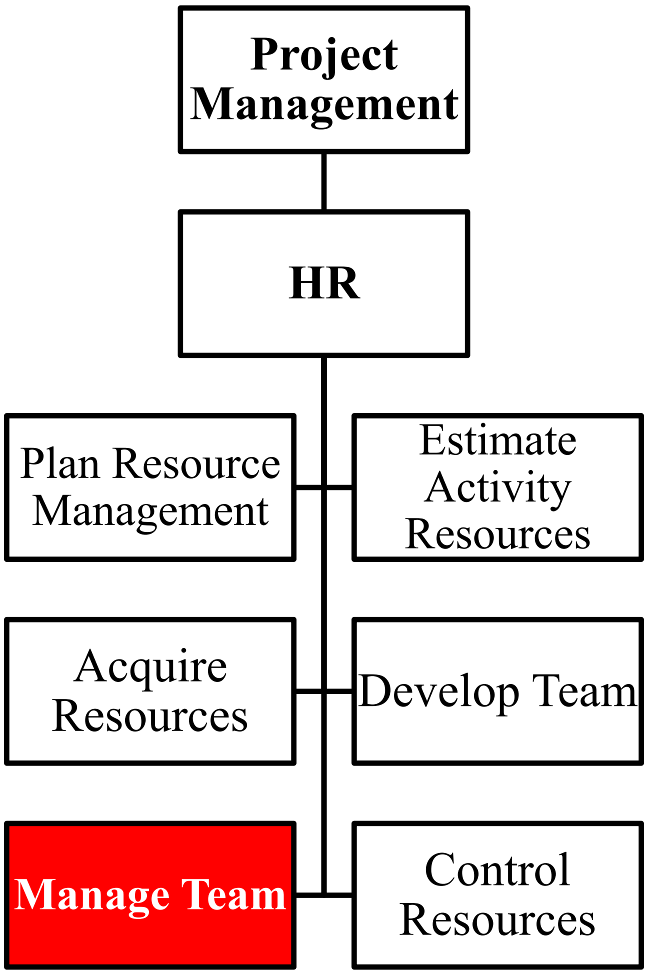
- Team performance assessments
- Change requests
- Project Management Plan updates
- Project Documents updates
- Enterprise environmental factors updates
- Organizational process assets updates

HR: 5. Manage Team



- It is the process of tracking team member performance, providing feedback, resolving issues, and managing team changes to optimize project performance.
- Managing the project team requires a variety of management skills for fostering teamwork and integrating the efforts of team members to create high-performance teams.
- Team management involves a combination of skills with special emphasis on communication, conflict management, negotiation, and leadership.

HR: 5. Manage Team



Input

- Project management plan
- Project documents
- staff assignments
- Work perf ormance reports
- Team Perfromance assessements
- Enterprise environmental factors
- Organizational processs assets

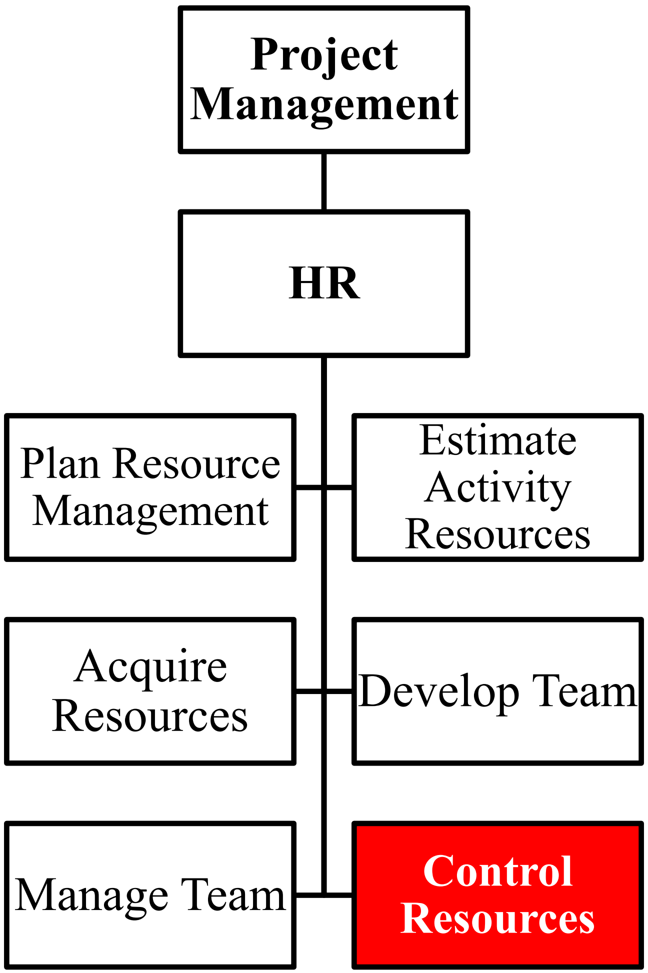
Tools & Techniques

- Interpersonal and team skills
- Project management information system

Output

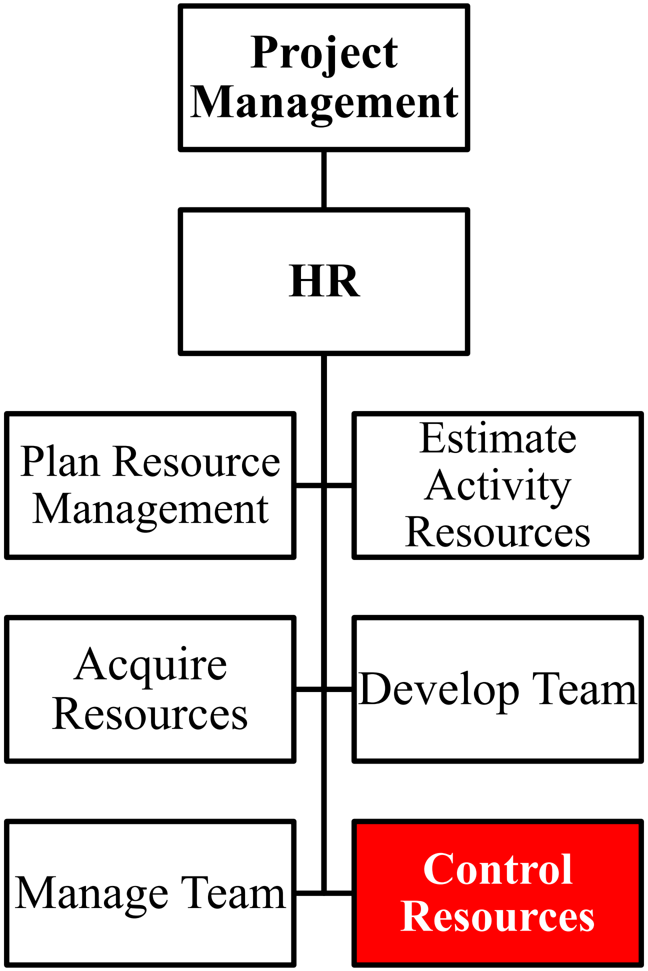
- Change requests
- Project management plan updates
- Project documents updates
- Enterprise environmental factors updates
- Organizational processs assets updates

HR: 6. Control Resources



- It is the process of ensuring that physical resources allocated to the project are available as planned, as well as monitoring the planned versus actual utilization of resources and taking corrective actions as necessary

HR: 6. Control Resources



Input

- Project management plan
- Project documents
- Work performance data
- Agreements
- Organizational process assets

Tools & Techniques

- Data analysis
- Problem solving
- Observation and conversation
- Interpersonal and team skills
- Project Management Information system

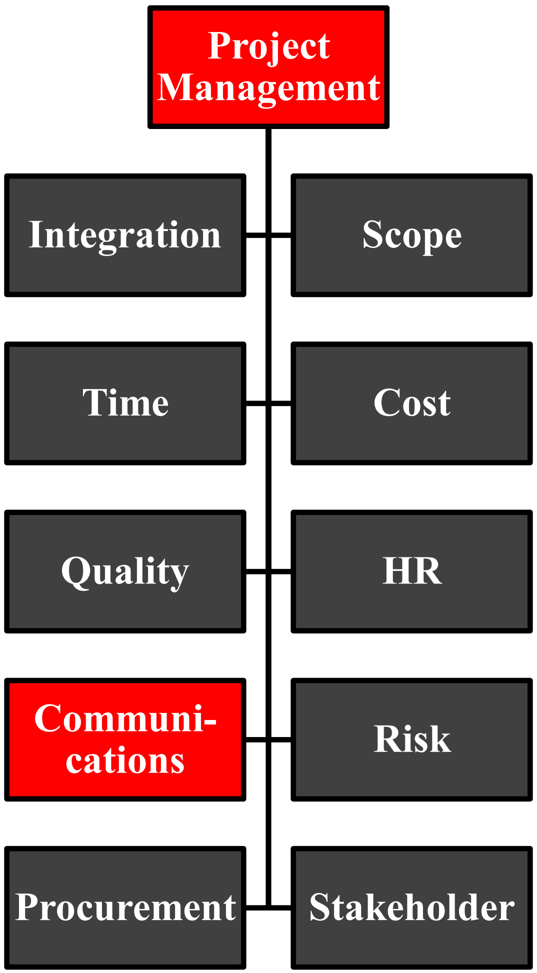
Output

- Work Performance Information
- Change Requests
- Project Management Plan updates
- Project Documents updates

Knowledge Areas

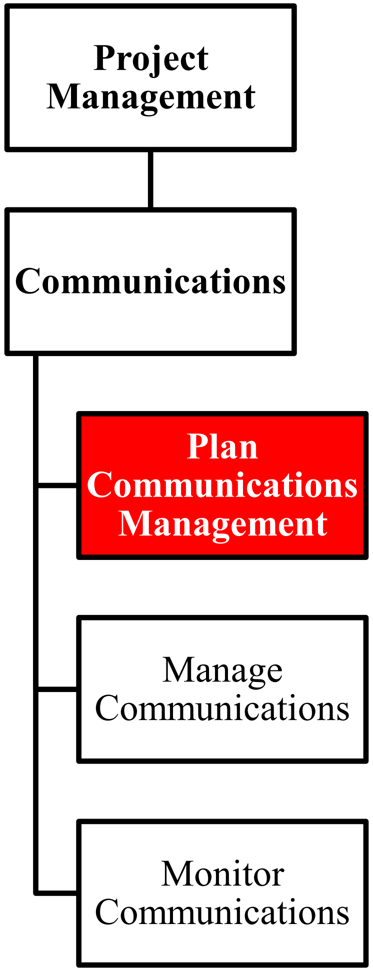
Project Communications

PMBOK : 7. Communications Management



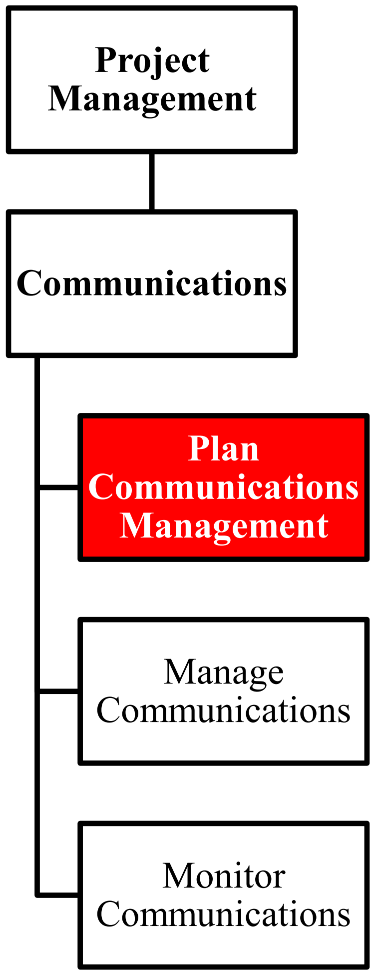
- Project Communications Management includes the processes for planning, collection, creation, distribution, storage, retrieval, management, control, monitoring, and the ultimate disposition of project information.
- Project managers spend most of their time communicating with team members and other project stakeholders.

Communications:
1. Plan Communications Management



- It is the process of developing an appropriate approach and plan for project communications based on stakeholder's information needs and requirements, and available organizational assets.
- Planning project communications is important to the ultimate success of any project.
- On most projects, communication planning is performed very early, such as during project management plan development.
- Effective communication means that the information is provided in the right format, at the right time, to the right audience, and with the right impact. Efficient communication means providing only the information that is needed.

Communications:
1. Plan Communications Management

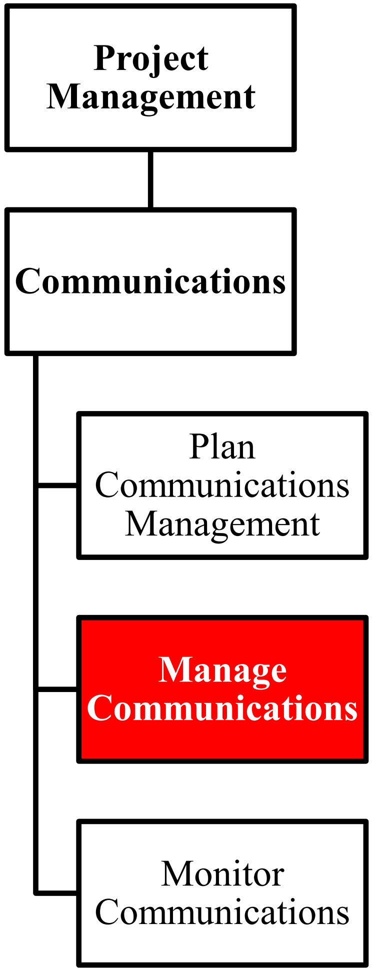


Input
- Project Charter
- Project management plan
- Project Documents
- Enterprise environmental factors
- Organizational process assets

Tools & Techniques
- Expert Judgement
- Communication requirements analysis
- Communication technology
- Communication models
- Communication methods
- Itnterpersonal and team skills
- Dat representation
- Meetings

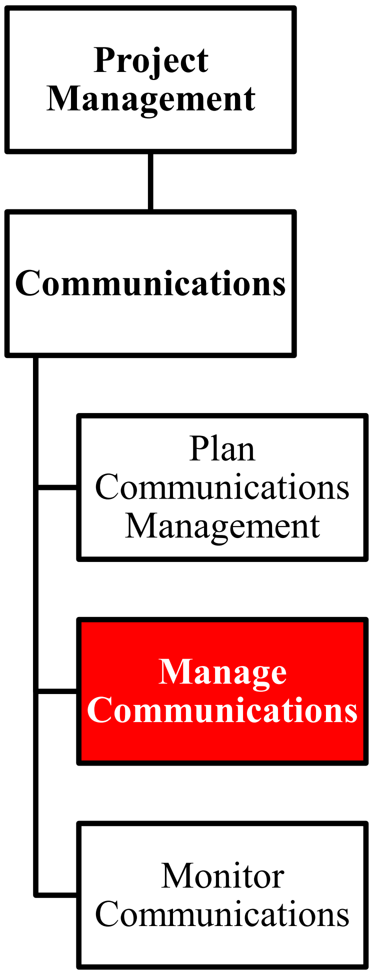
Output
- Communications management plan
- Project Management Plan updates
- Project documents updates

Communications:
2. Manage Communications



- It is the process of creating, collecting, distributing, storing, retrieving, and the ultimate disposition of project information in accordance to the communications management plan.
- This process goes beyond the distribution of relevant information and seeks to ensure that the information being communicated to project stakeholders has been appropriately generated, received and understood.

Communications:
2. Manage Communications



Input

- Project Management Plan
- Project documents
- Work performance reports
- Enterprise environmental factors
- Organizational process assets

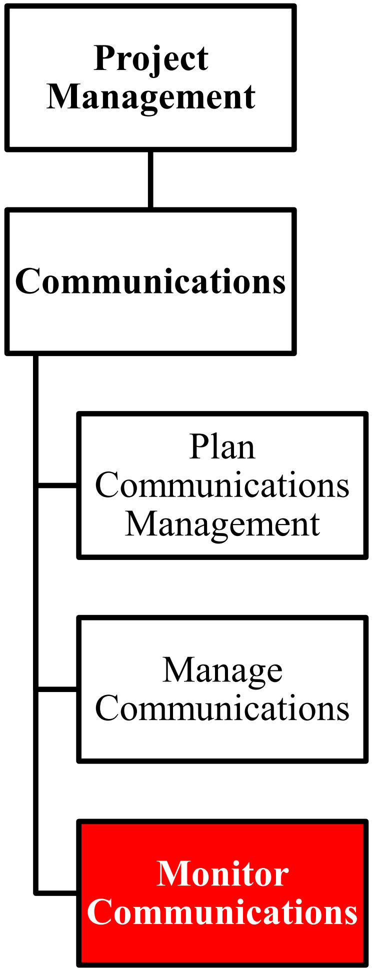
Tools & Techniques

- Communication technology
- Communication methods
- Communication skills
- Project Management Information system
- Project reporting
- Interpersonal and team skills
- Meetings

Output

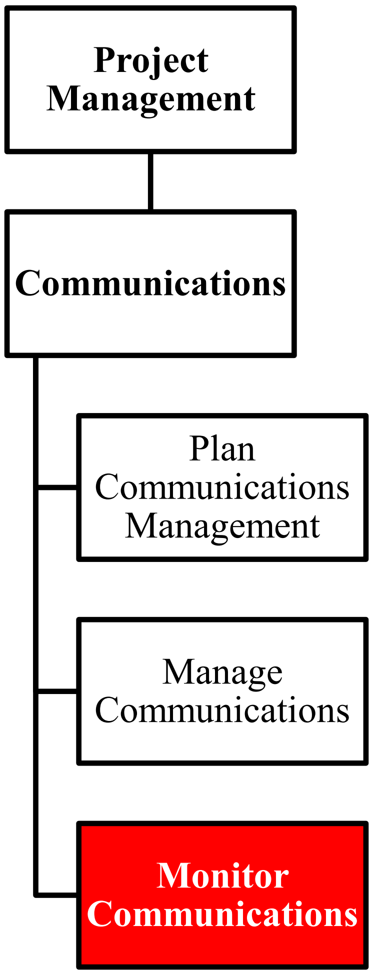
- Project communications
- Project management plan updates
- Project documents updates
- Organizational process assets updates

Communications:
3. Monitor Communications



- It is the process of monitoring and controlling communications throughout the entire project life cycle to ensure the information needs of the project stakeholders are met.
- Specific communication elements, such as issues or key performance indicators (e.g., actual vs. planned schedule, cost, and quality), may trigger an immediate revision, while others may not.
- The impact and repercussions of project communications should be carefully evaluated and controlled to ensure that the right message is delivered to the right audience at the right time

Communications:
3. Monitor Communications



Input

- Project management plan
- Project documents
- Work performance data
- Enterprise environmental factors
- Organizational process assets

Tools & Techniques

- Expert judgment
- Project management information system
- Data representation
- Interpersonal and team skills
- Meetings

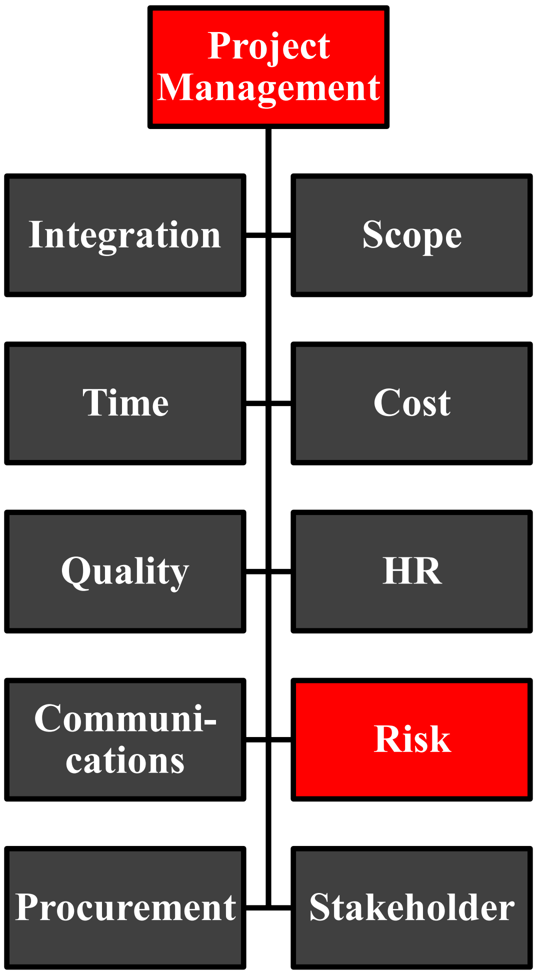
Output

- Work performance information
- Change requests
- Project management plan updates
- Project documents updates

Knowledge Areas

Project Risk

PMBOK : 8. Risk Management



- It aims to increase the likelihood and impact of positive events and decrease the likelihood and impact of negative events on the project.

Risk: 1. Plan Risk Management

- It defines how to conduct risk management activities for a project.
- The risk management plan is vital to communicate with and obtain agreement and support from all stakeholders
- Careful and explicit planning enhances the probability of success for risk management processes.
- The Plan Risk Management process should begin when a project is conceived and should be completed early during project planning.

Risk: 1. Plan Risk Management

Input

- Project charter
- Project Management Plan
- Project Documents
- Enterprise environmental factors
- Organizational process assets

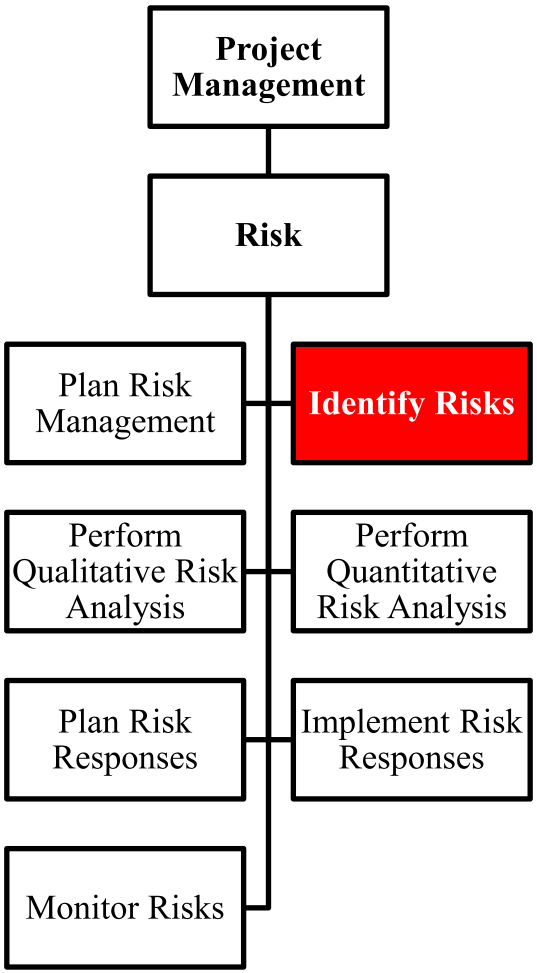
Tools & Techniques

- Expert Judgement
- Data Analysis
- Meetings

Output

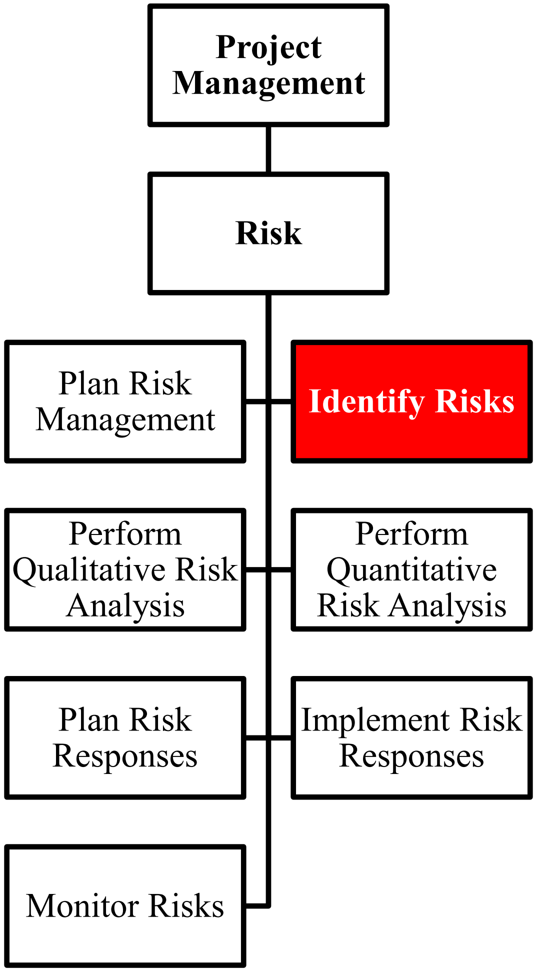
- Risk Management Plan

Risk: 2. Identify Risks



- It determines which risks may affect the project and documents their characteristics.
- Identifying risks is an iterative process, because new risks may evolve or become known as the project progresses.
- The risk statement should support the ability to compare the relative effect of one risk against others on the project.

Risk: 2. Identify Risks



Input

- Project Management Plan
- Project Documents
- Agreements
- Procurement documentation
- Enterprise environmental factors
- Organizational process assets

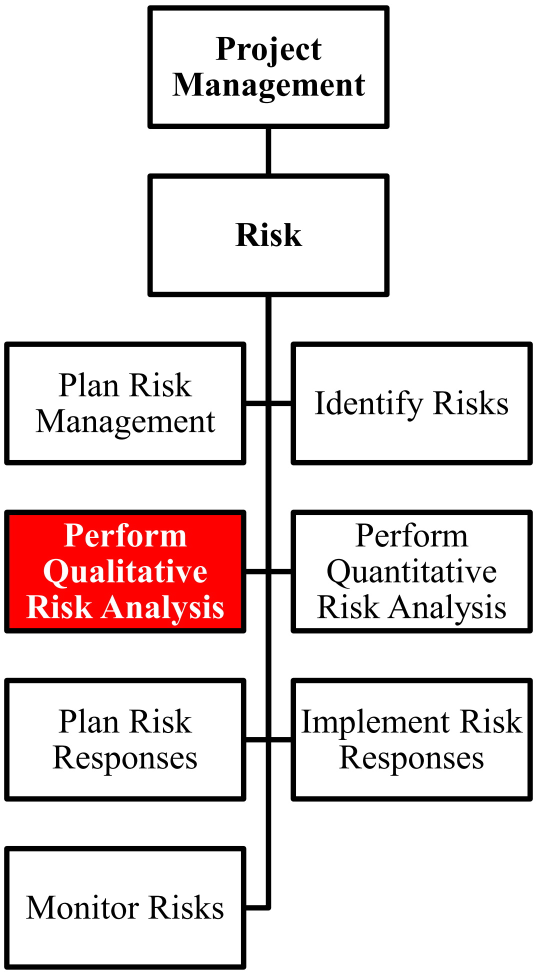
Tools & Techniques

- Expert judgment
- Data Gathering
- Data Analysis
- Interpersonal and team skills
- Prompt List
- Meetings

Output

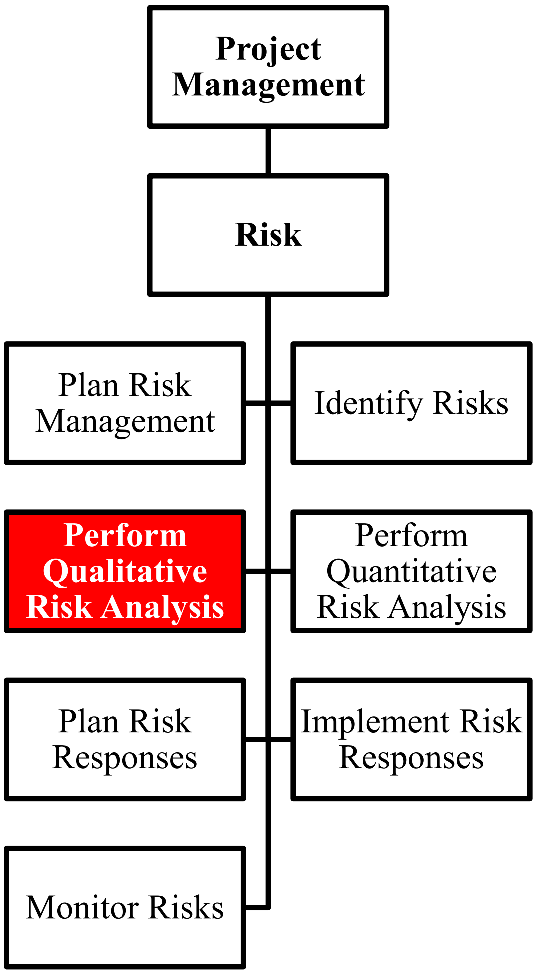
- Risk Register
- Risk Report
- Project Documents updates

Risk: 3. Perform Qualitative Risk Analysis



- It prioritizes risks for further analysis or action by assessing and combining their probability of occurrence and impact.
- It assesses the priority of identified risks using their relative likelihood of occurrence, the corresponding impact on project objectives if the risks occur, as well as other factors such as the time frame for response and the organization's risk tolerance
- An evaluation of the quality of the available information on project risks helps to clarify the assessment of the risk's importance to the project.
- The process is performed regularly throughout the project life cycle, as defined in the project's risk management plan.

Risk: 3. Perform Qualitative Risk Analysis



Input

- Project Management Plan
- Project Documents
- Enterprise environmental factors
- Organizational process assets

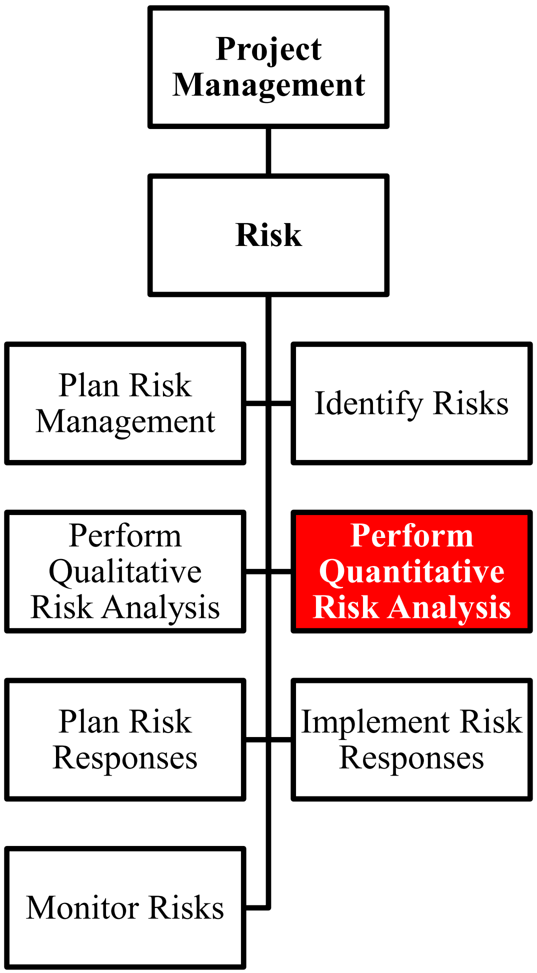
Tools & Techniques

- Expert judgment
- Data Gathering
- Data Analysis
- Interpersonal and team skills
- Risk Categorization
- Data Representation
- Meetings

Output

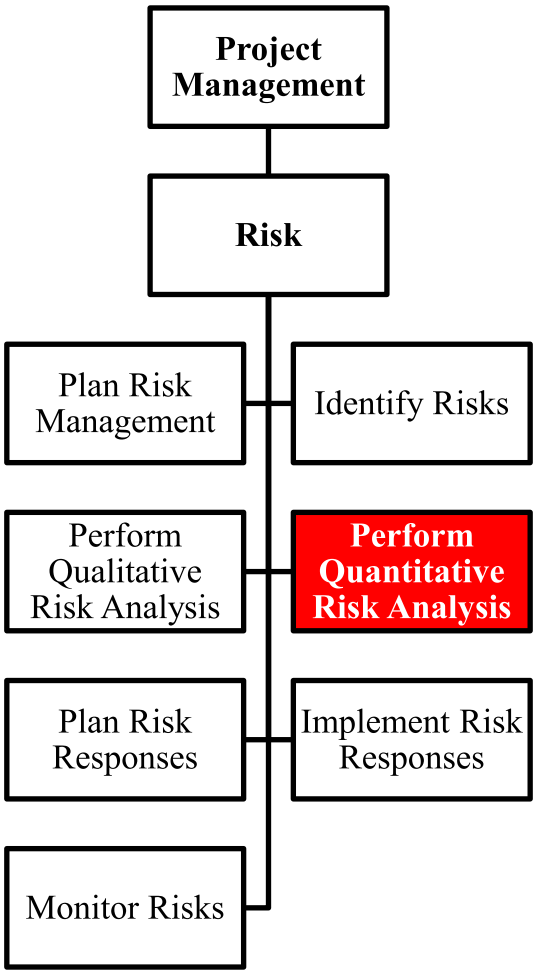
- Project documents updates

Risk: 4. Perform Quantitative Risk Analysis



- It numerically analyzes the effect of identified risks on overall project objectives.
- It generally follows the Perform Qualitative Risk Analysis process.
- In some cases, data are not sufficient to develop appropriate models. The project manager should exercise expert judgment to determine the need for and the viability of quantitative risk analysis.
- The process should be repeated, as needed.

Risk: 4. Perform Quantitative Risk Analysis



Input

- Project management plan
- .Project documents
- .Enterprise environmental factors
- .Organizational process assets

Tools & Techniques

- Expert judgment
- Data gathering and representation techniques
- Interpersonal and team skills
- Representation of uncertainty
- Data analysis

Output

- Project documents updates

Risk: 5. Plan Risk Responses

- It follows the Perform Quantitative Risk Analysis process (if used).
- It develops options and actions to enhance opportunities and to reduce threats to project objectives.
- Each risk response requires an understanding of the mechanism that will address the risk.
- It includes the identification and assignment of one person (the owner for risk response) to take responsibility for each agreed-to and funded risk response.
- The process can involve choosing alternative strategies, executing a contingency or fallback plan, taking corrective action, and modifying the project management plan.

Risk: 5. Plan Risk Responses

Input

- Project management plan
- Project documents
- Enterprise environmental factors
- Organizational process assets

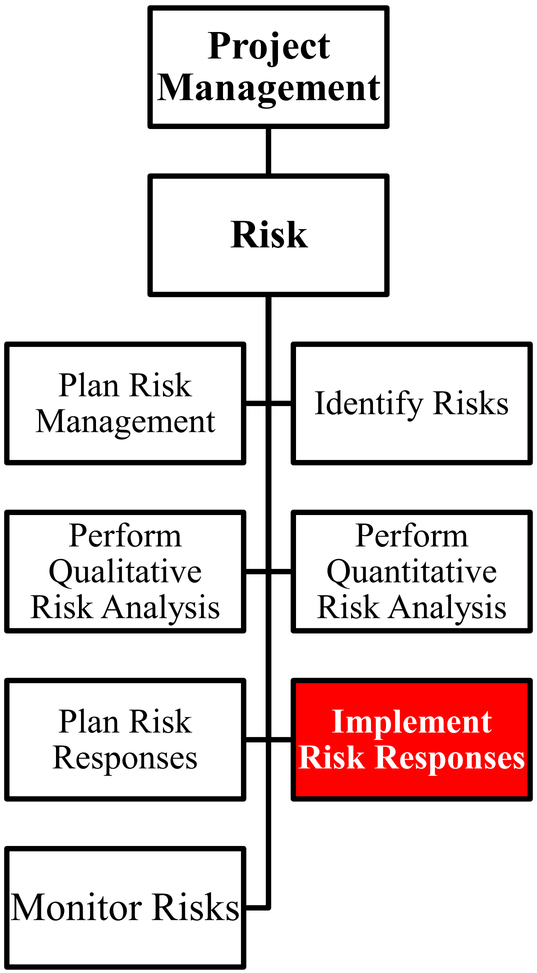
Tools & Techniques

- Expert judgment
- Data gathering
- Interpersonal and team skills
- Strategies for threats
- Strategies for opportunities
- Contingent response strategies
- Startegies for overall project risks
- Data analysis
- Decision makigng

Output

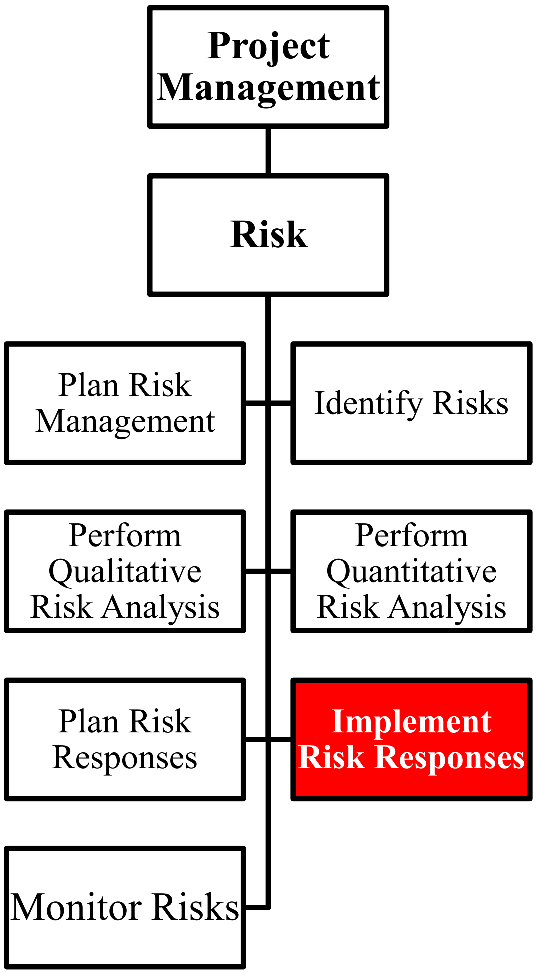
- Change requests
- Project management plan updates
- Project documents updates

Risk: 6. Implement Risk Responses



- It implements agreed risk response plans.
- It ensures that the agreed risk responses are executed
- This process is performed throughout the projects from the very beginning to the end
- The risk response owner reports periodically to the project manager on the effectiveness of the plan, any unanticipated effects, and any correction needed to handle the risk appropriately.

Risk: 6. Implement Risk Responses



Input

- Project management plan
- Project documents
- Organizational process assets

Tools & Techniques

- Expert judgment
- Interpersonal and team skills
- Project Management Information system

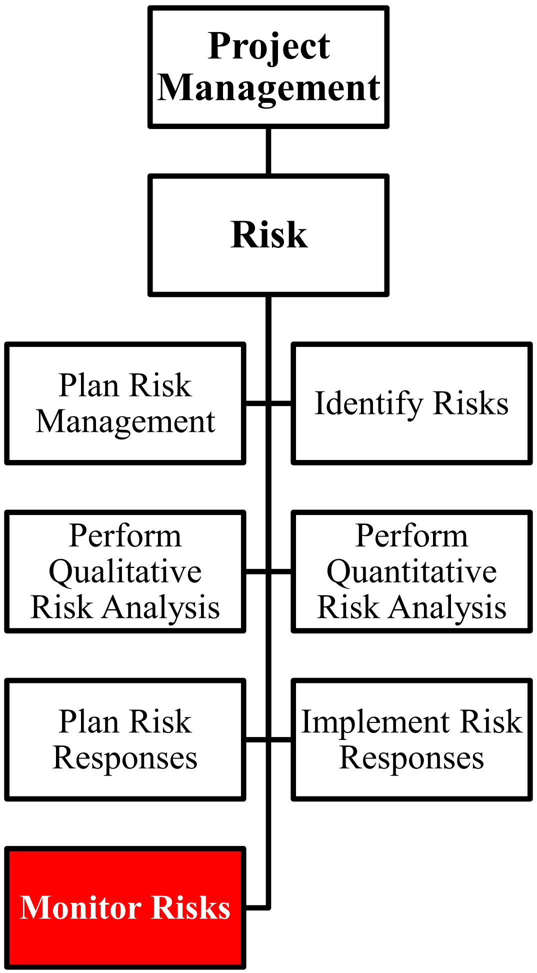
Output

- Change requests
- Project documents updates

Risk: 7. Monitor risks

- Planned risk responses in the risk register are executed during the life cycle of the project, but the project work should be continuously monitored for new, changing, and outdated risks.
- The process tracks identified risks, monitors residual risks, identifies new risks, and evaluates risk process effectiveness throughout the project.
- The process applies techniques, such as variance and trend analysis, which require the use of performance information generated during project execution.

Risk: 7. Monitor risks



Input

- Project management plan
- Project documents
- Work performance data
- Work performance reports

Tools & Techniques

- Data analysis
- Audits
- Meetings

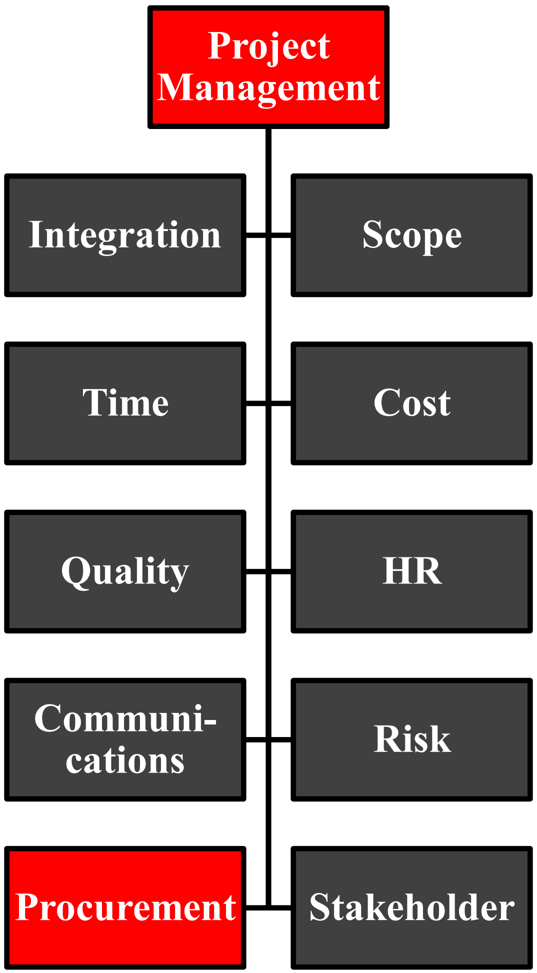
Output

- Work performance information
- Change requests
- Project management plan updates
- Project document updates
- Organizational process assets updates

Knowledge Areas

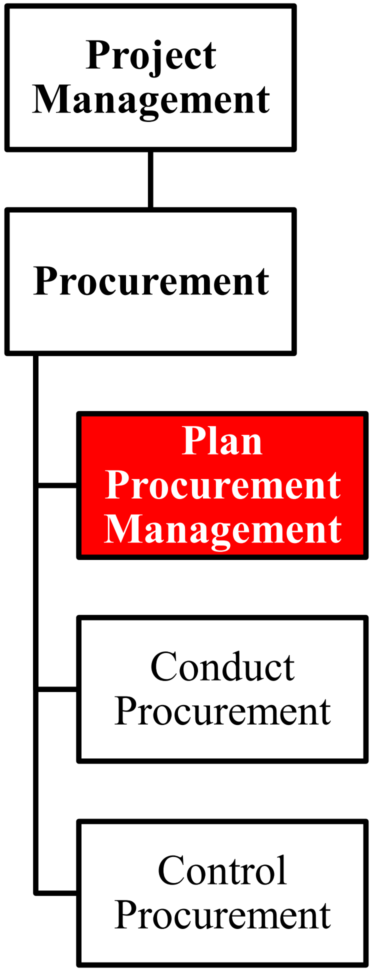
Project Procurement

PMBOK : 8. Procurement Management



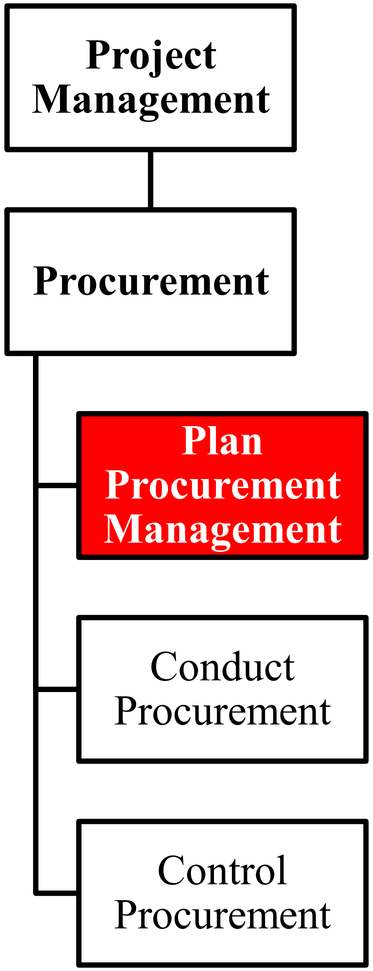
- Procurement Management includes
  - the processes necessary to purchase or acquire products, services, or results needed from outside the project team.
  - contract management and change control processes required to develop and administer contracts or purchase orders.
- In an external project, it includes controlling any contract issued by an outside organization (the buyer) that is acquiring deliverables from the project from the performing organization (the seller), and administering contractual obligations placed on the project team by the contract.
- The project manager generally is not authorized to sign contracts nor can be a procurement expert;however, he should be enough familiar with regulations to make intelligent and sound decisions

Procurement:1. Plan Procurement Management



- It documents project procurement decisions, specifies the procurement approach (e.g. bid, short list, etc.), and identifies the potential suppliers
- It determines whether to buy outside or acquire inside the organization
- This process is performed once only at predefined points of the project

Procurement: 1. Plan Procurement Management



Input

- Project charter
- Business documents
- Project management plan
- Project documents
- Enterprise environmental factors
- Organizational Process Assets

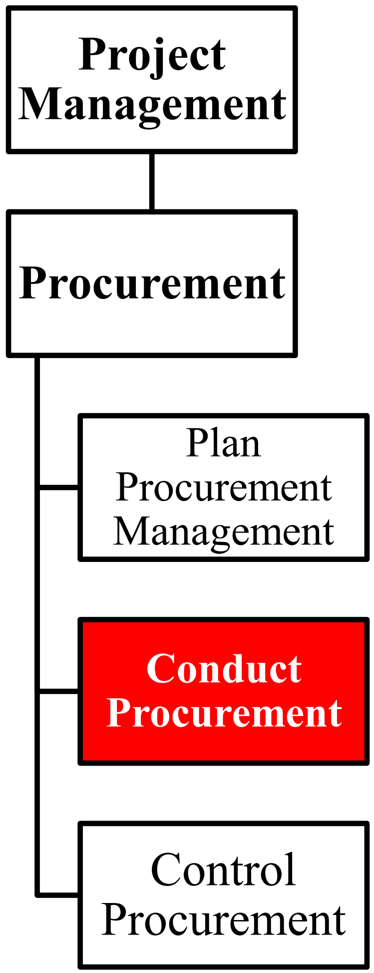
Tools & Techniques

- Expert judgment
- Data gathering
- Data analysis
- Source selection analysis
- Meetings

Output

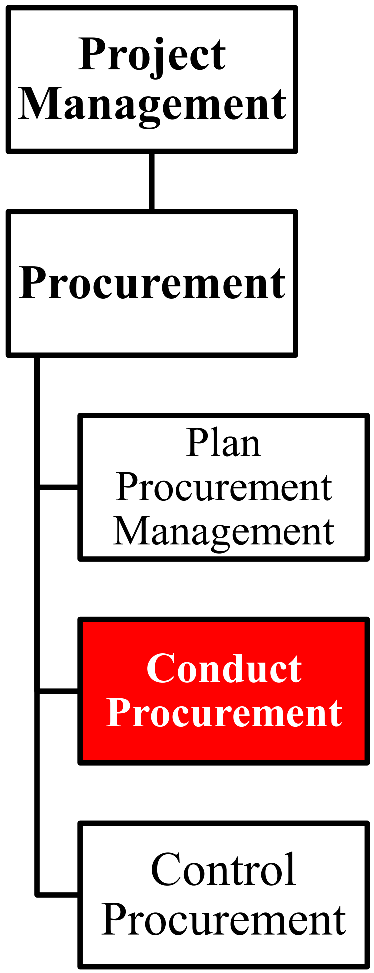
- Procurement management plan
- Procurement strategy
- Bid documents
- Procurement statement of work
- Source selection criteria
- Make or buy decisions
- Independent cost estimates
- Change requests
- Project documents updates
- Organizational process assets updates
- Project document updates
- Organizational process assets updates

Procurement: 2.Conduct Procurement



- It obtains the responses from suppliers, selects them, and awards contracts
- Thanks to the previous procurement planning phase suppliers can be better qualified
- The end outcome of the process are established agreements and contracts

Procurement: 2.Conduct Procurement



Input

- Project management plan
- Project documents
- Procurement documentation
- Seller proposals
- Enterprise environmental factors
- Organizational Process Assets

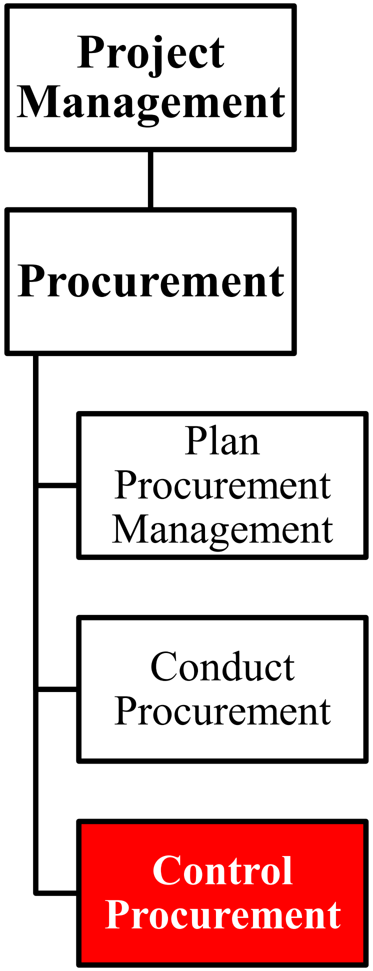
Tools & Techniques

- Expert judgment
- Advertising
- Bidder conferences
- Data analysis
- Interpersonal and team skills

Output

- Selected sellers
- Agreements
- Change requests
- Project documents updates
- Organizational process assets updates

Procurement: 3. Control Procurement



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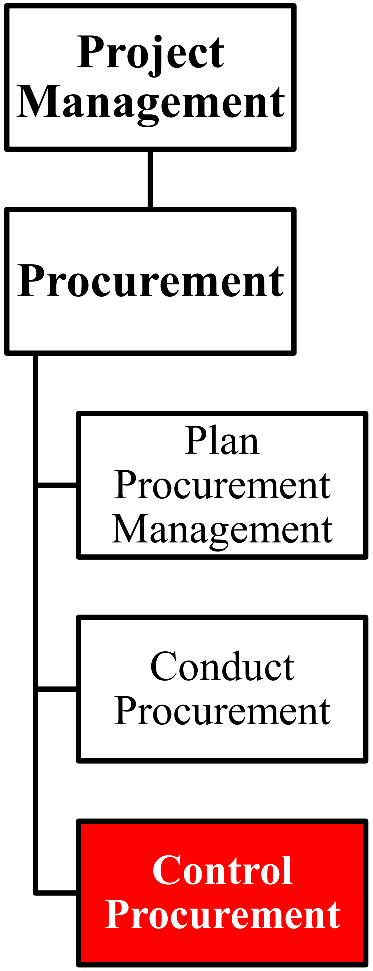
Procurement: 3. Control Procurement



Input

- Project management plan
- Project documents
- Agreements
- Procurement documentation
- Approved change requests
- Work performance data
- Enterprise environmental factors
- Organizational Process Assets

Tools & Techniques

- Expert judgment
- Claims administration
- Data analysis
- Inspection
- Audits

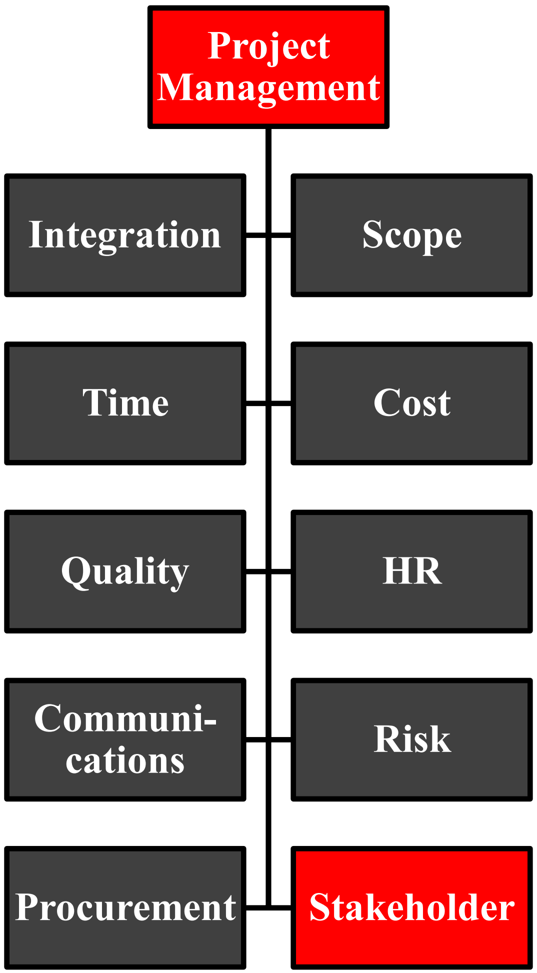
Output

- Closed procurements
- Work performance information
- Procurement documentation updates
- Change requests
- Project management plan updates
- Project documents updates
- Organizational process assets updates

Knowledge Areas

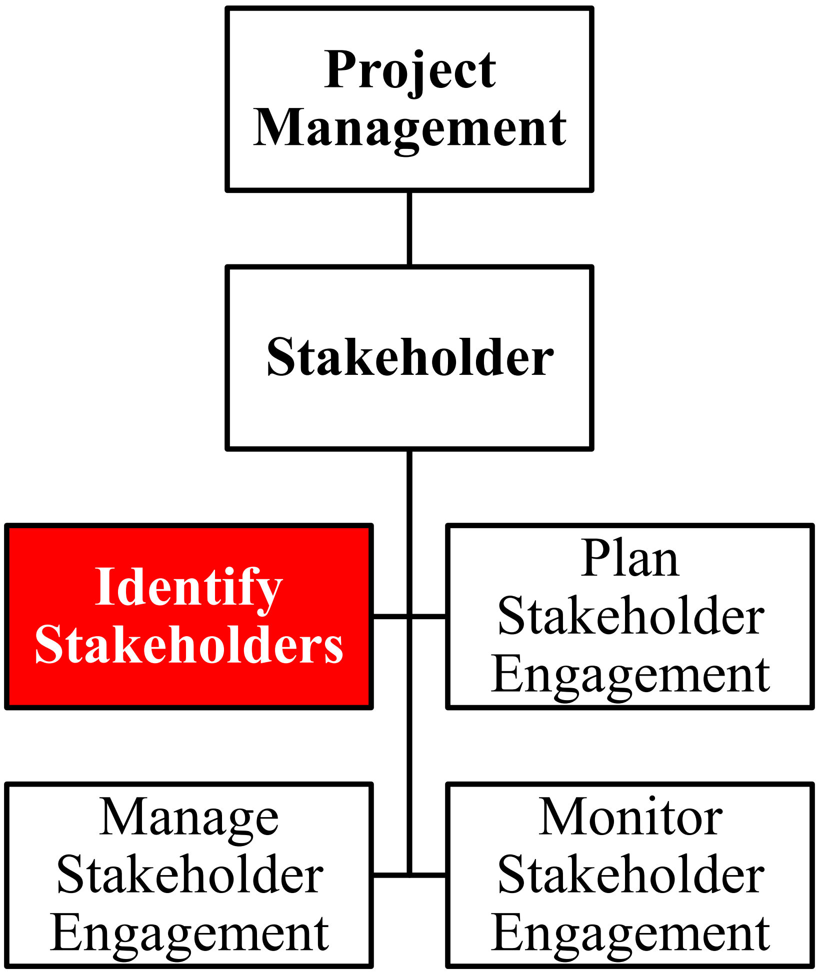
Project Stakeholder

PMBOK : 8. Stakeholder Management



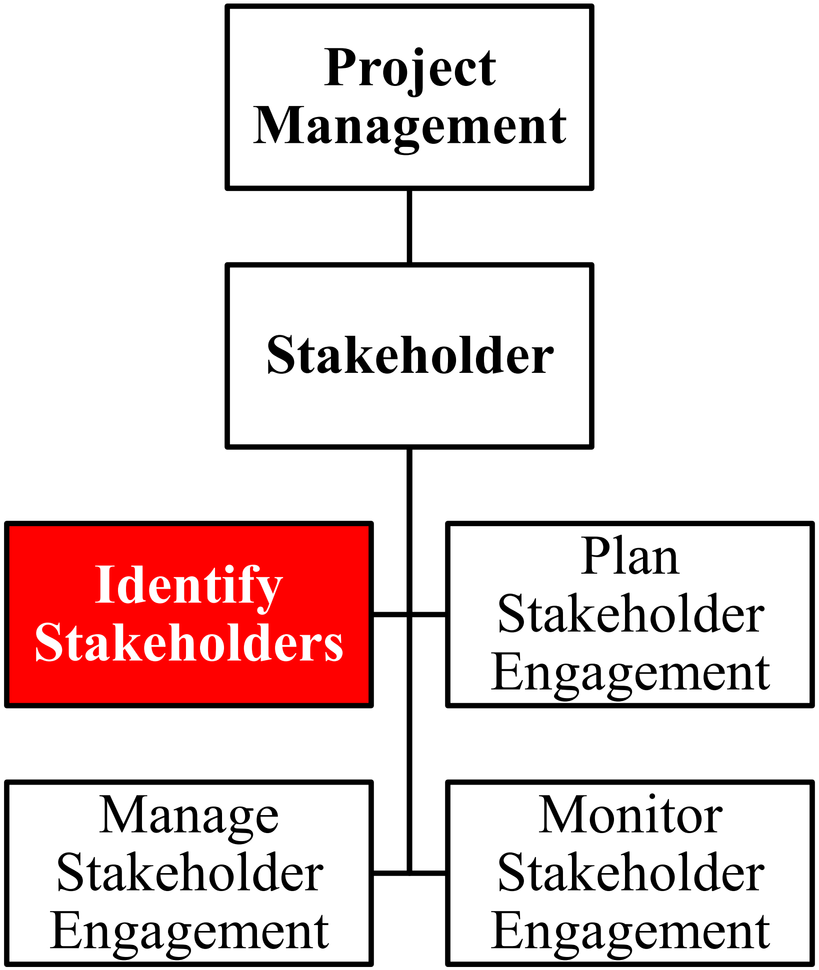
- Project Stakeholder Management includes the processes to identify the people, groups, or organizations that could impact or be impacted by the project, to analyze stakeholder expectations and their impact on the project, and to develop appropriate management strategies for effectively engaging stakeholders in project decisions and execution.
- Stakeholder management also focuses on continuous communication with stakeholders to understand their needs and expectations, addressing issues as they occur, managing conflicting interests and fostering appropriate stakeholder engagement in project decisions and activities.

Stakeholder: 1. Identify Stakeholders



- Project stakeholders are individuals, groups, or organizations who may affect, be affected by, or perceive themselves to be affected by the project.
- They include customers, sponsors, the performing organization, and the public who are actively involved in the project, or whose interests may be positively or negatively affected by the execution or completion of the project.
- The task identifies project stakeholders, analyzes and documents relevant information on their interests, involvement, interdependencies, influence, and potential impact on project success. Such initial assessment should be reviewed and updated regularly.

Stakeholder: 1. Identify Stakeholders



Input

- Project charter
- Business documents
- Project Management Plan
- Project documents
- Agreements
- Enterprise environmental factors
- Organizational process assets

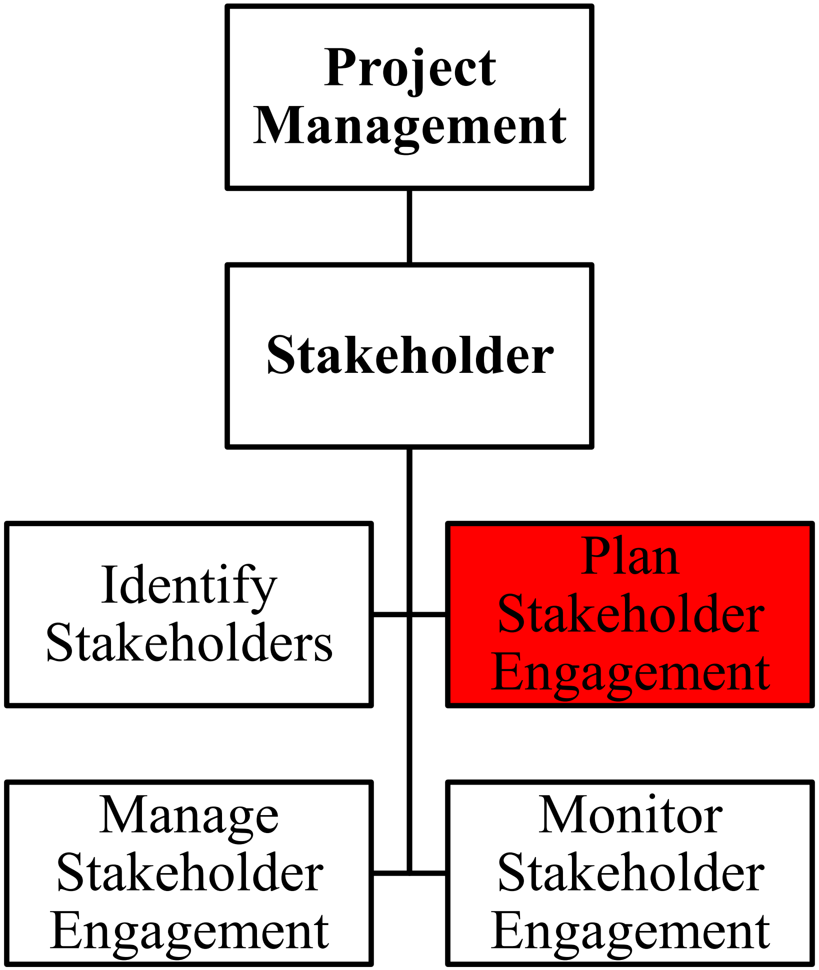
Tools & Techniques

- Expert Judgement
- Data Gathering
- Data Analysis
- Date Representaion
- Meetings

Output

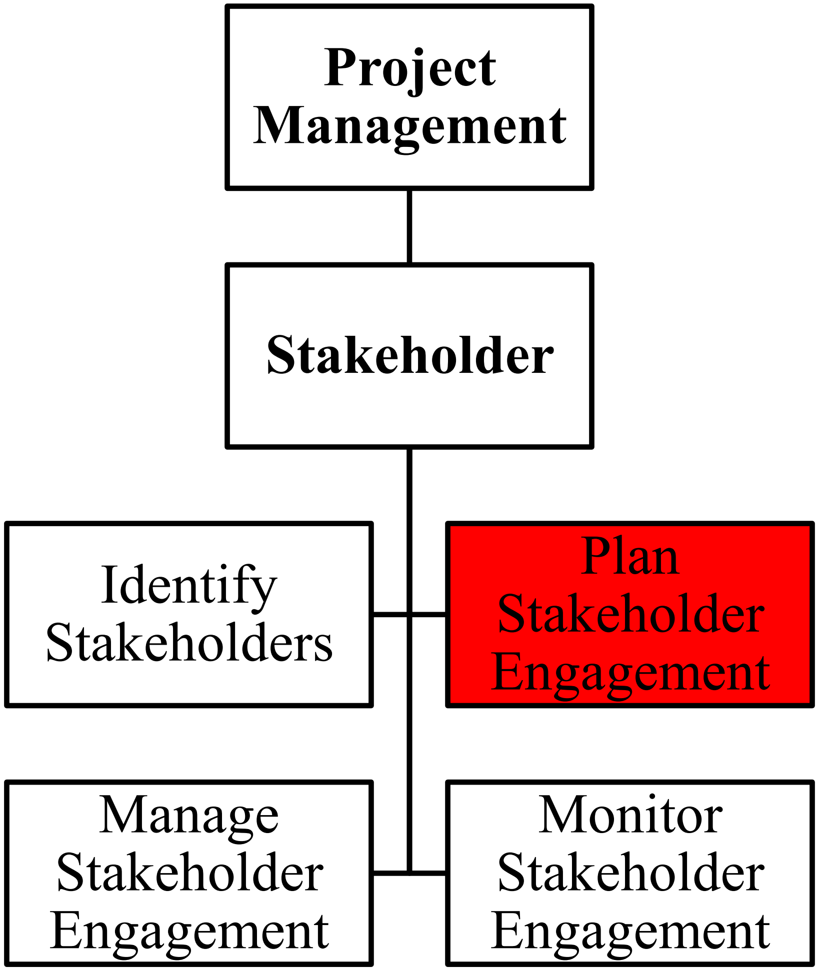
- Stakeholder Engagement Plan

Stakeholder: 2. Plan Stakeholder Engagement



- It is the process of developing appropriate management strategies to effectively engage stakeholders throughout the project life cycle, based on the analysis of their needs, interests, and potential impact on project success.
- Stakeholder management is about creation and maintenance of relationships between the project team and stakeholders, with the aim to satisfy their respective needs and requirements within project boundaries.
- This process generates the stakeholder management plan, which details how stakeholder management can be realized.
- As the project progresses, the stakeholder community and required level of engagement may change, therefore, stakeholder management planning is an iterative process.

Stakeholder: 2. Plan Stakeholder Engagement



Input

- Project charter
- Project management plan
- Project documents
- Agreements
- Enterprise environmental factors
- Organizational process assets

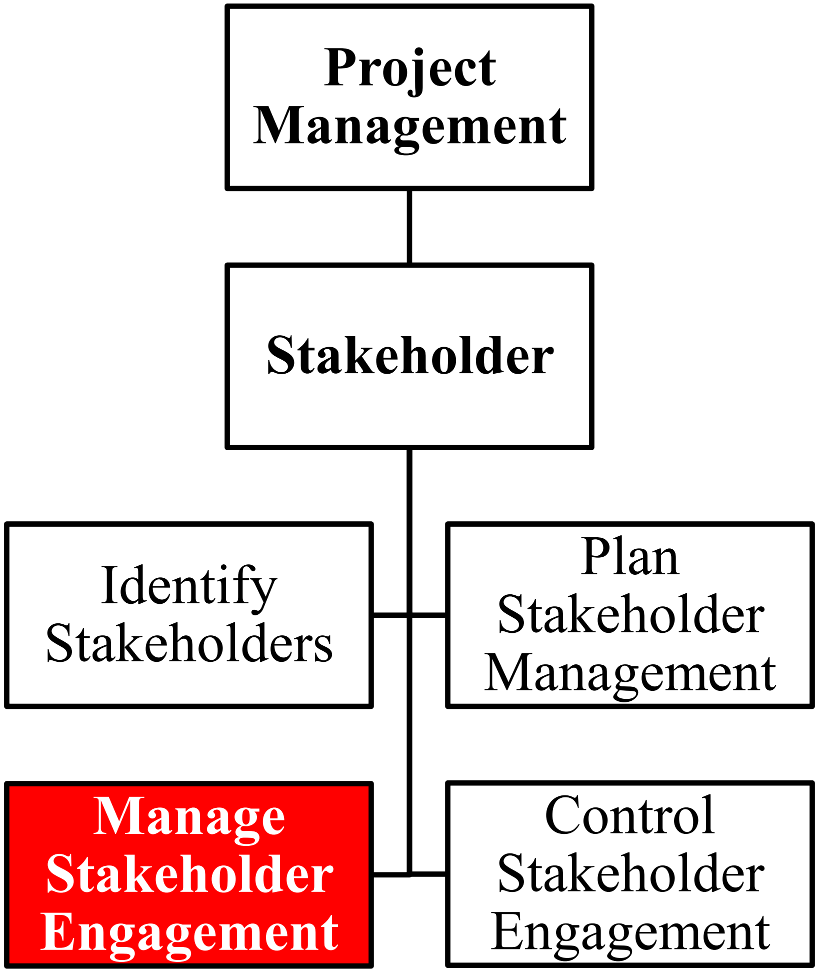
Tools & Techniques

- Expert Judgement
- Data Gathering
- Data Analysis
- Decision making
- Date Representaion
- Meetings

Output

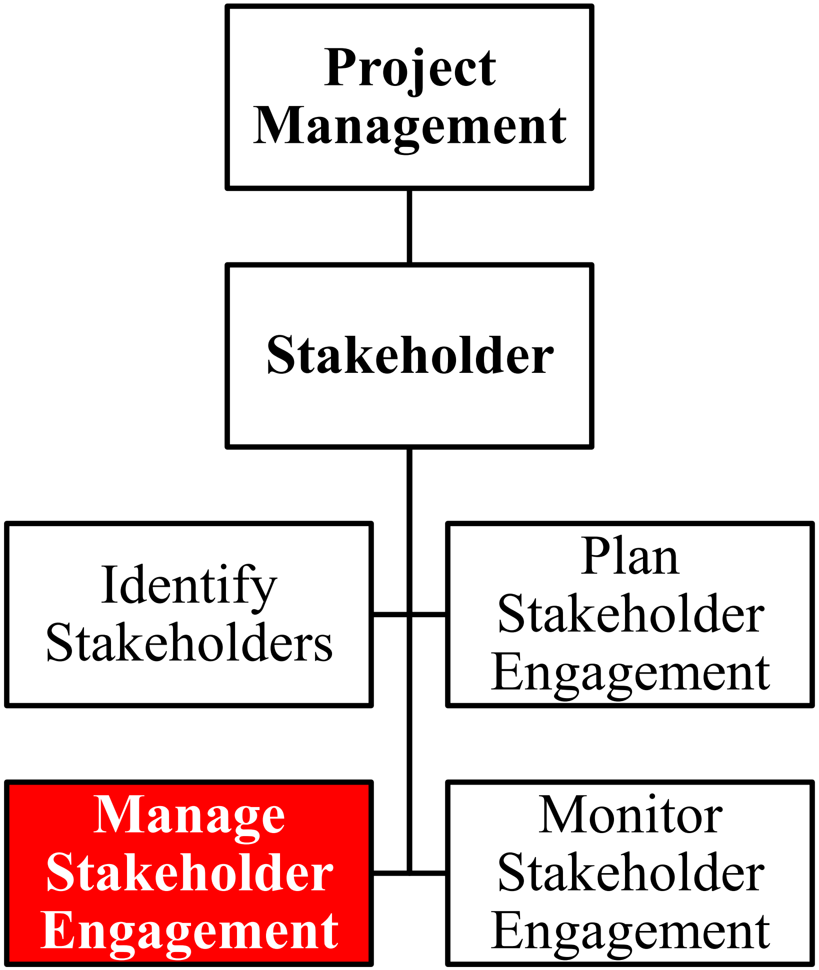
- Stakeholder Engagement Plan

Stakeholder:
3. Manage Stakeholder Engagement



- It is the process of communicating and working with stakeholders to meet their needs/expectations, address issues as they occur, and foster appropriate stakeholder engagement throughout the project life cycle.
- Managing stakeholder raises the probability of project success by ensuring that stakeholders clearly understand the project.
- Proactive actions can be taken to win support or minimize negative impacts.

Stakeholder:
3. Manage Stakeholder Engagement



Input

- Project management plan
- Project documents
- Enterprise environmental factors
- Organizational process assets

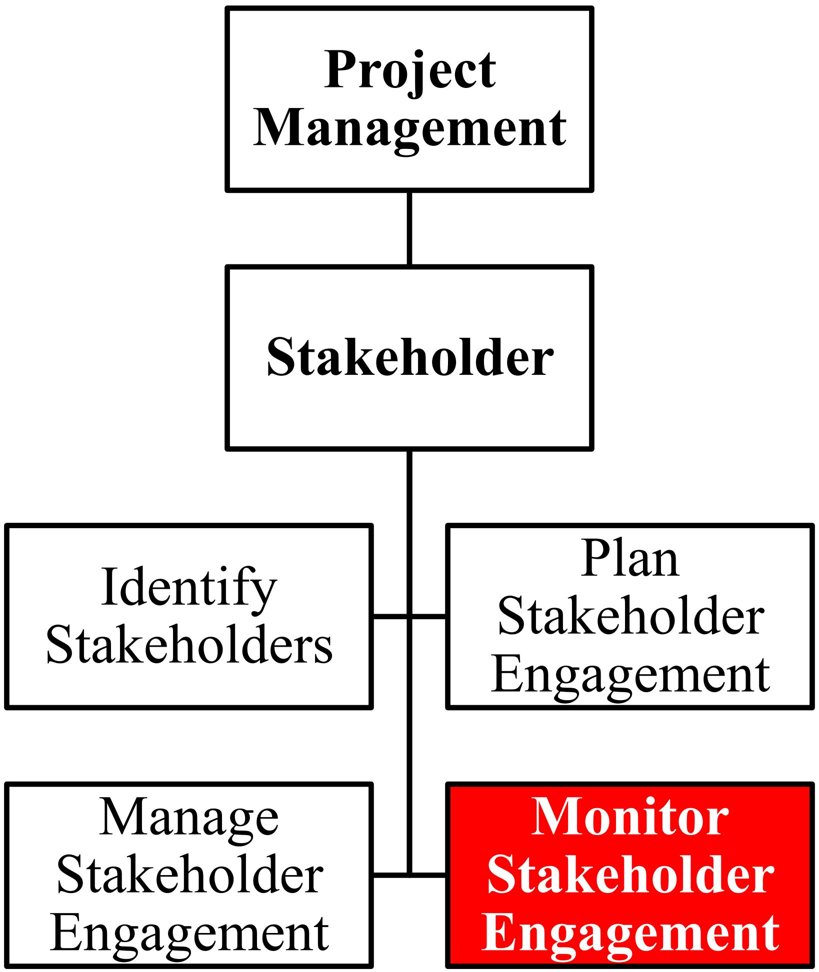
Tools & Techniques

- Expert judgement
- Communication skills
- Interpersonal and team skills
- Ground rules
- Meetings

Output

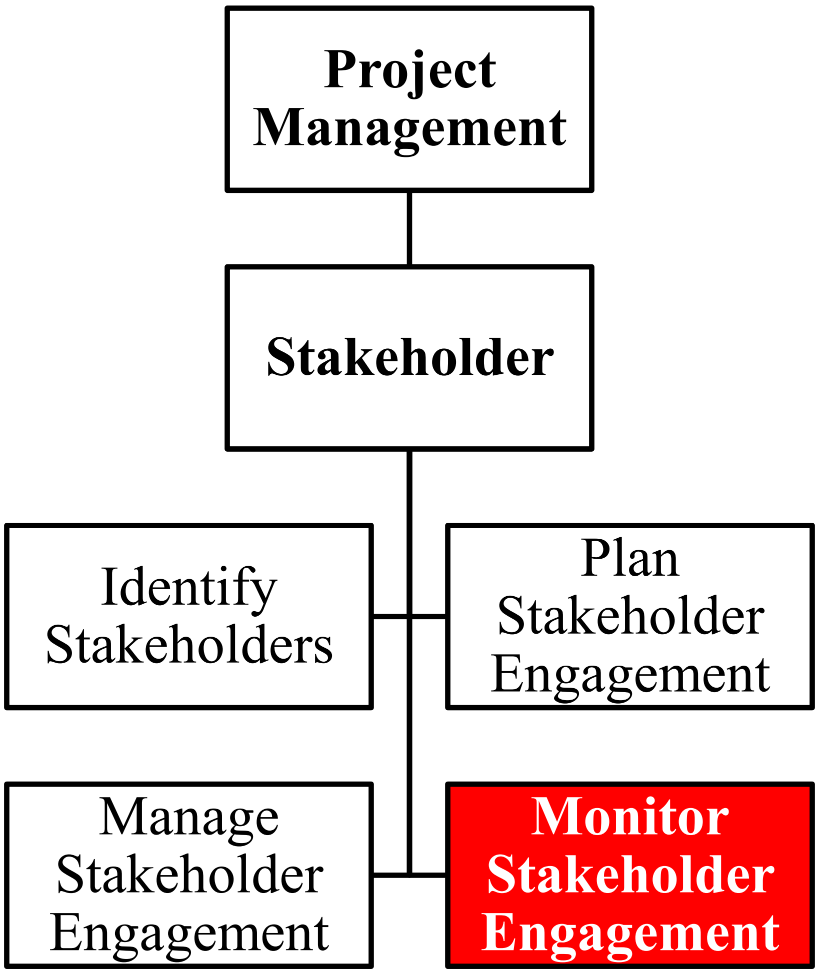
- Change requests
- Project management plan updates
- Project documents updates

Stakeholder:
4. Monitor Stakeholder Engagement



- It is the process of monitoring overall project stakeholder relationships and adjusting strategies and plans for engaging stakeholders.
- Stakeholder engagement activities are included in the stakeholder management plan and are executed during the life cycle of the project.
- Stakeholder engagement should be continuously controlled.

Stakeholder:
4. Monitor Stakeholder Engagement



Input

- Project management plan
- Project documents
- Work performance data
- Enterprise environmental factors
- Organizational process assets

Tools & Techniques

- Data Analysis
- Decision making
- Date Representaion
- Communication sklls
- Interpersonal and team sklls
- Meetings

Output

- Work performance information
- Change requests
- Project management plan updates
- Project documents updates